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Reducing CD33 expression to selectively protect therapeutic cells

Abstract

Systems and methods to selectively protect therapeutic cells by reducing CD33 expression in the therapeutic cells and targeting non-therapeutic cells with an anti-CD33 therapy. The selective protection results in the enrichment of the therapeutic cells while simultaneously targeting any diseased, malignant and/or non-therapeutic CD33 expressing cells within a subject.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a U.S. National Phase Application based on International Patent Application no. PCT/US2019/050859, filed Sep. 12, 2019, which claims priority to and the benefit of the earlier filing date of U.S. Provisional Application No. 62/730,164, filed Sep. 12, 2018, each of which is incorporated by reference herein in its entirety as if fully set forth herein.

REFERENCE TO SEQUENCE LISTING

(1) The Sequence Listing associated with this application is provided in text format in lieu of a paper copy and is hereby incorporated by reference into the specification. The name of the text file containing the Sequence Listing is 2G51591_ST25.txt. The text file is 224 KB, was created on Feb. 25, 2021, and is being submitted electronically via EFS-Web.

FIELD OF THE DISCLOSURE

(2) The current disclosure provides systems and methods to selectively protect therapeutic cells by reducing CD33 expression in the therapeutic cells and targeting non-therapeutic or unmodified native cells with an anti-CD33 therapy. The selective protection results in the enrichment of the therapeutic cells while simultaneously targeting any diseased, malignant and/or non-therapeutic CD33 expressing cells within a subject.

BACKGROUND OF THE DISCLOSURE

(3) Hematopoietic stem cells (HSC) are stem cells that can give rise to all blood cell types such as the white blood cells of the immune system (e.g., virus-fighting T cells and antibody-producing B cells), platelets, and red blood cells. The therapeutic administration of HSC can be used to treat a variety of adverse conditions including immune deficiency diseases, non-malignant blood disorders, cancers, infections, and radiation exposure (e.g., cancer treatment, accidental, or attack-based).

(4) As particular examples of conditions that can be treated with HSC, more than 80 primary immune deficiency diseases are recognized by the World Health Organization. These diseases are characterized by an intrinsic defect in the immune system in which, in some cases, the body is unable to produce any or enough antibodies against infection. In other cases, cellular defenses to fight infection fail to work properly. Typically, primary immune deficiencies are inherited disorders.

(5) One example of a primary immune deficiency is Fanconi anemia (FA). FA is an inherited blood disorder that leads to bone marrow (BM) failure. It is characterized, in part, by a deficient DNA-repair mechanism. At least 20% of patients with FA develop cancers such as acute myeloid leukemias and cancers of the skin, liver, gastrointestinal tract, and gynecological system. The skin and gastrointestinal tumors are usually squamous cell carcinomas. The average age of FA patients who develop cancer is 15 years for leukemia, 16 years for liver tumors, and 23 years for other tumors.

- (6) X-linked severe combined immunodeficiency (SCID-X1) is both a cellular and humoral immune depletion caused by mutations in the common gamma chain gene (γ C), which result in the absence of T and natural killer (NK) lymphocytes and the presence of nonfunctional B lymphocytes. SCID-X1 is fatal in the first two years of life unless the immune system is reconstituted, for example, through bone marrow transplant (BMT) or cell and gene therapy.
- (7) Secondary, or acquired, immune deficiencies are not the result of inherited genetic abnormalities, but rather occur in individuals in which the immune system is compromised by factors outside the immune system. Examples include trauma, viruses, chemotherapy, toxins, and pollution. Acquired immunodeficiency syndrome (AIDS) is an example of a secondary immune deficiency disorder caused by a virus, the human immunodeficiency virus (HIV), in which a depletion of T lymphocytes renders the body unable to fight infection.
- (8) Immune deficiencies, blood cancers, and other blood-related disorders can be treated by a BMT or by administering hematopoietic cells. In some instances, the hematopoietic cells can be genetically modified to provide a functioning gene that the patient lacks. In each of these scenarios, however, it is important to remove a patient's existing hematopoietic system, so that diseased cells do not remain following treatment. Removing a patient's existing hematopoietic system is most often accomplished utilizing a process referred to as conditioning.
- (9) Traditionally, conditioning has involved the delivery of maximally tolerated doses of chemotherapeutic agents with nonoverlapping toxicities, with or without radiation. Current conditioning regimens involve total body irradiation (TBI) and/or cytotoxic drugs. These regimens are non-targeted, genotoxic, and have multiple short- and long-term adverse effects such as an increased risk of developing DNA repair disorders, interstitial pneumonitis, idiopathic pulmonary fibrosis, reduced lung pulmonary function, renal damage, sinusoidal obstruction syndrome (SOS), infertility, cataract formation, hyperthyroidism, thyroiditis, and secondary cancers. Besides morbidity, these regimens are also associated with significant mortality. Therefore, methods to reduce or eliminate the need for conditioning in these patients is desperately needed.
- (10) CD33 is a protein that is expressed on normal hematopoietic cells as they mature. Thus, therapeutic cells administered as a treatment for immune deficiencies or other blood-related disorders express or begin to express CD33. CD33, however, is also widely expressed on malignant cells in patients with myeloid neoplasms, such as acute myeloid leukemia (AML). Accordingly, CD33 represents a cellular marker for both administered therapeutic cells and unwanted non-treated, cancerous, and/or malignant cells within a patient.
- (11) Because CD33 is a target to kill diseased and/or unwanted cells, there has been great interest in developing therapeutic antibodies directed at CD33. However, because CD33 is also expressed on normal immune cells and other non-malignant cells, treatments that target it have created what are referred to as significant “on-target, but off-leukemia” or “on-target, off-tumor” effects.” Such effects include suppression of the blood and immune system in the forms of severe thrombocytopenia, neutropenia, and monocytopenia in patients. For example, the CD33 antibody-drug conjugate (ADC) gemtuzumab ozogamicin (GO; MYLOTARG®, Pfizer, New York, NY) when given alone causes almost universal severe thrombocytopenia and neutropenia, and combined with conventional chemotherapy resulted in prolongation of cytopenias and increased non-relapse related mortality, in part due to fatal infections, in some clinical trials.

SUMMARY OF THE DISCLOSURE

- (12) The current disclosure provides systems and methods to protect beneficial therapeutic hematopoietic cells from anti-CD33 therapies while leaving residual diseased cells susceptible to anti-CD33 treatments. The systems and methods achieve this benefit by genetically modifying HSC to have reduced or eliminated expression of CD33, thus protecting them from anti-CD33 based therapies. In this manner, genetically modified therapeutic cells will not be harmed by concurrent or subsequent anti-CD33 therapies a patient may receive. However, pre-existing CD33-expressing cells in the patient and/or administered cells that lack the genetic modification will not

be protected, resulting in positive selection for the therapeutic cells over other cells.

(13) In particular embodiments, the HSC genetically modified to have reduced CD33 expression are also genetically modified for an additional therapeutic purpose. The genetic modification for an additional therapeutic purpose can provide a gene to treat a disorder such as an immune deficiency (e.g., Fanconi anemia, SCID, HIV), a cancer (e.g., leukemia, lymphoma, solid tumor), a blood-related disorder (e.g., sickle cell disease), a lysosomal storage disease (e.g., Pompe disease, Gaucher disease, Fabry disease, Mucopolysaccharidosis type I), or provide a therapeutic cassette that encodes a chimeric antigen receptor, engineered T-cell receptor, checkpoint inhibitor, or therapeutic antibody.

(14) In particular embodiments, when a therapeutic gene is provided in addition to reduced CD33 expression, the systems and methods disclosed herein can provide an important advance by ensuring that only cells that have been genetically modified with the therapeutic gene also have reduced CD33 expression that results in cellular protection. In particular embodiments, this advance is achieved by linking the therapeutic gene and a CD33 blocking molecule in a single intracellular delivery vehicle. In particular embodiments, the single intracellular delivery vehicle is a viral vector.

(15) In particular embodiments, the CD33 blocking molecule is an shRNA or siRNA CD33 blocking molecule combined with a therapeutic gene by inclusion within a common viral vector. In particular embodiments, the CD33 blocking molecule is shRNA referred to herein as shRNA4 encoded by SEQ ID NO: 8 or shRNA5 encoded by SEQ ID NO: 9. In particular embodiments, the viral vector is a lentiviral vector, a foamy viral vector, or an adenoviral vector.

(16) In particular embodiments, the systems and methods described herein further provide systems and methods to reduce or eliminate the need for genotoxic conditioning. In particular embodiments, the systems and methods allow the targeting and removal of any remaining CD33-expressing cells following conditioning in preparation for a bone marrow transplant or administration of therapeutic cells (e.g., genetically-modified therapeutic cells). In particular embodiments, the systems and methods clear the bone marrow niche and allow for further expansion of gene-corrected cells. In particular embodiments, the systems and methods deplete residual disease-related cells. The therapeutically administered cells with reduced CD33 expression are protected from the CD33-targeting and are able to reconstitute the patient's blood and immune systems. In combination, the approach can eliminate residual non-modified CD33-expressing cells, resulting in a completely corrected hematopoiesis, and minimizing risks of future myeloid malignancy after gene therapy or allogeneic transplantation.

(17) In particular embodiments, the genetically-modified therapeutic cells described herein are administered alone or in combination with a CD33-targeting treatment, such as an anti-CD33 antibody, an anti-CD33 immunotoxin, an anti-CD33 antibody-drug conjugate, an anti-CD33 antibody-radioisotope conjugate, an anti-CD33 bispecific antibody, an anti-CD33 bispecific immune cell activating antibody, an anti-CD33 trispecific antibody, and/or an anti-CD33 chimeric antigen receptor (CAR) or T cell receptor (TCR) modified immune cell.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) Many of the drawings submitted herein are better understood in color. Applicant considers the color versions of the drawings as part of the original submission and reserves the right to present color images of the drawings in later proceedings.

(2) FIGS. 1A, 1B. Protein sequences for CD33. (1A) Annotated sequence alignments for CD33 proteins from *Macaca fascicularis* (SEQ ID NO: 1), *Homo sapiens* (SEQ ID NO: 2), and *Mus musculus* (SEQ ID NO: 3). (1B) Protein sequences for full-length CD33 (SEQ ID NO: 4) and

CD33.sup.ΔE2 (SEQ ID NO: 5).

(3) FIG. 2. Schematic of antibody targeting of CD33.

(4) FIG. 3. Schematic of the different shRNA target sites within the CD33 coding region and sequences of the shRNA (SEQ ID NOs: 6-10) as well as of the respective siRNA sequences targeting CD33 (SEQ ID NOs: 11-15).

(5) FIG. 4. Efficient shRNA-mediated CD33 knockdown in the AML cell line ML1 (acute myeloblastic leukemia). ML1 cells were transduced with increasing multiplicities of infection (MOI) of control or shRNA-containing lentiviral vectors, and surface CD33 expression measured after 1 week. ML1 cells transduced with lentiviral vector including shRNA4 or shRNA5 show a dose dependent reduction in CD33 expression.

(6) FIG. 5. Efficient knockdown of CD33 in human CD34⁺ hematopoietic stem and progenitor cells (HSPCs) using shRNA4, shRNA5, or a control vector. CD33 surface expression was measured by flow cytometry on day 9 after transduction.

(7) FIG. 6. Time course of CD33 surface expression in human CD34⁺ HSPCs treated with shRNA4 (triangle), shRNA5 (circle), or control vector (square) lentiviral vectors.

(8) FIG. 7. Efficient knockdown of CD33 in ML1 cells treated with shRNA1, shRNA4, shRNA5, and a control vector was assessed using flow cytometry. These cells were subsequently tested for sensitivity to GO treatment in FIG. 8.

(9) FIG. 8. Gemtuzumab ozogamicin (GO)-induced toxicity in ML1 cells left untreated or treated with an empty vector, shRNA1, shRNA4, or shRNA5. Cytotoxicity was measured by DAPI (4',6-diamidino-2-phenylindole) staining after exposure to increasing concentrations of GO for 3 days. Frequency of DAPI positive cells is shown on the top of each bar. Transduction with lentiviral vector including shRNA4 or shRNA5 protected cells from the cytotoxic effects of GO.

(10) FIGS. 9A-9C. Efficient transduction and knockdown of CD33 in human CD34⁺ cells using shRNA4 lentiviral vector. Cells were transduced with control (empty vector) or shRNA4 lentiviral vectors and cultured for several days post transduction. (9A) Green fluorescent protein (GFP) expression was used as a marker for transduction to determine the transduction efficiency. (9B) CD33 expression was measured by flow cytometry over time for the total cell population. (9C) CD33 expression was measured by flow cytometry over time for the GFP⁺ cell fraction.

(11) FIG. 10. In vitro selection for CD33 shRNA-modified cells in human CD34⁺ cells following one or two rounds of GO treatment. Cells were transduced with control (empty) or shRNA 4 lentiviral vectors (shRNA), cultured for several days post transduction and treated with GO for 6 hours at the indicated times (arrow). GFP expression was measured to assess the frequency of gene modified cells.

(12) FIGS. 11A, 11B. Engraftment of shRNA-modified human CD34⁺ cells in the mouse xenotransplantation model. (11A) Table showing the two experimental groups in which mice were transplanted with either human CD34⁺ cells modified with the control or shRNA 4 lentiviral vectors or a pLL 'empty' vector. Cells were transplanted at a dose of 0.5 million cells per mouse. 6 mice per group were transplanted and 2 per group were treated with GO after stable engraftment. (11B) Timeline of transplantation experiment with blood collections to monitor engraftment and times of GO administration.

(13) FIGS. 12A, 12B. Comparable engraftment of control-(circle) and shRNA-(square) modified human CD34⁺ cells in mice. (12A) Human cell engraftment as determined by human CD45⁺ expression from peripheral blood of transplanted mice at different time points post-transplantation. (12B) Engraftment of gene-modified cells as determined by GFP expression within human CD45⁺ cells in both control (circle) and shRNA (square) modified human CD34⁺ cells.

(14) FIGS. 13A, 13B. shRNA-mediated CD33 knockdown is maintained in vivo. (13A) Frequency of CD14⁺ monocyte within human CD45⁺ cells measured in peripheral blood of all engrafted mice over time. Arrows show times of GO treatment in 2 mice per cohort. (13B) Frequency of CD33 expression within CD14⁺/GFP⁺ cells in both animal cohorts. Arrows show times of GO treatment

in 2 mice per cohort.

(15) FIGS. **14A, 14B.** In vivo selection for CD33 shRNA-modified cells following two rounds of GO treatment in engrafted mice. (**14A**) Weekly measurement of the frequency of gene modified cells as determined by GFP⁺ human CD45⁺ cells in two mice from the control group after two rounds of GO administered in vivo and separated by 5 weeks. (**14B**) Weekly measurement of the frequency of gene modified cells as determined by GFP⁺ human CD45⁺ cells in two mice from the CD33 shRNA group after two rounds of GO administered in vivo and separated by 5 weeks.

(16) FIGS. **15A, 15B.** (**15A**) Schematic of proposed viral vector. This exemplary viral vector results in expression of both a therapeutic gene (TG) and a GFP reporter driven by unique constitutive promoters alongside an shRNA sequence under the control of the U6 transcription promoter. Exemplary promoters include phosphoglycerate kinase (PGK) and elongation factor-1 α (EF1 α). (**15B**) Schematics of additional TG viral vectors into which a CD33 blocking molecule can be incorporated. Exemplary viral vectors include the validated therapeutic gene with lentiviral (LV) vector, foamy viral (FV) vector, and foamy viral vector with enhanced GFP (eGFP). Vectors also include long terminal repeats (LTR) at both the 5' and 3' ends and in particular embodiments, a Woodchuck Hepatitis Virus Posttranscriptional Regulatory Element (wpre).

(17) FIG. **16.** Plasmid map of the lentilox 3.7 lentiviral vector used for delivery and screening of CD33 shRNA activity.

(18) FIG. **17.** Schematic of exemplary clinical application of CD33 shRNA for gene therapy treatment.

(19) FIG. **18.** DNA sequence of lentilox 3.7 lentiviral vector (pLL37) used for cloning and delivery of CD33 shRNA (SEQ ID NO: 16).

(20) FIG. **19.** DNA sequence of FANCA lentiviral vector used for cloning CD33 shRNA (SEQ ID NO: 17).

(21) FIG. **20.** Additional supporting sequences (SEQ ID NOs: 18, 19, 117-155).

DETAILED DESCRIPTION

(22) Hematopoietic stem cells (HSC) are stem cells that can give rise to all blood cell types such as the white blood cells of the immune system (e.g., virus-fighting T cells and antibody-producing B cells), platelets, and red blood cells. The therapeutic administration of HSC can be used to treat a variety of adverse conditions including immune deficiency diseases, blood disorders, malignant cancers, infections, and radiation exposure (e.g., cancer treatment, accidental, or attack-based). As examples, more than 80 primary immune deficiency diseases are recognized by the World Health Organization. These diseases are characterized by an intrinsic defect in the immune system in which, in some cases, the body is unable to produce any or enough antibodies against infection. In other cases, cellular defenses to fight infection fail to work properly. Typically, primary immune deficiencies are inherited disorders.

(23) One example of a primary immune deficiency is Fanconi anemia (FA). FA is an inherited blood disorder that leads to bone marrow (BM) failure. It is characterized, in part, by a deficient DNA-repair mechanism. At least 20% of patients with FA develop cancers such as acute myeloid leukemias, and cancers of the skin, liver, gastrointestinal tract, and gynecological system. The skin and gastrointestinal tumors are usually squamous cell carcinomas. The average age of patients who develop cancer is 15 years for leukemia, 16 years for liver tumors, and 23 years for other tumors.

(24) X-linked severe combined immunodeficiency (SCID-X1) is both a cellular and humoral immune depletion caused by mutations in the common gamma chain gene (γ C), which result in the absence of T and natural killer (NK) lymphocytes and the presence of nonfunctional B lymphocytes. SCID-X1 is fatal in the first two years of life unless the immune system is reconstituted, for example, through bone marrow transplant (BMT) or gene therapy.

(25) Secondary, or acquired, immune deficiencies are not the result of inherited genetic abnormalities, but rather occur in individuals in which the immune system is compromised by factors outside the immune system. Examples include trauma, viruses, chemotherapy, toxins, and

pollution. Acquired immunodeficiency syndrome (AIDS) is an example of a secondary immune deficiency disorder caused by a virus, the human immunodeficiency virus (HIV), in which a depletion of T lymphocytes renders the body unable to fight infection.

(26) FA, SCID, and other immune deficiencies or blood disorders as well as viral infections and cancer can be treated by a bone marrow transplant (BMT) or by administering hematopoietic cells that have been genetically modified to provide a functioning gene that the patient lacks.

Therapeutic genes that can treat FA and SCID are described below. Therapeutic genes can also provide enzymes that are currently used for Enzyme replacement therapies (ERT) for lysosomal storage diseases such as Pompe disease (acid alpha-glucosidase), Gaucher disease (glucocerebrosidase), Fabry disease (alpha-galactosidase A), and Mucopolysaccharidosis type I (alpha-L-Iduronidase); blood-related cardiovascular diseases (e.g. familial apolipoprotein E deficiency and atherosclerosis (ApoE)); viral infections by expression of viral decoy receptors (e.g. for HIV-soluble CD4, or broadly neutralizing antibodies (bNAbs)) for HIV, chronic HCV, or HBV infections; and cancer (e.g. controlled expression of monoclonal antibodies (e.g. trastuzumab) or checkpoint inhibitors (e.g. aPDL1). Other additional uses are described in more detail elsewhere herein.

(27) In these treatment scenarios, it is important to remove a patient's existing hematopoietic system to avoid leaving diseased cells behind following the treatment. This is in part because, if left, the diseased, residual cells can lead to malignancies later in life.

(28) Currently, conditioning is used to remove a patient's existing hematopoietic system. All of the currently used conditioning regimens, however, whether myeloablative or nonmyeloablative, rely on the use of alkylating chemotherapy drugs and/or radiation such as involve total body irradiation (TBI) and/or cytotoxic drugs. Aside from any potential remaining residual cells, these conditioning regimens are also independently associated with an increased risk of developing malignancies, especially in DNA repair disorders like FA. These regimens are non-targeted, genotoxic, and have multiple short- and long-term adverse effects (La Nasa, et al., Bone Marrow Transplant. 2005; 36:971-975 and Chen, et al., Blood. 2006; 107:3764-3771) such as an increased risk of developing DNA repair disorders, interstitial pneumonitis, idiopathic pulmonary fibrosis, reduced lung pulmonary function, renal damage, sinusoidal obstruction syndrome (SOS), infertility, cataract formation, hyperthyroidism, and thyroiditis (Gyurkocza, et al., Blood. 2014; 124:344-353). Not only do these regimens result in impaired immune function, but they are associated with significant morbidity and mortality (Armitage. N Engl J Med. 1996; 330:827-837). Therefore, methods to reduce or eliminate the need for conditioning in these patients are desperately needed.

(29) CD33 is primarily displayed on maturing and mature cells of the myeloid lineage, including multipotent myeloid precursors. CD33 is not found on pluripotent hematopoietic stem cells or non-blood cells. Consistent with its role as a myeloid differentiation antigen, CD33 is widely expressed on malignant cells in patients with myeloid neoplasms, particularly acute myeloid leukemia (AML), where it is displayed on at least a subset of the leukemia blasts in almost all cases and possibly leukemia stem cells in some. Because of this expression pattern, there has been great interest in developing therapeutic antibodies directed at CD33. While unconjugated monoclonal CD33 antibodies proved ineffective in patients with AML, several randomized trials with the CD33 antibody-drug conjugate (ADC) gemtuzumab ozogamicin (GO) have demonstrated improved survival in some AML patients. This establishes the value of antibodies in this disease and validates CD33 as the first and so far only therapeutic target for AML immunotherapy. This benefit of GO in randomized trials led to regulatory re-approval by the U.S. Food & Drug Administration (FDA) in 2017 for the treatment of newly-diagnosed as well as relapsed or refractory CD33-expressing AML. In 2018, GO was also approved by the European Medicines Agency (EMA) in Europe for the treatment of patients ≥ 15 years in combination with intensive chemotherapy for the treatment of newly-diagnosed de novo AML.

(30) The expression of CD33 on maturing and mature cells of the myeloid lineage leads to

significant on-target, off-leukemia effects of CD33-targeted immunotherapy, manifesting primarily as severe thrombocytopenia, neutropenia, and monocytopenia. For example, with GO monotherapy given at standard dose, grade 3/4 toxicities include invariable myelosuppression. When combined with conventional chemotherapy, GO has resulted in prolongation of cytopenias and increased non-relapse mortality, in part due to more frequent fatal infections, in some clinical trials. Some non-randomized studies similarly reported substantially increased hematologic toxicities with the use of GO together with conventional chemotherapeutics, indicating a narrow therapeutic window. Several CD33-targeting therapeutics, including newer-generation ADCs (SGN-CD33A, IMGN779), bispecific antibodies (AMG330, AMG673, AMV-564), and CAR-modified T-cells have entered clinical testing and are more potent than GO. Among these recently developed investigational agents, most advanced in development was SGN-CD33A, with clinical data from early-phase clinical trials indicating not only anti-leukemia efficacy but also the potential to cause prolonged cytopenias and life-threatening sequelae (e.g., bleeding, infection). The latter problem is, perhaps best exemplified by the premature termination of the CASCADE trial (phase 3 trial testing SGN-CD33A addition to DNA methyltransferase inhibitor) because of an increase in deaths including fatal infections with SGN-CD33A. In fact, partly because of these results, SGN-CD33A is no longer currently pursued as a clinical therapeutic. While no robust data are available yet for many of the other newer-generation anti-CD33 therapeutics, highly effective elimination of CD33-positive cells is expected to cause very prolonged cytopenias and increase risks of infection and bleeding with such potent CD33-targeted immunotherapies.

(31) The experience with GO and SGN-CD33A suggests that clinically-relevant toxicity of CD33-targeted immunotherapy could be minimized in the presence of normal hematopoietic cells that do not display or have reduced expression of the CD33 antigen.

(32) The current disclosure provides systems and methods to protect beneficial therapeutic hematopoietic cells from anti-CD33 therapies while leaving residual diseased cells susceptible to anti-CD33 treatments. The systems and methods achieve this benefit by genetically modifying HSC to have reduced or eliminated expression of CD33, thus protecting them from anti-CD33 based therapies. Importantly, CD33 knockout in HSC does not impair functional, multilineage hematopoiesis, and yields cells resistant to CD33-targeting immunotherapy. In this manner, genetically modified cells will not be harmed by concurrent or subsequent anti-CD33 therapies a patient may receive. Thus, the systems and methods disclosed herein can be used to improve therapies involving blood BMT, autologous cell therapies, and treatments for diseases associated with cellular expression of CD33.

(33) In particular embodiments, the HSC genetically modified to have reduced CD33 expression are also genetically modified for an additional therapeutic purpose. The genetic modification for an additional therapeutic purpose can provide a gene to treat a disorder such as an immune deficiency (e.g., Fanconi anemia, SCID, HIV), a blood cancer (e.g., leukemia, lymphoma), a blood-related disorder (e.g., sickle cell disease), or a lysosomal storage disease (e.g., Pompe disease, Gaucher disease, Fabry disease, Mucopolysaccharidosis type I). Additional examples of conditions that can be treated with the systems and methods disclosed herein are described below.

(34) In particular embodiments, the systems and methods reduce or eliminate the need for genotoxic conditioning. In particular embodiments, the systems and methods allow for the simultaneous targeting and removal of any remaining CD33-expressing diseased cells following conditioning in preparation for a bone marrow transplant or administration of genetically-modified therapeutic cells. In particular embodiments, the systems and methods clear the niche and allow for further expansion of gene-corrected cells. In particular embodiments, the systems and methods deplete residual disease-related cells. The therapeutically administered cells with reduced CD33 expression are selectively protected from the CD33-targeting molecules. Thus, the systems and methods provide a selective protective advantage to the genetically modified cells as they reconstitute the patient's blood and immune systems while also allowing the continued use of anti-

CD33 therapies to target remaining, diseased and/or malignant CD33-expressing cells within a subject as well as any administered cells lacking the intended genetic modification. In combination, the approaches disclosed herein can eliminate CD33-expressing cells, resulting in a completely gene-corrected hematopoiesis, and minimizing risks of future myeloid malignancy after gene therapy or allogeneic transplantation.

(35) Importantly, and as indicated, the design of the systems and methods disclosed herein provide further embodiments that not only genetically modify cells to be protected from a CD33-targeting agent but also include a therapeutic gene. In particular embodiments, the present disclosure provides for ensuring that only cells that have been genetically modified with the therapeutic gene also have reduced CD33 expression that results in cellular protection. In particular embodiments, this advance is achieved by combining the CD33 blocking molecule and a therapeutic gene in a single intracellular delivery vehicle. In particular embodiments, the single intracellular delivery vehicle is a viral vector.

(36) In particular embodiments, the CD33 blocking molecule is an shRNA or siRNA CD33 blocking molecule. In particular embodiments, the cell is genetically modified using a common viral vector. In particular embodiments, the CD33 blocking molecule is shRNA referred to herein as shRNA4 and encoded by SEQ ID NO: 8 or shRNA5 encoded by SEQ ID NO: 9.

(37) In particular embodiments, the viral vector is a lentiviral vector, a foamy viral vector, or an adenoviral vector and the CD33 blocking molecule is shRNA encoded by a sequence selected from SEQ ID NO: 8 or SEQ ID NO: 9 and the therapeutic gene treats FA, SCID, Pompe disease, Gaucher disease, Fabry disease, Mucopolysaccharidosis type I, familial apolipoprotein E deficiency and atherosclerosis (ApoE), viral infections, and cancer. Other additional uses are described in more detail elsewhere herein.

(38) Particular embodiments utilize a plasmid containing SEQ ID NOs: 18 and 19 which provides a destination plasmid into which sequences encoding active RNA interference (RNAi) sequences can be cloned. In particular embodiments, SEQ ID NO: 8 and/or SEQ ID NO: 9 are cloned between SEQ ID NO: 18 and SEQ ID NO: 19. Cloned between refers to a nucleotide sequence in line with and between the first and third sequence. In particular embodiments, the sequence cloned between two other sequences is immediately adjacent to the two other sequences. In particular embodiments, the sequence cloned between two other sequences is within 5,000, 1,000, 500, or 100 bp of both other sequences.

(39) In particular embodiments, the viral vector is a lentiviral vector, a foamy viral vector, or an adenoviral vector, the CD33 blocking molecule is shRNA encoded by SEQ ID NO: 8 or SEQ ID NO: 9 and the therapeutic gene treats FA. In particular embodiments the therapeutic gene is FANCA. Particular embodiments utilize SEQ ID NO: 17 which provides a Fanconi destination plasmid into which sequences encoding active RNA interference (RNAi) sequences can be cloned. In particular embodiments, SEQ ID NO: 8 and/or SEQ ID NO: 9 are cloned into SEQ ID NO: 17. In particular embodiments, SEQ ID NO: 8 and/or SEQ ID NO: 9 are cloned between SEQ ID NO: 18 and SEQ ID NO: 19 with a therapeutic gene that treats an immune deficiency or cancer.

(40) In particular embodiments, the viral vector is a lentiviral vector, a foamy viral vector, or an adenoviral vector, the CD33 blocking molecule is shRNA encoded by SEQ ID NO: 8 or SEQ ID NO: 9 and the therapeutic gene treats SCID. In particular embodiments, the therapeutic gene is γ C.

(41) In particular embodiments, the genetically-modified cells described herein are administered in combination with a treatment to target CD33-expressing cells using a CD33-targeting agent, such as an anti-CD33 antibody, an anti-CD33 immunotoxin (e.g., an antibody linked to a plant and/or bacterial toxin), an anti-CD33 antibody-drug conjugate (e.g., an antibody bound to a small molecule toxin), an anti-CD33 antibody-radioimmunoconjugate, an anti-CD33 bispecific antibody, an anti-CD33 bispecific antibody that binds CD33 and an immune activating epitope on an immune cell (e.g., a BiTE® (Amgen, Munich, Germany)), an anti-CD33 trispecific antibody, and/or an anti-CD33 CAR or TCR-modified T-cell.

(42) Aspects of the current disclosure are now described in more supporting detail as follows: (I) HSC and HSPC Populations; (II) CD33 Blocking Molecules; (III) Optional Therapeutic Genes; (IV) Delivery and Expression of CD33 Blocking Molecules and Optional Therapeutic Genes; (V) CD33-Targeting Agents; (VI) Cell Formulations and CD33-Targeting Agent Compositions; (VII) Methods of Use; (VIII) Reference Levels Derived from Control Populations; (IX) Exemplary Embodiments; (X) Experimental Examples; and (XI) Closing Paragraphs.

(I) HSC AND HSPC POPULATIONS

(43) As indicated, HSC are stem cells that can give rise to all blood cell types such as the white blood cells of the immune system (e.g., virus-fighting T cells and antibody-producing B cells), platelets, and red blood cells. In particular embodiments, HSC can be identified and/or sorted by the following marker profiles: CD34+; Lin-CD34+CD38-CD45RA-CD90+CD49f+ (HSC1); and CD34+CD38-CD45RA-CD90- CD49f+ (HSC2). Human HSC1 can be identified by the following profiles: CD34+/CD38-/CD45RA-/CD90+ or CD34+/CD45RA-/CD90+ and mouse LT-HSC can be identified by Lin-Sca1+ckit+CD150+CD48-Flt3-CD34- (where Lin represents the absence of expression of any marker of mature cells including CD3, CD4, CD8, CD11b, CD11c, NK1.1, Gr1, and TER119). In particular embodiments, HSC are identified by a CD164+ profile. In particular embodiments, HSC are identified by a CD34+/CD164+ profile. In particular embodiments, the CD34+/CD45RA-/CD90+ HSC population is selected. For additional information regarding HSC marker profiles, see WO2017/218948.

(44) HSC differentiate into HSPC. HSPC can self-renew or can differentiate into (i) myeloid progenitor cells which ultimately give rise to monocytes and macrophages, neutrophils, basophils, eosinophils, erythrocytes, megakaryocytes/platelets, or dendritic cells; or (ii) lymphoid progenitor cells which ultimately give rise to T-cells, B-cells, and lymphocyte-like cells called natural killer cells (NK-cells). For a general discussion of hematopoiesis and HSPC differentiation, see Chapter 17, Differentiated Cells and the Maintenance of Tissues, Alberts et al., 1989, Molecular Biology of the Cell, 2nd Ed., Garland Publishing, New York, NY; Chapter 2 of Regenerative Medicine, Department of Health and Human Services, Aug. 5, 2006, and Chapter 5 of Hematopoietic Stem Cells, 2009, Stem Cell Information, Department of Health and Human Services.

(45) HSPC can be positive for a specific marker expressed in increased levels on HSPC relative to other types of hematopoietic cells. For example, such markers include CD34, CD43, CD45RO, CD45RA, CD59, CD90, CD109, CD117, CD133, CD166, HLA DR, or a combination thereof. Also, the HSPC can be negative for an expressed marker relative to other types of hematopoietic cells. For example, such markers include Lin, CD38, or a combination thereof. Preferably, HSPC are CD34+.

(46) HSC and HSPC sources include umbilical cord blood, placental blood, bone marrow and peripheral blood (see U.S. Pat. Nos. 5,004,681; 7,399,633; and 7,147,626; Craddock et al., Blood. 90(12):4779-4788 (1997); Jin et al., Bone Marrow Transplant. 42(9):581-588 (2008); Jin et al., Bone Marrow Transplant. 42(7):455-459 (2008); Pelus, Curr. Opin. Hematol. 15(4):285-292 (2008); Papayannopoulou et al., Blood. 91:2231-2239 (1998); Tricot et al., Haematologica. 93(11):1739-1742 (2008); and Weaver et al., Bone Marrow Transplant. 27: S23-S29 (2001)), as well as fetal liver, and embryonic stem cells (ESC) and induced pluripotent stem cells (iPSCs) that can be differentiated into HSC. Methods regarding collection, anti-coagulation and processing, etc. of blood and tissue samples are well known in the art. See, for example, Alsever et al., J. Med. 41:126 (1941); De Gowin, et al., J. Am. Med. Assoc. 114:850 (1940); Smith, et al., J. Thorne. Cardiovasc. Surg. 38:573 (1959); Rous and Turner, J. Exp. Med. 23(2): 219-237 (1916); and Hum, Calif. Med. 108(3):218-224 (1968). Stem cell sources of HSC and HSPC also include aortal-gonadal-mesonephros derived cells, lymph, liver, thymus, and spleen from age-appropriate donors. All collected stem cell sources of HSC and HSPC can be screened for undesirable components and discarded, treated, or used according to accepted current standards at the time. These stem cell sources can be steady state/naïve or primed with mobilizing or growth factor agents.

(47) In order to avoid surgical procedures to perform a bone marrow harvest to isolate HSC or HSPC, approaches that harvest stem cells from the peripheral blood can be preferred. Mobilization is a process whereby stem cells are stimulated out of the bone marrow (BM) niche into the peripheral blood (PB), and likely proliferate in the PB. Mobilization allows for a larger frequency of stem cells within the PB minimizing the number of days of apheresis, reaching target number collection of stem cells, and minimizing discomfort to the donor. Agents that enhance mobilization can either enhance proliferation in the PB, or enhance migration from the BM to PB, or both. Mobilizing agents include cytotoxic drugs, cytokines, and/or small molecules. A historically used regimen is a combination of cyclophosphamide (Cy) plus granulocyte-colony stimulating factor (G-CSF) (Bonig et al., Stem Cells. 27(4):836-837 (2009)). Additional mobilizing agents include alpha4-integrin blockade with anti-functional antibodies and CXCR4 blockade with the small-molecule inhibitor plerixafor (also referred to as AMD3100). Plerixafor is a bicyclam molecule that specifically and reversibly blocks SDF-1 binding to CXCR4. Another protocol is the combined regimen of granulocyte-macrophage colony stimulating factor (GM-CSF) or G-CSF with plerixafor. In certain embodiments, plerixafor is used as a single agent for mobilization of HSPCs. Plerixafor is also known commercially under the trade names Mozobil, Revixil, UMK121, AMD3000, AMD3100, GZ316455, JM3100, and SDZSID791. In particular embodiments, the mobilizing agent is C4, a CXC chemokine ligand for the CXCR2 receptor. GRObeta rapidly mobilizes short- and long-term repopulating cells in mice and/or monkeys and synergistically enhances mobilization responses with G-CSF (Pelus and Fukuda, Exp. Hematol. 34(8):1010-1020 (2006)). Furthermore, GRObeta can be combined with antagonists of VLA4 to synergistically increase circulating HSPC numbers (Karpova et al., Blood. 129(21):2939-2949 (2017)).

(48) HSC and/or HSPC can be collected and isolated from a sample using any appropriate technique. Appropriate collection and isolation procedures include magnetic separation; fluorescence activated cell sorting (FACS; Williams et al., Dev. Biol. 112(1):126-134 (1985); Lu et al., Exp. Hematol. 14(10):955-962 (1986); Lu et al., Blood. 68(1):126-133 (1986)); nanosorting based on fluorophore expression; affinity chromatography; cytotoxic agents joined to a monoclonal antibody or used in conjunction with a monoclonal antibody, e.g., complement and cytotoxins; "panning" with antibody attached to a solid matrix; selective agglutination using a lectin such as soybean (Reisner et al., Lancet. 2(8208-8209): 1320-1324 (1980)); immunomagnetic bead-based sorting or combinations of these techniques, etc. These techniques can also be used to assay for successful engraftment or manipulation of hematopoietic cells in vivo, for example for gene transfer, genetic editing or cell population expansion.

(49) In particular embodiments, it is important to remove contaminating cell populations that would interfere with isolation of the intended cell population, such as red blood cells. Removing includes both biochemical and mechanical methods to remove the undesired cell populations. Examples include lysis of red blood cells using detergents, hetastarch, hetastarch with centrifugation, cell washing, cell washing with density gradient, Ficoll-hypaque, Sepx, Optipress, Filters, and other protocols that have been used both in the manufacture of HSC and/or gene therapies for research and therapeutic purposes.

(50) In particular embodiments, a sample can be processed to select/enrich for CD34+ cells using anti-CD34 antibodies directly or indirectly conjugated to magnetic particles in connection with a magnetic cell separator, for example, the CliniMACS® Cell Separation System (Miltenyi Biotec, Bergisch Gladbach, Germany). See also, sec. 5.4.1.1 of U.S. Pat. No. 7,399,633 which describes enrichment of CD34+ HSC/HSPC from 1-2% of a normal bone marrow cell population to 50-80% of the population. HSC can also be selected to achieve the HSC profiles noted above, such as CD34+/CD45RA-/CD90+ or CD34+/CD38-/CD45RA-/CD90+.

(51) Similarly, HSPC expressing CD43, CD45RO, CD45RA, CD59, CD90, CD109, CD117, CD133, CD166, HLA DR, or a combination thereof, can be enriched for using antibodies against these antigens. U.S. Pat. No. 5,877,299 describes additional appropriate hematopoietic antigens

that can be used to isolate, collect, and enrich HSPC cells from samples.

(52) Following isolation and/or enrichment, HSC or HSPC can be expanded in order to increase the number of HSC/HSPC. Isolation and/or expansion methods are described in, for example, U.S. Pat. Nos. 7,399,633 and 5,004,681; U.S. Patent Publication No. 2010/0183564; International Patent Publication Nos. (WO) WO2006/047569; WO2007/095594; WO 2011/127470; and WO 2011/127472; Varnum-Finney et al., 1993, *Blood* 101:1784-1789; Delaney et al., 2005, *Blood* 106:2693-2699; Ohishi et al., 2002, *J. Clin. Invest.* 110:1165-1174; Delaney et al., 2010, *Nature Med.* 16(2): 232-236; and Chapter 2 of *Regenerative Medicine*, Department of Health and Human Services, August 2006, and the references cited therein. Each of the referenced methods of collection, isolation, and expansion can be used in particular embodiments of the disclosure.

(53) Particular methods of expanding HSC/HSPC include expansion with a Notch agonist. For information regarding expansion of HSC/HSPC using Notch agonists, see sec. 5.1 and 5.3 of U.S. Pat. Nos. 7,399,633; 5,780,300; 5,648,464; 5,849,869; and 5,856,441; WO 1992/119734; Schlondorfi and Blobel, 1999, *J. Cell Sci.* 112:3603-3617; Olkkonen and Stenmark, 1997, *Int. Rev. Cytol.* 176:1-85; Kopan et al., 2009, *Cell* 137:216-233; Rebay et al., 1991, *Cell* 67:687-699 and Jarriault et al., 1998, *Mol. Cell. Biol.* 18:7423-7431.

(54) Additional culture conditions can include expansion in the presence of one or more growth factors, such as: angiopoietin-like proteins (Angptls, e.g., Angptl2, Angptl3, Angptl7, Angptl5, and Mfap4); erythropoietin; fibroblast growth factor-1 (FGF-1); Flt-3 ligand (Flt-3L); G-CSF; GM-CSF; insulin growth factor-2 (IGF-2); interleukin-3 (IL-3); interleukin-6 (IL-6); interleukin-7 (IL-7); interleukin-11 (IL-11); stem cell factor (SCF; also known as the c-kit ligand or mast cell growth factor); thrombopoietin (TPO); and analogs thereof (wherein the analogs include any structural variants of the growth factors having the biological activity of the naturally occurring growth factor; see, e.g., WO 2007/1145227 and U.S. Patent Publication No. 2010/0183564).

(55) As a particular example for expanding HSC/HSPC, the cells can be cultured on a plastic tissue culture dish containing immobilized Delta ligand and fibronectin and 50 ng/ml of each of SCF, Flt-3L and TPO.

(II) CD33 BLOCKING MOLECULES

(56) As indicated, CD33 blocking molecules are provided to selectively protect therapeutic cells from CD33-targeting therapies. In particular embodiments, interfering RNA molecules that are homologous to target mRNA can lead to its degradation, a process referred to as RNA interference (RNAi) (Carthew, *Curr. Opin. Cell. Biol.* 13: 244-248 (2001)). RNAi occurs in cells naturally to remove foreign RNAs (e.g., viral RNAs). Natural RNAi proceeds via fragments cleaved from free double-strand RNA (dsRNA) which direct the degradative mechanism to other similar RNA sequences. Alternatively, RNAi can be manufactured, for example, to silence the expression of target genes. Exemplary RNAi molecules include small hairpin RNA (shRNA, also referred to as short hairpin RNA) and small interfering RNA (siRNA).

(57) Without limiting the disclosure, and without being bound by theory, RNA interference is typically a two-step process. In the first step, the initiation step, input dsRNA is digested into 21-23 nucleotide (nt) siRNA, probably by the action of Dicer, a member of the ribonuclease (RNase) III family of dsRNA-specific ribonucleases, which processes (cleaves) dsRNA (introduced directly or via a transgene or a virus) in an ATP-dependent manner. Successive cleavage events degrade the RNA to 19-21 base pair (bp) duplexes (siRNA), each with 2-nucleotide 3' overhangs (Hutvagner and Zamore, *Curr. Opin. Genet. Dev.* 12: 225-232 (2002); Bernstein, *Nature* 409:363-366 (2001)).

(58) In an effector step, the siRNA duplexes bind to a nuclease complex to form the RNA-induced silencing complex (RISC). An ATP-dependent unwinding of the siRNA duplex is required for activation of the RISC. The active RISC then targets the homologous transcript by base pairing interactions and typically cleaves the mRNA into 12 nucleotide fragments from the 3' terminus of the siRNA (Hutvagner and Zamore, *Curr. Opin. Genet. Dev.* 12: 225-232 (2002); Hammond et al., *Nat. Rev. Gen.* 2:110-119 (2001); Sharp, *Genes. Dev.* 15:485-490 (2001)). Research indicates that

each RISC contains a single siRNA and an RNase (Hutvagner and Zamore, *Curr. Opin. Genet. Dev.* 12: 225-232 (2002)).

(59) Because of the remarkable potency of RNAi, an amplification step within the RNAi pathway has been suggested. Amplification could occur by copying of the input dsRNAs which would generate more siRNAs, or by replication of the siRNAs formed. Alternatively or additionally, amplification could be effected by multiple turnover events of the RISC (Hutvagner and Zamore, *Curr. Opin. Genet. Dev.* 12: 225-232 (2002); Hammond et al., *Nat. Rev. Gen.* 2:110-119 (2001); Sharp, *Genes. Dev.* 15:485-490 (2001)). RNAi is also described in Tuschl, *Chem. Biochem.* 2: 239-245 (2001); Cullen, *Nat. Immunol.* 3:597-599 (2002); and Brantl, *Biochem. Biophys. Act.* 1575:15-25 (2002).

(60) Synthesis of RNAi molecules suitable for use with the present disclosure can be performed as follows. First, an mRNA sequence can be scanned downstream of the start codon of targeted CD33. Occurrence of each AA and the 3' adjacent 19 nucleotides is recorded as potential siRNA target sites. In particular embodiments, the siRNA target sites can be selected from the open reading frame, as untranslated regions (UTRs) are richer in regulatory protein binding sites. UTR-binding proteins and/or translation initiation complexes may interfere with binding of the siRNA endonuclease complex (Tuschl, *Chem. Biochem.* 2: 239-245 (2001)). It will be appreciated though, that siRNAs directed at untranslated regions may also be effective, as demonstrated for Glyceraldehyde 3-phosphate dehydrogenase (GAPDH) wherein siRNA directed at the 5' UTR mediated a 90% decrease in cellular GAPDH mRNA and completely abolished protein level. Second, potential target sites can be compared to an appropriate genomic database using any sequence alignment software, such as the Basic Local Alignment Search Tool (BLAST) software available from the National Center for Biotechnology Information (NCBI) server. Putative target sites which exhibit significant homology to other coding sequences can be filtered out.

(61) Qualifying target sequences can be selected as templates for siRNA synthesis. Selected sequences can include those with low G/C content as these have been shown to be more effective in mediating gene silencing as compared to those with G/C content higher than 55%. Several target sites can be selected along the length of the target gene for evaluation. For better evaluation of the selected siRNAs, a negative control can be used. Negative control siRNA can include the same nucleotide composition as the siRNAs but lack significant homology to the genome. Thus, a scrambled nucleotide sequence of the siRNA may be used, provided it does not display any significant homology to other genes.

(62) A sense strand is designed based on the sequence of the selected portion. The antisense strand is routinely the same length as the sense strand and includes complementary nucleotides. In particular embodiments, the strands are fully complementary and blunt-ended when aligned or annealed. In other embodiments, the strands align or anneal such that 1-, 2- or 3-nucleotide overhangs are generated, i.e., the 3' end of the sense strand extends 1, 2 or 3 nucleotides further than the 5' end of the antisense strand and/or the 3' end of the antisense strand extends 1, 2 or 3 nucleotides further than the 5' end of the sense strand. Overhangs can include nucleotides corresponding to the target gene sequence (or complement thereof). Alternatively, overhangs can include deoxyribonucleotides, for example deoxythymines (dT_s), or nucleotide analogs, or other suitable non-nucleotide material.

(63) To facilitate entry of the antisense strand into RISC (and thus increase or improve the efficiency of target cleavage and silencing), the base pair strength between the 5' end of the sense strand and 3' end of the antisense strand can be altered, e.g., lessened or reduced. In particular embodiments, the base-pair strength is less due to fewer G:C base pairs between the 5' end of the first or antisense strand and the 3' end of the second or sense strand than between the 3' end of the first or antisense strand and the 5' end of the second or sense strand. In particular embodiments, the base pair strength is less due to at least one mismatched base pair between the 5' end of the first or antisense strand and the 3' end of the second or sense strand. Preferably, the mismatched base pair

is selected from the group including G:A, C:A, C:U, G:G, A:A, C:C and U:U. In another embodiment, the base pair strength is less due to at least one wobble base pair, e.g., G:U, between the 5' end of the first or antisense strand and the 3' end of the second or sense strand. In another embodiment, the base pair strength is less due to at least one base pair including a rare nucleotide, e.g., inosine (I). In particular embodiments, the base pair is selected from the group including an I:A, I:U and I:C. In yet another embodiment, the base pair strength is less due to at least one base pair including a modified nucleotide. In particular embodiments, the modified nucleotide is selected from, for example, 2-amino-G, 2-amino-A, 2,6-diamino-G, and 2,6-diamino-A.

(64) ShRNAs are single-stranded polynucleotides with a hairpin loop structure. The single-stranded polynucleotide has a loop segment linking the 3' end of one strand in the double-stranded region and the 5' end of the other strand in the double-stranded region. The double-stranded region is formed from a first sequence that is hybridizable to a target sequence, such as a polynucleotide encoding CD33, and a second sequence that is complementary to the first sequence, thus the first and second sequence form a double stranded region to which the linking sequence connects the ends of to form the hairpin loop structure. The first sequence can be hybridizable to any portion of a polynucleotide encoding CD33. The double-stranded stem domain of the shRNA can include a restriction endonuclease site.

(65) Transcription of shRNAs is initiated at a polymerase III (Pol III) promoter and is thought to be terminated at position 2 of a 4-5-thymine transcription termination site. Upon expression, shRNAs are thought to fold into a stem-loop structure with 3' UU-overhangs; subsequently, the ends of these shRNAs are processed, converting the shRNAs into siRNA-like molecules of 21-23 nucleotides (Brummelkamp et al., *Science*. 296(5567): 550-553 (2002); Lee et al., *Nature Biotechnol.* 20(5): 500-505 (2002); Miyagishi and Taira, *Nature Biotechnol.* 20(5): 497-500 (2002); Paddison et al., *Genes & Dev.* 16(8): 948-958 (2002); Paul et al., *Nature Biotechnol.* 20(5): 505-508 (2002); Sui, *Proc. Natl. Acad. Sci. USA.* 99(6): 5515-5520 (2002); Yu et al., *Proc. Natl. Acad. Sci. USA.* 99(9): 6047-6052 (2002)).

(66) The stem-loop structure of shRNAs can have optional nucleotide overhangs, such as 2-bp overhangs, for example, 3' UU overhangs. While there may be variation, stems typically range from 15 to 49, 15 to 35, 19 to 35, 21 to 31 bp, or 21 to 29 bp, and the loops can range from 4 to 30 bp, for example, 4 to 23 bp. In particular embodiments, shRNA sequences include 45-65 bp; 50-60 bp; or 51, 52, 53, 54, 55, 56, 57, 58, or 59 bp. In particular embodiments, shRNA sequences include 52 or 55 bp. In particular embodiments siRNAs have 15-25 bp. In particular embodiments siRNAs have 16, 17, 18, 19, 20, 21, 22, 23, or 24 bp. In particular embodiments siRNAs have 19 bp. The skilled artisan will appreciate, however, that siRNAs having a length of less than 16 nucleotides or greater than 24 nucleotides can also function to mediate RNAi. Longer RNAi agents have been demonstrated to elicit an interferon or Protein kinase R (PKR) response in certain mammalian cells which may be undesirable. Preferably the RNAi agents do not elicit a PKR response (i.e., are of a sufficiently short length). However, longer RNAi agents may be useful, for example, in situations where the PKR response has been downregulated or dampened by alternative means.

(67) Particular embodiments utilize one or more of SEQ ID NOs: 6-15 to encode CD33 blocking molecules. In particular embodiments CD33 blocking molecules are encoded by SEQ ID NO: 8 and/or SEQ ID NO: 9. Because SEQ ID NO: 8 and SEQ ID NO: 9 encode CD33 blocking molecules these sequences can also be considered CD33 blocking molecules.

(III) OPTIONAL THERAPEUTIC GENES

(68) Particular examples of therapeutic genes and/or gene products to treat immune deficiencies can include genes associated with FA including: FancA, FancB, FancC, FancD1 (BRCA2), FancD2, FancE, FancF, FancG, FancI, FancJ (BRIP1), FancL, FancM, FancN (PALB2), FancO (RAD51C), FancP (SLX4), FancQ (ERCC4), FancR (RAD51), FancS (BRCA1), FancT (UBE2T), FancU (XRCC2), FancV (MAD2L2), and FancW (RFXD3). Exemplary genes and proteins

associated with FA include: *Homo sapiens* FANCA coding sequence; *Homo sapiens* FANCC coding sequence; *Homo sapiens* FANCE coding sequence; *Homo sapiens* FANCF coding sequence; *Homo sapiens* FANCG coding sequence; *Homo sapiens* FANCA AA; *Homo sapiens* FANCC AA; *Homo sapiens* FANCE AA; *Homo sapiens* FANCF AA; and *Homo sapiens* FANCG AA.

(69) Particular examples of therapeutic genes and/or gene products to treat immune deficiencies can include genes associated with SCID including: γ C, JAK3, IL7RA, RAG1, RAG2, DCLRE1C, PRKDC, LIG4, NHEJ1, CD3D, CD3E, CD3Z, CD3G, PTPRC, ZAP70, LCK, AK2, ADA, PNP, WHN, CHD7, ORAI1, STIM1, CORO1A, CIITA, RFXANK, RFX5, RFXAP, RMRP, DKC1, TERT, TINF2, DCLRE1B, and SLC46A1. Exemplary genes and proteins associated with SCID include: exemplary codon optimized Human γ C DNA; exemplary native Human γ C DNA; exemplary native canine γ C DNA; exemplary human γ C AA; and exemplary native canine γ C AA (91% conserved with human). Exemplary genes and proteins associated with SCID include: *Homo sapiens* JAK3 coding sequence; *Homo sapiens* PNP coding sequence; *Homo sapiens* ADA coding sequence; *Homo sapiens* RAG1 coding sequence; *Homo sapiens* RAG2 coding sequence; *Homo sapiens* JAK3 AA; *Homo sapiens* PNP AA; *Homo sapiens* ADA AA; *Homo sapiens* RAG1 AA; and *Homo sapiens* RAG2 AA.

(70) Additional exemplary therapeutic genes can include or encode for clotting and/or coagulation factors such as factor VIII (FVIII), FVII, von Willebrand factor (VWF), FI, FII, FV, FX, FXI, and FXIII).

(71) Additional examples of therapeutic genes and/or gene products include those that can provide a therapeutically effective response against diseases related to red blood cells and clotting. In particular embodiments, the disease is a hemoglobinopathy like thalassemia, or a SCD/trait. Exemplary therapeutic genes include F8 and F9.

(72) Particular examples of therapeutic genes and/or gene products include γ -globin; soluble CD40; CTLA; Fas L; antibodies to CD4, CD5, CD7, CD52, etc.; antibodies to IL1, IL2, IL6; an antibody to TCR specifically present on autoreactive T cells; IL4; IL10; IL12; IL13; IL1Ra, sIL1RI, sIL1RII; sTNFRI; sTNFRII; antibodies to TNF; P53, PTPN22, and DRB1*1501/DQB1*0602; globin family genes; WAS; phox; dystrophin; pyruvate kinase (PK); CLN3; ABCD1; arylsulfatase A (ARSA); SFTPB; SFTPC; NLX2.1; ABCA3; GATA1; ribosomal protein genes; TERC; CFTR; LRRK2; PARK2; PARK7; PINK1; SNCA; PSEN1; PSEN2; APP; SOD1; TDP43; FUS; ubiquilin 2; C9ORF72 and other therapeutic genes described herein.

(73) Particular embodiments include inserting or altering a gene selected from ABLI, AKT1, APC, ARSB, BCL11A, BLC1, BLC6, BRCA1, BRIP1, C46, CAS9, C-CAM, CBFAI, CBL, CCR5, CD19, CDA, C-MYC, CRE, CSCR4, CSFIR, CTS-I, CYB5R3, DCC, DHFR, DLL1, DMD, EGFR, ERBA, ERBB, EBRB2, ETSI, ETS2, ETV6, FCC, FGR, FOX, FUSI, FYN, GALNS, GLB1, GNS, GUSB, HBB, HBD, HBE1, HBG1, HBG2, HCR, HGSNAT, HOXB4, HRAS, HYAL1, ICAM-1, iCaspase, IDUA, IDS, JUN, KLF4, KRAS, LYN, MCC, MDM2, MGMT, MLL, MMACI, MYB, MEN-I, MEN-II, MYC, NAGLU, NANOG, NF-1, NF-2, NKX2.1, NOTCH, OCT4, p16, p2I, p27, p57, p73, PALB2, RAD51C, ras, at least one of RPL3 through RPL40, RPLP0, RPLP1, RPLP2, at least one of RPS2 through RPS30, RPSA, SGSH, SLX4, SOX2, VHL, and WT-I.

(74) In addition to therapeutic genes and/or gene products, the transgene can also encode for therapeutic molecules, such as checkpoint inhibitor reagents, chimeric antigen receptor molecules specific to one or more cellular antigen (e.g. cancer antigen), and/or T-cell receptor specific to one or more cellular antigen (e.g. cancer antigen).

(IV) DELIVERY AND EXPRESSION OF CD33 BLOCKING MOLECULES AND OPTIONAL THERAPEUTIC Genes

(75) Cells can be genetically modified to express CD33 blocking molecules and optionally a therapeutic gene using any method known in the art.

(76) Particular embodiments use a genetic construct or vector to deliver and express CD33 blocking molecules and optional therapeutic genes in cells. A genetic construct is an artificially produced combination of nucleotides to express particular intended molecules.

(77) A “vector” is a nucleic acid molecule that is capable of transporting another nucleic acid molecule, such as a gene encoding a CD33 blocking molecule and optionally a therapeutic gene. Vectors include, e.g., plasmids, cosmids, viruses, and phage. Viral vectors refer to nucleic acid molecules that include virus-derived nucleic acid elements that facilitate transfer and expression of non-native genes within a cell. In particular embodiments, viral-mediated genetic modification can utilize, for example, retroviral vectors, lentiviral vectors, foamy viral vectors, adenoviral vectors, adeno-associated viral vectors, alpharetroviral vectors or gammaretroviral vectors. In particular embodiments, retroviral vectors (see Miller, et al., 1993, *Meth. Enzymol.* 217:581-599) can be used. In these embodiments, the gene to be expressed is cloned into the retroviral vector for its delivery into cells. In particular embodiments, a retroviral vector includes all of the cis-acting sequences necessary for the packaging and integration of the viral genome in the target cell, i.e., (a) a long terminal repeat (LTR), or portions thereof, at each end of the vector; (b) primer binding sites for negative and positive strand DNA synthesis; and (c) a packaging signal, necessary for the incorporation of genomic RNA into virions. More detail about retroviral vectors can be found in Boesen, et al., 1994, *Biotherapy* 6:291-302; Clowes, et al., 1994, *J. Clin. Invest.* 93:644-651; Kiem, et al., 1994, *Blood* 83:1467-1473; Salmons and Gunzberg, 1993, *Human Gene Therapy* 4:129-141; and Grossman and Wilson, 1993, *Curr. Opin. in Genetics and Devel.* 3:110-114.

(78) Lentiviral vectors or “lentivirus” refers to a genus of retroviruses that are capable of infecting dividing and non-dividing cells and typically produce high viral titers. Lentiviral vectors have been employed in gene therapy for a number of diseases. For example, hematopoietic gene therapies using lentiviral vectors or gammaretroviral vectors have been used for x-linked adrenoleukodystrophy and β -thalassemia. Several examples of lentiviruses include HIV (including HIV type 1, and HIV type 2); equine infectious anemia virus; feline immunodeficiency virus (FIV); bovine immune deficiency virus (BIV); and simian immunodeficiency virus (SIV).

(79) In particular embodiments, other retroviral vectors can be used in the practice of the methods of the invention. These include, e.g., vectors based on human foamy virus (HFV) or other viruses in the Spumavirus genera.

(80) Foamy viruses (FVs) are the largest retroviruses known today and are widespread among different mammals, including all non-human primate species, however are absent in humans. This complete apathogenicity qualifies FV vectors as ideal gene transfer vehicles for genetic therapies in humans and clearly distinguishes FV vectors as gene delivery system from HIV-derived and also gammaretrovirus-derived vectors.

(81) FV vectors are also suitable for gene therapy applications because they can (1) accommodate large transgenes (>9 kb), (2) transduce slowly dividing cells efficiently, and (3) integrate as a provirus into the genome of target cells, thus enabling stable long-term expression of the transgene(s). FV vectors do need cell division for the pre-integration complex to enter the nucleus, however the complex is stable for at least 30 days and still infective. The intracellular half-life of the FV pre-integration complex is comparable to the one of lentiviruses and significantly higher than for gammaretroviruses, therefore FVs are also, similar to lentivirus vectors, able to transduce rarely dividing cells. FV vectors are natural self-inactivating vectors and characterized by the fact that they seem to have hardly any potential to activate neighboring genes. In addition, FV vectors can enter any cells known (although the receptor is not identified yet) and infectious vector particles can be concentrated 100-fold without loss of infectivity due to a stable envelope protein. FV vectors achieve high transduction efficiency in pluripotent hematopoietic stem cells and have been used in animal models to correct monogenetic diseases such as leukocyte adhesion deficiency (LAD) in dogs and FA in mice. FV vectors are also used in preclinical studies of β -thalassemia.

(82) Point mutations can be made in FVs to render them integration incompetent. For example,

foamy viruses can be rendered integration incompetent by introducing point mutations into the highly conserved DD35E catalytic core motif of the foamy virus integrase sequence. See, for example, Deyle D R et al. (2010) *J. Virol.* 84(18): 9341-9349. As another example, an FV vector can be rendered integration deficient by introducing point mutations into the Pol gene of the FV vector. FIG. 20 shows FV Pol coding sequence (SEQ ID NO: 104) and FV Pol amino acid sequence (SEQ ID NO: 105) with indicated nucleotides or amino acid residues, respectively, that can be mutated to render the FV vector integration deficient.

(83) In particular embodiments, adenoviruses (e.g., adenovirus 5 (Ad5), adenovirus 35 (Ad35), adenovirus 11 (Ad11), adenovirus 26 (Ad26), adenovirus 48 (Ad48), adenovirus 50 (Ad50), Ad5/35++, and helper-dependent forms thereof (e.g., helper-dependent Ad5/35++ or helper dependent Ad35), adeno-associated viruses (AAV; see, e.g., U.S. Pat. No. 5,604,090), and alphaviruses can be used. See Kozarsky and Wilson, 1993, *Current Opinion in Genetics and Development* 3:499-503, Rosenfeld, et al., 1991, *Science* 252:431-434; Rosenfeld, et al., 1992, *Cell* 68:143-155; Mastrangeli, et al., 1993, *J. Clin. Invest.* 91:225-234; Walsh, et al., 1993, *Proc. Soc. Exp. Biol. Med.* 204:289-300; and Lundstrom, 1999, *J. Recept. Signal Transduct. Res.* 19: 673-686. Additional examples of viral vectors include those derived from cytomegaloviruses (CMV), flaviviruses, herpes viruses (e.g., herpes simplex), influenza viruses, papilloma viruses (e.g., human and bovine papilloma virus; see, e.g., U.S. Pat. No. 5,719,054), poxviruses, vaccinia viruses, modified vaccinia Ankara (MVA), NYVAC, or strains derived therefrom. Other examples include avipox vectors, such as a fowlpox vectors (e.g., FP9) or canarypox vectors (e.g., ALVAC and strains derived therefrom). As indicated, helper dependent forms of viral vectors may also be used.

(84) Methods of using retroviral and lentiviral viral vectors and packaging cells for transducing mammalian host cells with viral particles including desired transgenes are described in, e.g., U.S. Pat. No. 8,119,772; Walchli, et al., 2011, *PLoS One* 6:327930; Zhao, et al., 2005, *J. Immunol.* 174:4415; Engels, et al., 2003, *Hum. Gene Ther.* 14:1155; Frecha, et al., 2010, *Mol. Ther.* 18:1748; and Verhoeven, et al., 2009, *Methods Mol. Biol.* 506:97. Retroviral and lentiviral vector constructs and expression systems are also commercially available.

(85) Although viral vectors are useful in the co-delivery of a CD33 blocking molecule and a therapeutic gene to ensure that only cells expressing the therapeutic gene are protected, other vectors or targeted genetic engineering approaches may also be utilized. The CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats)/Cas (CRISPR-associated protein) nuclease system is an engineered nuclease system used for genetic engineering that is based on a bacterial system. Information regarding CRISPR-Cas systems and components thereof are described in, for example, U.S. Pat. Nos. 8,697,359, 8,771,945, 8,795,965, 8,865,406, 8,871,445, 8,889,356, 8,889,418, 8,895,308, 8,906,616, 8,932,814, 8,945,839, 8,993,233 and 8,999,641 and applications related thereto; and WO2014/018423, WO2014/093595, WO2014/093622, WO2014/093635, WO2014/093655, WO2014/093661, WO2014/093694, WO2014/093701, WO2014/093709, WO2014/093712, WO2014/093718, WO2014/145599, WO2014/204723, WO2014/204724, WO2014/204725, WO2014/204726, WO2014/204727, WO2014/204728, WO2014/204729, WO2015/065964, WO2015/089351, WO2015/089354, WO2015/089364, WO2015/089419, WO2015/089427, WO2015/089462, WO2015/089465, WO2015/089473 and WO2015/089486, WO2016205711, WO2017/106657, WO2017/127807 and applications related thereto.

(86) Particular embodiments utilize zinc finger nucleases (ZFNs) as gene editing agents. ZFNs are a class of site-specific nucleases engineered to bind and cleave DNA at specific positions. ZFNs are used to introduce double stranded breaks (DSBs) at a specific site in a DNA sequence which enables the ZFNs to target unique sequences within a genome in a variety of different cells. For additional information regarding ZFNs and ZFNs useful within the teachings of the current disclosure, see, e.g., U.S. Pat. Nos. 6,534,261; 6,607,882; 6,746,838; 6,794,136; 6,824,978; 6,866,997; 6,933,113; 6,979,539; 7,013,219; 7,030,215; 7,220,719; 7,241,573; 7,241,574; 7,585,849; 7,595,376; 6,903,185; 6,479,626; US 2003/0232410 and US 2009/0203140 as well as

Gaj et al., *Nat Methods*, 2012, 9(8):805-7; Ramirez et al., *Nucl Acids Res*, 2012, 40(12):5560-8; Kim et al., *Genome Res*, 2012, 22(7): 1327-33; Urnov et al., *Nature Reviews Genetics*, 2010, 11:636-646; Miller, et al. *Nature Biotechnol.* 25, 778-785 (2007); Bibikova, et al. *Science* 300, 764 (2003); Bibikova, et al. *Genetics* 161, 1169-1175 (2002); Wolfe, et al. *Annu. Rev. Biophys. Biomol. Struct.* 29, 183-212 (2000); Kim, et al. *Proc. Natl. Acad. Sci. USA.* 93, 1156-1160 (1996); and Miller, et al. *The EM BO journal* 4, 1609-1614 (1985).

(87) Particular embodiments can use transcription activator like effector nucleases (TALENs) as gene editing agents. TALENs refer to fusion proteins including a transcription activator-like effector (TALE) DNA binding protein and a DNA cleavage domain. TALENs are used to edit genes and genomes by inducing double DSBs in the DNA, which induce repair mechanisms in cells. Generally, two TALENs must bind and flank each side of the target DNA site for the DNA cleavage domain to dimerize and induce a DSB. For additional information regarding TALENs, see U.S. Pat. Nos. 8,440,431; 8,440,432; 8,450,471; 8,586,363; and 8,697,853; as well as Joung and Sander, *Nat Rev Mol Cell Biot*, 2013, 14(I):49-55; Beurdeley et al., *Nat Commun*, 2013, 4: 1762; Scharenberg et al., *Curr Gene Ther*, 2013, 13(4):291-303; Gaj et al., *Nat Methods*, 2012, 9(8):805-7; Miller, et al. *Nature biotechnology* 29, 143-148 (2011); Christian, et al. *Genetics* 186, 757-761 (2010); Boch, et al. *Science* 326, 1509-1512 (2009); and Moscou, & Bogdanove, *Science* 326, 1501 (2009).

(88) Particular embodiments can utilize MegaTALs as gene editing agents. MegaTALs have a single chain rare-cleaving nuclease structure in which a TALE is fused with the DNA cleavage domain of a meganuclease. Meganucleases, also known as homing endonucleases, are single peptide chains that have both DNA recognition and nuclease function in the same domain. In contrast to the TALEN, the megaTAL only requires the delivery of a single peptide chain for functional activity.

(89) Other methods of gene delivery include use of artificial chromosome vectors such as mammalian artificial chromosomes (Vos, *Curr. Opin. Genet. Dev.* 8(3): 351-359, 1998) and yeast artificial chromosomes (YAC); liposomes (Tarahovsky and Ivanitsky, 1998, *Biochemistry (Mosc)* 63:607-618); ribozymes (Branch and Klotman, 1998, *Exp. Nephrol.* 6:78-83); and triplex DNA (Chan and Glazer, 1997, *J. Mol. Med.* 75:267-282). YAC are typically used when the inserted nucleic acids are too large for more conventional vectors (e.g., greater than 12 kb).

(90) When targeted genome editing approaches are utilized, genes can be inserted within genomic safe harbors. Genomic safe harbor sites are intragenic or extragenic regions of the genome that are able to accommodate the predictable expression of newly integrated DNA without adverse effects on the host cell. A useful safe harbor must permit sufficient transgene expression to yield desired levels of the encoded molecule. A genomic safe harbor site also must not alter cellular functions. Methods for identifying genomic safe harbor sites are described in Sadelain et al., *Nature Reviews* (2012); 12:51-58; and Papapetrou et al., *Nat Biotechnol.* (2011) January; 29(1):73-8. In particular embodiments, a genomic safe harbor site meets one or more (one, two, three, four, or five) of the following criteria: (i) distance of at least 50 kb from the 5' end of any gene, (ii) distance of at least 300 kb from any cancer-related gene, (iii) within an open/accessible chromatin structure (measured by DNA cleavage with natural or engineered nucleases), (iv) location outside a gene transcription unit and (v) location outside ultraconserved regions (UCRs), microRNA or long non-coding RNA of the genome.

(91) In particular embodiments, a genomic safe harbor meets criteria described herein and also demonstrates a 1:1 ratio of forward:reverse orientations of lentiviral integration further demonstrating the loci does not impact surrounding genetic material.

(92) Particular genomic safe harbors sites include CCR5, HPRT, AAVS1, Rosa and albumin. See also, e.g., U.S. Pat. Nos. 7,951,925 and 8,110,379; U.S. Publication Nos. 20080159996; 201000218264; 20120017290; 20110265198; 20130137104; 20130122591; 20130177983 and 20130177960 for additional information and options for appropriate genomic safe harbor integration sites.

(93) The vectors and genetic engineering approaches described herein are used to deliver genes to cells for expression. Delivery can utilize any appropriate technique, such as transfection, electroporation, microinjection, lipofection, calcium phosphate mediated transfection, infection with a viral or bacteriophage vector including the gene sequences, cell fusion, chromosome-mediated gene transfer, microcell-mediated gene transfer, spheroplast fusion, in vivo nanoparticle-mediated delivery, etc. Numerous techniques are known in the art for the introduction of foreign genes into cells (see e.g., Loeffler and Behr, 1993, Meth. Enzymol. 217:599-618; Cohen, et al., 1993, Meth. Enzymol. 217:618-644; Cline, 1985, Pharmac. Ther. 29:69-92) and may be used, provided that the necessary developmental and physiological functions of the recipient cells are not unduly disrupted. The technique can provide for the stable transfer of the gene to the cell, so that the gene is expressible by the cell and, in certain instances, preferably heritable and expressible by its cell progeny.

(94) The term “gene” refers to a nucleic acid sequence (used interchangeably with polynucleotide or nucleotide sequence) that encodes one or more CD33 blocking molecules and optionally one or more therapeutic proteins as described herein. Gene sequences encoding the molecule can be DNA or RNA. As appropriate for the given context, these nucleic acid sequences may be a DNA strand sequence that is transcribed into RNA or an RNA sequence that is translated into protein. The nucleic acid sequences include both the full-length nucleic acid sequences as well as non-full-length sequences derived from the full-length protein. The gene sequence can be readily prepared by synthetic or recombinant methods.

(95) The definition of a gene includes various sequence polymorphisms; mutations; degenerate codons of the native sequence; sequences that may be introduced to provide codon preference in a specific cell type (e.g., codon optimized for expression in mammalian cells); and/or sequence variants wherein such alterations do not substantially affect the function of the encoded molecule. The term further can include all introns and other DNA sequences spliced from an mRNA transcript, along with variants resulting from alternative splice sites. Portions of complete gene sequences are referenced throughout the disclosure as is understood by one of ordinary skill in the art. Nucleotide sequences encoding other sequences disclosed herein can be readily determined by one of ordinary skill in the art.

(96) The term “gene” may include not only coding sequences but also coding sequences operably linked to each other and relevant regulatory sequences such as promoters, enhancers, and termination regions. For example, there can be a functional linkage between a regulatory sequence and an exogenous nucleic acid sequence resulting in expression of the latter. For another example, a first nucleic acid sequence can be operably linked with a second nucleic acid sequence when the first nucleic acid sequence is placed in a functional relationship with the second nucleic acid sequence. For instance, a promoter is operably linked to a coding sequence if the promoter affects the transcription or expression of the coding sequence. Generally, operably linked DNA sequences are contiguous and, where necessary or helpful, join coding regions into a common reading frame.

(97) These regulatory sequences can be eukaryotic or prokaryotic in nature. In particular embodiments, the regulatory sequence can result in the constitutive expression of the CD33 blocking molecule and optionally one or more therapeutic proteins upon entry of the vector into the cell. Alternatively, the regulatory sequences can include inducible sequences. Inducible regulatory sequences are well known to those skilled in the art and are those sequences that require the presence of an additional inducing factor to result in expression of the one or more molecules. Examples of suitable regulatory sequences include binding sites corresponding to tissue-specific transcription factors based on endogenous nuclear proteins, sequences that direct expression in a specific cell type, the lac operator, the tetracycline operator and the steroid hormone operator. Any inducible regulatory sequence known to those of skill in the art may be used.

(98) In particular embodiments, the PGK promoter is used to drive expression of a CD33 blocking molecule and optionally a therapeutic gene. In particular embodiments, the PGK promoter is

derived from the human gene encoding phosphoglycerate kinase (PGK). In particular embodiments, the PGK promoter includes binding sites for the Rap1p, AbfIp, and/or GcrIp transcription factors. In particular embodiments, the PGK promoter includes 500 base pairs: Start (0); StyI (21); NspI-SphI (40); BpmI-Eco57MI (52); BaeGI-Bme1580I (63); AgeI (111); BsmBI-SpeI (246); BssS α I (252); BlpI (274); BsrDI (285); StuI (295); BglI (301); EaeI (308); AlwNI (350); EcoO109I-PpuMI (415); BspEI (420); BsmI (432); EarI (482); End (500). In particular embodiments, a PGK promoter includes SEQ ID NO: 106 in FIG. 20.

(99) In particular embodiments, RNA polymerase III (also called Pol III) promoters can be used to drive expression of a CD33 blocking molecule and optionally a therapeutic gene. Pol III transcribes DNA to synthesize ribosomal 5S rRNA, tRNA, and other small RNAs. The Pol III promoters generally have well-defined initiation and stop sites and their transcripts lack poly(A) tails. The termination signal for these promoters is defined by the polythymidine tract, and the transcript is typically cleaved after the second uridine. Cleavage at this position generates a 3' UU overhang in the expressed shRNA, which is similar to the 3' overhangs of synthetic siRNAs.

(100) siRNA molecules can be transcribed from expression vectors which can facilitate stable expression of the siRNA transcripts once introduced into a host cell. These vectors are engineered to express shRNAs, which can be processed in vivo into siRNA molecules capable of carrying out gene-specific silencing (Brummelkamp et al., Science 296:550-553 (2002); Paddison et al., Genes Dev. 16:948-958 (2002); Paul et al., Nature Biotech. 20: 505-508 (2002); Yu et al., Proc. Natl. Acad. Sci. USA 99:6047-6052 (2002)).

(101) In particular embodiments, a suitable siRNA expression vector encodes the sense and antisense siRNA under the regulation of separate Pol III promoters (Miyagishi and Taira, Nature Biotech. 20:497-500 (2002)). The siRNA, generated by this vector also includes a five thymidine (T5) termination signal.

(102) In particular embodiments, the promoter will drive expression of the CD33 blocking molecule. In particular embodiments, the promoter will drive expression of the CD33 blocking molecule and a therapeutic gene. In particular embodiments, the promoter will be oriented in such a way that results in expression of the CD33 blocking molecule and therapeutic gene driven by the promoter.

(103) Additional exemplary promoters are known in the art and include galactose inducible promoters, pGAL1, pGAL1-10, pGal4, and pGal10; cytochrome c promoter, pCYC1; and alcohol dehydrogenase 1 promoter, pADH1, EF1alpha.

(104) In particular embodiments, the efficiency of integration, the size of the DNA sequence that can be integrated, and the number of copies of a DNA sequence that can be integrated into a genome can be improved by using transposons. Transposons or transposable elements include a short nucleic acid sequence with terminal repeat sequences upstream and downstream. Active transposons can encode enzymes that facilitate the excision and insertion of nucleic acid into a target DNA sequence.

(105) A number of transposable elements have been described in the art that facilitate insertion of nucleic acids into the genome of vertebrates, including humans. Examples include Sleeping Beauty® (Regents of the University of Minnesota, Minneapolis, MN) (e.g., derived from the genome of salmonid fish); piggyBac® (Poseida Therapeutics, Inc. San Diego CA) (e.g., derived from lepidopteran cells and/or the *Myotis lucifugus*); mariner (e.g., derived from *Drosophila*); frog prince (e.g., derived from *Rana pipiens*); Tol2 (e.g., derived from medaka fish); TcBuster (e.g., derived from the red flour beetle *Tribolium castaneum*) and spinON.

(106) In particular embodiments, vectors provide cloning sites to facilitate transfer of the polynucleotide sequences. Such vector cloning sites include at least one restriction endonuclease recognition site positioned to facilitate excision and insertion, in reading frame, of polynucleotide segments. Any of the restriction sites known in the art can be utilized. Most commercially available vectors already contain multiple cloning site (MCS) or polylinker regions. In addition, genetic

engineering techniques useful to incorporate new and unique restriction sites into a vector are known and routinely practiced by persons of ordinary skill in the art. A cloning site can involve as few as one restriction endonuclease recognition site to allow for the insertion or excision of a single polynucleotide fragment. More typically, two or more restriction sites are employed to provide greater control of for example, insertion (e.g., direction of insert), and greater flexibility of operation (e.g., the directed transfer of more than one polynucleotide fragment). Multiple restriction sites can be the same or different recognition sites.

(107) In particular embodiments, the gene sequence encoding any of these sequences can have one or more restriction enzyme sites at the 5' and/or 3' ends of the coding sequence in order to provide for easy excision and replacement of the gene sequence encoding the sequence with another gene sequence encoding a different sequence. In particular embodiments, each of the restriction sites is unique in the vector and different from the other restriction sites. In particular embodiments, each of the restriction sites are identical to the other restriction sites.

(108) In particular embodiments, for expression of shRNAs within cells, vectors containing either the polymerase III H1-RNA or U6 promoter, a cloning site for the stem-looped RNA insert, and a 4 5-thymidine transcription termination signal can be employed.

(109) In particular embodiments, the pSUPER vector which contains polymerase-III H1-RNA gene promoter with a well-defined start of transcription and a termination signal including five thymidines in a row (T5) (Brummelkamp et al., *Science* 296:550-553 (2002)) is used. The cleavage of the transcript at the termination site is at a site following the second uridine, thus yielding a transcript which resembles the ends of synthetic siRNAs, which also contain nucleotide overhangs. siRNA is cloned such that it includes the sequence of interest. The resulting transcript folds back on itself to form a stem-loop structure, which mediates CD33 RNAi.

(110) In particular embodiments, nucleotide sequences encoding one or more of SEQ ID NOs: 6-15 are cloned between SEQ ID NO: 18 and SEQ ID NO: 19 with a therapeutic gene.

(111) In particular embodiments, the nucleic acid is stably integrated into the genome of a cell. In particular embodiments, the nucleic acid is stably maintained in a cell as a separate, episomal segment.

(112) For additional information regarding procedures for genetic modification, see Anderson, *Science* 256:808-813 (1992); Nabel & Feigner, *TIBTECH* 11:211-217 (1993); Mitani & Caskey, *TIBTECH* 11:162-166 (1993); Dillon, *TIBTECH* 11:167-175 (1993); Miller, *Nature* 357:455-460 (1992); Van Brunt, *Biotechnology* 6(10):1149-1154 (1988); Vigne, *Restorative Neurology and Neuroscience* 8:35-36 (1995); Kremer & Perricaudet, *British Medical Bulletin* 51(1):31-44 (1995); Haddada et al., in *Current Topics in Microbiology and Immunology* Doerfler and Böhm (eds) (1995); and Yu et al., *Gene Therapy* 1:13-26 (1994).

(V) CD33-TARGETING AGENTS

(113) Particular embodiments include targeting any residual and/or non-therapeutic cells that express CD33 with the use of a CD33-targeting agent. A CD33-targeting agent refers to a molecule, cell, drug, or combination thereof that targets CD33-expressing cells for cell death or to inhibit cell growth. Examples of CD33-targeting agents include anti-CD33 antibodies, anti-CD33 immunotoxins, anti-CD33 antibody-drug conjugates, anti-CD33 antibody-radioisotope conjugates), anti-CD33 multispecific antibodies (e.g. anti-CD33 bispecific antibodies, anti-CD33 bispecific antibodies that bind CD33 and an immune activating epitope on an immune cell (e.g., a CD3 as in BiTE®), anti-CD33 trispecific antibodies), and/or genetically modified cells expressing an anti-CD33 CAR or an engineered TCR. Each of these types of CD33-targeting agents include a binding domain that binds CD33, and most (except certain antibody forms) also include a linker.

Accordingly, CD33 binding domains are described first and general description of linkers is provided next. Following this description of CD33 binding domains and linkers, more particular information regarding the different CD33-targeting agents are provided.

(114) CD33-targeting agents can bind different forms and/or epitopes of CD33. For example, full

length CD33 (CD33FL) is a transmembrane glycoprotein that is characterized by an amino-terminal, membrane-distant V-set immunoglobulin (Ig)-like domain and a membrane-proximal C2-set Ig-like domain in its extracellular portion. In addition to CD33FL, a splice variant that misses exon 2 (CD33ΔE2) has also been identified. Thus, CD33 refers to any native, mature CD33 which results from processing of a CD33 precursor protein in a cell (FIGS. 1A, 1B).

(115) (V-a) CD33 Binding Domains.

(116) Binding domains include any substance that binds to CD33 to form a complex. The choice of binding domain can depend upon the type and number of CD33 markers that define the surface of a target cell or the type of selected CD33-targeting agent. Examples of binding domains include cellular marker ligands, receptor ligands, antibodies, antibody binding domains, peptides, peptide aptamers, receptors (e.g., T cell receptors), or combinations and engineered fragments or formats thereof.

(117) Antibodies are one example of binding domains and include whole antibodies or binding fragments of an antibody, e.g., Fv, Fab, Fab', F(ab')₂, and single chain (sc) forms and fragments thereof that bind specifically CD33. Antibodies or antigen binding fragments can include all or a portion of polyclonal antibodies, monoclonal antibodies, human antibodies, humanized antibodies, synthetic antibodies, non-human antibodies, recombinant antibodies, chimeric antibodies, bispecific antibodies, mini bodies, and linear antibodies.

(118) Antibodies are produced from two genes, a heavy chain gene and a light chain gene. Generally, an antibody includes two identical copies of a heavy chain, and two identical copies of a light chain. Within a variable heavy chain and variable light chain, segments referred to as complementary determining regions (CDRs) dictate epitope binding. Each heavy chain has three CDRs (i.e., CDRH1, CDRH2, and CDRH3) and each light chain has three CDRs (i.e., CDRL1, CDRL2, and CDRL3). CDR regions are flanked by framework residues (FR).

(119) In particular embodiments, the CD33 binding domain can be derived from or include hP67.6 which is an anti-CD33 antibody used in the ADC, GO. In particular embodiments, the light chain of hP67.6 includes:

(120) TABLE-US-00001 (SEQ ID NO: 39)

MSVPTQVLGLLLLWLTDARCDIQLTQSPSTLSASVGDRVTITCRASESL
DNYGIRFLTWFQQKPGKAPKLLMYAASNQGSQVPSRFSQSGSGTEFTLT
ISSLPDDFATYYCQQTKVEPWSFGQGTKVEVKRT

(121) and the heavy chain of hP67.6 includes:

(122) TABLE-US-00002 (SEQ ID NO: 40)

MEWSWVFLFFLSVTTGVHSEVQLVQSGAEVKKPGSSVKVSCKASGYTIT
DSNIHWVRQAPGQSLEWIGYIYPYNGGTDYNQKFKNRATLTVDNPTNTA
YMELSSLRSEDTDFYYCVNGNPWLAYWGQGTLLTVSSASTKGP.

(123) In particular embodiments, the hP67.6 binding domain includes a variable light chain including a CDRL1 sequence including QSPSTLSASV (SEQ ID NO: 41), a CDRL2 sequence including DNYGIRFLTWFQQKPG (SEQ ID NO: 42), and a CDRL3 sequence including FTLTSSL (SEQ ID NO: 43). In particular embodiments, the hP67.6 binding domain includes a variable heavy chain including a CDRH1 sequence including VQSGAEVKKPG (SEQ ID NO: 44), a CDRH2 sequence including DSNIHWV (SEQ ID NO: 45), and a CDRH3 sequence including LTVDNPTNT (SEQ ID NO: 46).

(124) In particular embodiments, the CD33 binding domain can be derived from or include h2H12EC which is the anti-CD33 antibody used in the ADC, SGN-CD33A. In particular embodiments, the h2H12EC binding domain includes a variable light chain including a CDRL1 sequence including NYDIN (SEQ ID NO: 98), a CDRL2 sequence including WIYPGDGSTKYNEKFKA (SEQ ID NO: 99), and a CDRL3 sequence including GYEDAMDY (SEQ ID NO: 100). In particular embodiments, the h2H12EC binding domain includes a variable heavy chain including a CDRH1 sequence including KASQDINSYLS (SEQ ID NO: 101), a

CDRH2 sequence including RANRLVD (SEQ ID NO: 102), and a CDRH3 sequence including LQYDEFPLT (SEQ ID NO: 103).

(125) Additional examples of anti-CD33 antibody heavy and light chains, as well as specific CDRs, include those described in U.S. Pat. No. 7,557,198. For instance, in particular embodiments, a light chain of a representative anti-CD33 antibody includes:

(126) TABLE-US-00003 (SEQ ID NO: 47)

NIMLTQSPSSLAVSAGEKVTMSCKSSQSVFFSSSQKNYLAWYQQIPGQS
PKLLIYWASTRESGVPDRFTGSGSGTDFTLTISSVQSEDLAIYYCHQYL
SSRTFGGGTKLEIKR

(127) and a heavy chain of this representative anti-CD33 antibody includes:

(128) TABLE-US-00004 (SEQ ID NO: 48)

QVQLQQPGAIEVVKPGASVKMSCKASGYTFTSYIHWIKQTPGQGLEWVG
VIYPGNDDISYNQKFKGKATLTADKSSTTAYMQLSSLTSEDSAVYYCAR
EVLRYFDVWGAGTTVTVSS.

(129) Additional examples of anti-CD33 antibody heavy and light chains, as well as specific CDRs, include those described in U.S. Pat. No. 7,557,198. In particular embodiments, the CD33 binding domain includes a variable light chain including a CDRL1 sequence including SYIHW (SEQ ID NO: 49), a CDRL2 sequence including VIYPGNDDISYNQKFXG (SEQ ID NO: 50) wherein X is K or Q, and a CDRL3 sequence including EVLRYFDV (SEQ ID NO: 51). In particular embodiments, the CD33 binding domain includes a variable heavy chain including a CDRH1 sequence including KSSQSVFFSSSQKNYLA (SEQ ID NO: 52), a CDRH2 sequence including WASTRES (SEQ ID NO: 53), and a CDRH3 sequence including HQYLSST (SEQ ID NO: 54).

(130) In some instances, it is beneficial for the binding domain to be derived from the same species it will ultimately be used in. For example, for use in humans, it may be beneficial for the antigen binding domain to include a human antibody, humanized antibody, or a fragment or engineered form thereof. Antibodies from human origin or humanized antibodies have lowered or no immunogenicity in humans and have a lower number of non-immunogenic epitopes compared to non-human antibodies. Antibodies and their engineered fragments will generally be selected to have a reduced level or no antigenicity in human subjects.

(131) In particular embodiments, the binding domain includes a humanized antibody or an engineered fragment thereof. In particular embodiments, a non-human antibody is humanized, where one or more amino acid residues of the antibody are modified to increase similarity to an antibody naturally produced in a human or fragment thereof. These nonhuman amino acid residues are often referred to as “import” residues, which are typically taken from an “import” variable domain. As provided herein, humanized antibodies or antibody fragments include one or more CDRs from nonhuman immunoglobulin molecules and framework regions wherein the amino acid residues including the framework are derived completely or mostly from human germline. In one aspect, the antigen binding domain is humanized. A humanized antibody can be produced using a variety of techniques known in the art, including CDR-grafting (see, e.g., European Patent No. EP 239,400; WO 91/09967; and U.S. Pat. Nos. 5,225,539, 5,530,101, and 5,585,089), veneering or resurfacing (see, e.g., EP 592,106 and EP 519,596; Padlan, 1991, Molecular Immunology, 28(4/5):489-498; Studnicka et al., 1994, Protein Engineering, 7(6):805-814; and Roguska et al., 1994, PNAS, 91:969-973), chain shuffling (see, e.g., U.S. Pat. No. 5,565,332), and techniques disclosed in, e.g., US 2005/0042664, US 2005/0048617, U.S. Pat. Nos. 6,407,213, 5,766,886, WO 9317105, Tan et al., J. Immunol., 169:1119-25 (2002), Caldas et al., Protein Eng., 13(5):353-60 (2000), Morea et al., Methods, 20(3):267-79 (2000), Baca et al., J. Biol. Chem., 272(16): 10678-84 (1997), Roguska et al., Protein Eng., 9(10):895-904 (1996), Couto et al., Cancer Res., 55 (23 Supp):5973s-5977s (1995), Couto et al., Cancer Res., 55(8):1717-22 (1995), Sandhu J S, Gene, 150(2):409-10 (1994), and Pedersen et al., J. Mol. Biol., 235(3):959-73 (1994). Often, framework residues in the framework regions will be substituted with the corresponding residue from the CDR

donor antibody to alter, for example improve, cellular marker binding. These framework substitutions are identified by methods well-known in the art, e.g., by modeling of the interactions of the CDR and framework residues to identify framework residues important for cellular marker binding and sequence comparison to identify unusual framework residues at particular positions. (See, e.g., U.S. Pat. No. 5,585,089; and Riechmann et al., 1988, *Nature*, 332:323).

(132) Antibodies with binding domains that specifically bind CD33 can be prepared using methods of obtaining monoclonal antibodies, methods of phage display, methods to generate human or humanized antibodies, or methods using a transgenic animal or plant engineered to produce antibodies as is known to those of ordinary skill in the art (see, for example, U.S. Pat. Nos. 6,291,161 and 6,291,158). Phage display libraries of partially or fully synthetic antibodies are available and can be screened for an antibody or fragment thereof that can bind to CD33. For example, binding domains may be identified by screening a Fab phage library for Fab fragments that specifically bind CD33 (see Hoet et al., *Nat. Biotechnol.* 23:344, 2005). Phage display libraries of human antibodies are also available. Additionally, traditional strategies for hybridoma development using CD33 as an immunogen in convenient systems (e.g., mice, HuMAb mouse® (GenPharm Intl. Inc., Mountain View, CA), TC mouse® (Kirin Pharma Co. Ltd., Tokyo, JP), KM-mouse® (Medarex, Inc., Princeton, NJ), llamas, chicken, rats, hamsters, rabbits, etc.) can be used to develop binding domains. Once identified, the amino acid sequence of the antibody and gene sequence encoding the antibody can be isolated and/or determined.

(133) As indicated, antibodies can be used as whole antibodies or binding fragments thereof, e.g., Fv, Fab, Fab', F(ab')₂, and single chain (sc) forms and fragments thereof that specifically bind CD33.

(134) In some instances, scFvs can be prepared according to methods known in the art (see, for example, Bird et al., (1988) *Science* 242:423-426 and Huston et al., (1988) *Proc. Natl. Acad. Sci. USA* 85:5879-5883). ScFv molecules can be produced by linking VH and VL regions of an antibody together using flexible polypeptide linkers. If a short polypeptide linker is employed (e.g., between 5-10 amino acids) intrachain folding is prevented. Interchain folding is also required to bring the two variable regions together to form a functional epitope binding site. For examples of linker orientations and sizes see, e.g., Hollinger et al. 1993 *Proc Natl Acad. Sci. U.S.A.* 90:6444-6448, US 2005/0100543, US 2005/0175606, US 2007/0014794, and WO2006/020258 and WO2007/024715. More particularly, linker sequences that are used to connect the VL and VH of an scFv are generally five to 35 amino acids in length. In particular embodiments, a VL-VH linker includes from five to 35, ten to 30 amino acids or from 15 to 25 amino acids. Variation in the linker length may retain or enhance activity, giving rise to superior efficacy in activity studies. scFV are commonly used as the binding domains of CAR discussed below.

(135) Additional examples of antibody-based binding domain formats include scFv-based grababodies and soluble VH domain antibodies. These antibodies form binding regions using only heavy chain variable regions. See, for example, Jespers et al., *Nat. Biotechnol.* 22:1161, 2004; Cortez-Retamozo et al., *Cancer Res.* 64:2853, 2004; Baral et al., *Nature Med.* 12:580, 2006; and Barthelemy et al., *J. Biol. Chem.* 283:3639, 2008.

(136) In particular embodiments, a VL region in a binding domain of the present disclosure is derived from or based on a VL of a known monoclonal antibody and contains one or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10) insertions, one or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10) deletions, one or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10) amino acid substitutions (e.g., conservative amino acid substitutions), or a combination of the above-noted changes, when compared with the VL of the known monoclonal antibody. An insertion, deletion or substitution may be anywhere in the VL region, including at the amino- or carboxy-terminus or both ends of this region, provided that each CDR includes zero changes or at most one, two, or three changes and provided a binding domain containing the modified VL region can still specifically bind its target with an affinity similar to the wild type binding domain.

(137) In particular embodiments, a binding domain VH region of the present disclosure can be derived from or based on a VH of a known monoclonal antibody and can contain one or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10) insertions, one or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10) deletions, one or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10) amino acid substitutions (e.g., conservative amino acid substitutions or non-conservative amino acid substitutions), or a combination of the above-noted changes, when compared with the VH of a known monoclonal antibody. An insertion, deletion or substitution may be anywhere in the VH region, including at the amino- or carboxy-terminus or both ends of this region, provided that each CDR includes zero changes or at most one, two, or three changes and provided a binding domain containing the modified VH region can still specifically bind its target with an affinity similar to the wild type binding domain.

(138) In particular embodiments, a binding domain includes or is a sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, at least 99.5%, or 100% identical to an amino acid sequence of a light chain variable region (VL) or to a heavy chain variable region (VH), or both, wherein each CDR includes zero changes or at most one, two, or three changes, from a monoclonal antibody or fragment or derivative thereof that specifically binds to a cellular marker of interest.

(139) An alternative source of binding domains includes sequences that encode random peptide libraries or sequences that encode an engineered diversity of amino acids in loop regions of alternative non-antibody scaffolds, such as single chain (sc) T-cell receptor (scTCR) (see, e.g., Lake et al., *Int. Immunol.* 11:745, 1999; Maynard et al., *J. Immunol. Methods* 306:51, 2005; U.S. Pat. No. 8,361,794), fibrinogen domains (see, e.g., Weisel et al., *Science* 230:1388, 1985), Kunitz domains (see, e.g., U.S. Pat. No. 6,423,498), designed ankyrin repeat proteins (DARPs; Binz et al., *J. Mol. Biol.* 332:489, 2003 and Binz et al., *Nat. Biotechnol.* 22:575, 2004), fibronectin binding domains (adnectins or monobodies; Richards et al., *J. Mol. Biol.* 326:1475, 2003; Parker et al., *Protein Eng. Des. Selec.* 18:435, 2005 and Hackel et al. (2008) *J. Mol. Biol.* 381:1238-1252), cysteine-knot miniproteins (Vita et al., 1995, *Proc. Nat'l. Acad. Sci. (USA)* 92:6404-6408; Martin et al., 2002, *Nat. Biotechnol.* 21:71, 2002 and Huang et al. (2005) *Structure* 13:755, 2005), tetratricopeptide repeat domains (Main et al., *Structure* 11:497, 2003 and Cortajarena et al., *ACS Chem. Biol.* 3:161, 2008), leucine-rich repeat domains (Stumpp et al., *J. Mol. Biol.* 332:471, 2003), lipocalin domains (see, e.g., WO 2006/095164, Beste et al., *Proc. Nat'l. Acad. Sci. (USA)* 96:1898, 1999 and Schönfeld et al., *Proc. Nat'l. Acad. Sci. (USA)* 106:8198, 2009), V-like domains (see, e.g., US 2007/0065431), C-type lectin domains (Zelensky and Gready, *FEBS J.* 272:6179, 2005; Beavil et al., *Proc. Nat'l. Acad. Sci. (USA)* 89:753, 1992 and Sato et al., *Proc. Nat'l. Acad. Sci. (USA)* 100:7779, 2003), mAb2 or Fc-region with antigen binding domain (Fcab™ (F-Star Biotechnology, Cambridge UK; see, e.g., WO 2007/098934 and WO 2006/072620), armadillo repeat proteins (see, e.g., Madhurantakam et al., *Protein Sci.* 21: 1015, 2012; WO 2009/040338), affilin (Ebersbach et al., *J. Mol. Biol.* 372: 172, 2007), affibody, avimers, knottins, fynomers, atrimers, cytotoxic T-lymphocyte associated protein-4 (Weidle et al., *Cancer Gen. Proteo.* 10:155, 2013), or the like (Nord et al., *Protein Eng.* 8:601, 1995; Nord et al., *Nat. Biotechnol.* 15:772, 1997; Nord et al., *Euro. J. Biochem.* 268:4269, 2001; Binz et al., *Nat. Biotechnol.* 23:1257, 2005; Boersma and Plückthun, *Curr. Opin. Biotechnol.* 22:849, 2011).

(140) Peptide aptamers include a peptide loop (which is specific for a cellular marker) attached at both ends to a protein scaffold. This double structural constraint increases the binding affinity of peptide aptamers to levels comparable to antibodies. The variable loop length is typically 8 to 20 amino acids and the scaffold can be any protein that is stable, soluble, small, and non-toxic. Peptide aptamer selection can be made using different systems, such as the yeast two-hybrid system (e.g., Gal4 yeast-two-hybrid system), or the LexA interaction trap system.

(141) In particular embodiments, a binding domain is a scTCR including V α / β and C α / β chains (e.g., V α -C α , V β -C β , V α -V β) or including a V α -C α , V β -C β , V α -V β pair specific for a CD33 peptide-MHC complex.

(142) In particular embodiments, engineered binding domains include V α , V β , C α , or C β regions derived from or based on a V α , V β , C α , or C β and includes one or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10) insertions, one or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10) deletions, one or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10) amino acid substitutions (e.g., conservative amino acid substitutions or non-conservative amino acid substitutions), or a combination of the above-noted changes, when compared with the referenced V α , V β , C α , or C β . An insertion, deletion or substitution may be anywhere in a V.sub.L, V.sub.H, V α , V β , C α , or C β region, including at the amino- or carboxy-terminus or both ends of these regions, provided that each CDR includes zero changes or at most one, two, or three changes and provides a target binding domain containing a modified V α , V β , C α , or C β region can still specifically bind its target with an affinity and action similar to wild type.

(143) In particular embodiments, engineered binding domains include a sequence that is at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, at least 99.5%, or 100% identical to an amino acid sequence of a known or identified binding domain, wherein each CDR includes zero changes or at most one, two, or three changes, from a known or identified binding domain or fragment or derivative thereof that specifically binds to the targeted cellular marker.

(144) The precise amino acid sequence boundaries of a given CDR or FR can be readily determined using any of a number of well-known schemes, including those described by: Kabat et al. (1991) "Sequences of Proteins of Immunological Interest," 5th Ed. Public Health Service, National Institutes of Health, Bethesda, Md. (Kabat numbering scheme); Al-Lazikani et al. (1997) J Mol Biol 273: 927-948 (Chothia numbering scheme); Maccallum et al. (1996) J Mol Biol 262: 732-745 (Contact numbering scheme); Martin et al. (1989) Proc. Natl. Acad. Sci., 86: 9268-9272 (AbM numbering scheme); Lefranc M P et al. (2003) Dev Comp Immunol 27(1): 55-77 (IMGT numbering scheme); and Honegger and Pluckthun (2001) J Mol Biol 309(3): 657-670 ("Aho" numbering scheme). The boundaries of a given CDR or FR may vary depending on the scheme used for identification. For example, the Kabat scheme is based on structural alignments, while the Chothia scheme is based on structural information. Numbering for both the Kabat and Chothia schemes is based upon the most common antibody region sequence lengths, with insertions accommodated by insertion letters, for example, "30a," and deletions appearing in some antibodies. The two schemes place certain insertions and deletions ("indels") at different positions, resulting in differential numbering. The Contact scheme is based on analysis of complex crystal structures and is similar in many respects to the Chothia numbering scheme. In particular embodiments, the antibody CDR sequences disclosed herein are according to Kabat numbering.

(V-B) LINKERS

(145) As indicated, many CD33-targeting agents include linkers. Linkers can be used to achieve different outcomes depending on the particular CD33-targeting agent under consideration. A linker can include any chemical moiety that is capable of linking portions of a CD33-targeting agent. Linkers can be flexible, rigid, or semi-rigid, depending on the desired function of the linker.

(146) For example, in particular embodiments, linkers provide flexibility and room for conformational movement between different components of CD33-targeting agents. Commonly used flexible linkers include linker sequence with the amino acids glycine and serine (Gly-Ser linkers). In particular embodiments, the linker sequence includes sets of glycine and serine repeats such as from one to ten repeats of (Gly.sub.xSer.sub.y).sub.n, wherein x and y are independently an integer from 0 to 10 provided that x and y are not both 0 and wherein n is an integer of 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10). Particular examples include (Gly₄Ser)_n (SEQ ID NO: 20), (Gly₃Ser)_n(Gly₄Ser)_n (SEQ ID NO: 21), (Gly₃Ser)_n(Gly₂Ser)_n (SEQ ID NO: 22), or (Gly₃Ser)_n(Gly₄Ser)₁ (SEQ ID NO: 23). In particular embodiments, the linker is (Gly₄Ser)₄ (SEQ ID NO: 24), (Gly₄Ser)₃ (SEQ ID NO: 25), (Gly₄Ser)₂ (SEQ ID NO: 26), (Gly₄Ser)₁ (SEQ ID NO: 27), (Gly₃Ser)₂ (SEQ ID NO: 28), (Gly₃Ser)₁ (SEQ ID NO: 29), (Gly₂Ser)₂ (SEQ ID NO: 30) or (Gly₂Ser)₁, GGSGGGSGSG (SEQ ID NO: 31), GGSGGGSGSG (SEQ ID NO: 32), or GGSGGGSG (SEQ

(147) In some situations, flexible linkers may be incapable of maintaining a distance or positioning of CD33-targeting agent components needed for a particular use. In these instances, rigid or semi-rigid linkers may be useful. Examples of rigid or semi-rigid linkers include proline-rich linkers. In particular embodiments, a proline-rich linker is a peptide sequence having more proline residues than would be expected based on chance alone. In particular embodiments, a proline-rich linker is one having at least 30%, at least 35%, at least 36%, at least 39%, at least 40%, at least 48%, at least 50%, or at least 51% proline residues. Particular examples of proline-rich linkers include fragments of proline-rich salivary proteins (PRPs).

(148) Spacer regions are a type of linker region that are used to create appropriate distances and/or flexibility from other linked components. In particular embodiments, the length of a spacer region can be customized for individual cellular markers on unwanted cells to optimize unwanted CD33-expressing cell recognition and destruction. The spacer can be of a length that provides for increased effectiveness of the CD33-targeting agent following CD33 binding, as compared to in the absence of the spacer. In particular embodiments, a spacer region length can be selected based upon the location of a cellular marker epitope, affinity of a binding domain for the epitope, and/or the ability of the CD33-targeting agent to mediate cell destruction following CD33 binding.

(149) Spacer regions typically include those having 10 to 250 amino acids, 10 to 200 amino acids, 10 to 150 amino acids, 10 to 100 amino acids, 10 to 50 amino acids, or 10 to 25 amino acids. In particular embodiments, a spacer region is 12 amino acids, 20 amino acids, 21 amino acids, 26 amino acids, 27 amino acids, 45 amino acids, or 50 amino acids.

(150) Exemplary spacer regions include all or a portion of an immunoglobulin hinge region. An immunoglobulin hinge region may be a wild-type immunoglobulin hinge region or an altered wild-type immunoglobulin hinge region. In certain embodiments, an immunoglobulin hinge region is a human immunoglobulin hinge region. As used herein, a “wild type immunoglobulin hinge region” refers to a naturally occurring upper and middle hinge amino acid sequences interposed between and connecting the CH1 and CH2 domains (for IgG, IgA, and IgD) or interposed between and connecting the CH1 and CH3 domains (for IgE and IgM) found in the heavy chain of an antibody.

(151) An immunoglobulin hinge region may be an IgG, IgA, IgD, IgE, or IgM hinge region. An IgG hinge region may be an IgG1, IgG2, IgG3, or IgG4 hinge region. Sequences from IgG1, IgG2, IgG3, IgG4 or IgD can be used alone or in combination with all or a portion of a CH2 region; all or a portion of a CH3 region; or all or a portion of a CH2 region and all or a portion of a CH3 region.

(152) Other examples of hinge regions used in fusion binding proteins described herein include the hinge region present in the extracellular regions of type 1 membrane proteins, such as CD8a, CD4, CD28 and CD7, which may be wild-type or variants thereof.

(153) In particular embodiments, a spacer region includes a hinge region that includes a type II C-lectin interdomain (stalk) region or a cluster of differentiation (CD) molecule stalk region. A “stalk region” of a type II C-lectin or CD molecule refers to the portion of the extracellular domain of the type II C-lectin or CD molecule that is located between the C-type lectin-like domain (CTLCD; e.g., similar to CTLCD of natural killer cell receptors) and the hydrophobic portion (transmembrane domain). For example, the extracellular domain of human CD94 (GenBank Accession No.

AAC50291.1) corresponds to amino acid residues 34-179, but the CTLCD corresponds to amino acid residues 61-176, so the stalk region of the human CD94 molecule includes amino acid residues 34-60, which are located between the hydrophobic portion (transmembrane domain) and CTLCD (see Boyington et al., *Immunity* 10:15, 1999; for descriptions of other stalk regions, see also Beavil et al., *Proc. Nat'l. Acad. Sci. USA* 89:153, 1992; and Figdor et al., *Nat. Rev. Immunol.* 2:11, 2002). These type II C-lectin or CD molecules may also have junction amino acids (described below) between the stalk region and the transmembrane region or the CTLCD. In another example, the 233 amino acid human NKG2A protein (GenBank Accession No. P26715.1) has a hydrophobic portion (transmembrane domain) ranging from amino acids 71-93 and an extracellular domain

ranging from amino acids 94-233. The CTLD includes amino acids 119-231 and the stalk region includes amino acids 99-116, which may be flanked by additional junction amino acids. Other type II C-lectin or CD molecules, as well as their extracellular ligand-binding domains, stalk regions, and CTLDs are known in the art (see, e.g., GenBank Accession Nos. NP 001993.2; AAH07037.1; NP 001773.1; AAL65234.1; CAA04925.1; for the sequences of human CD23, CD69, CD72, NKG2A, and NKG2D and their descriptions, respectively).

(154) In particular embodiments, a spacer region is (GGGGS)_n (SEQ ID NO: 20) wherein n is an integer including, 1, 2, 3, 4, 5, 6, 7, 8, 9, or more. In particular embodiments, the spacer region is (EAAAK)_n (SEQ ID NO: 36) wherein n is an integer including 1, 2, 3, 4, 5, 6, 7, 8, 9, or more.

(155) Junction amino acids can be a short oligo- or protein linker, preferably between 2 and 9 amino acids (e.g., 2, 3, 4, 5, 6, 7, 8, or 9 amino acids) in length to form the linker. In particular embodiments, a glycine-serine doublet can be used as a suitable junction amino acid linker. In particular embodiments, a single amino acid, e.g., an alanine, a glycine, can be used as a suitable junction amino acid.

(156) Linkers can be susceptible to cleavage (cleavable linker), such as, acid-induced cleavage, photo-induced cleavage, peptidase-induced cleavage, esterase-induced cleavage, and disulfide bond cleavage. Alternatively, linkers can be substantially resistant to cleavage (e.g., stable linker or noncleavable linker). In some aspects, the linker is a procharged linker, a hydrophilic linker, or a dicarboxylic acid-based linker.

(V-C) EXAMPLES OF CD33-TARGETING AGENTS

(157) CD33-targeting agents include molecules that result in the destruction of CD33-expressing cells. Examples of CD33-targeting agents include anti-CD33 antibodies; anti-CD33 immunotoxins; anti-CD33 antibody-drug conjugates; anti-CD33 antibody-radioisotope conjugates; anti-CD33 multispecific antibodies (e.g. bi- and trispecific antibodies); and/or immune cells expressing CARs or engineered TCRs that specifically bind CD33. Anti-CD33 antibodies are described above in relation to binding domains.

(V-C-I) ANTI-CD33 ANTIBODY CONJUGATES

(158) Anti-CD33 antibody conjugates are artificial molecules that include a molecule conjugated to a CD33 binding domain. Anti-CD33 antibody conjugates include anti-CD33 immunotoxins, ADCs, and radioisotope conjugates.

(159) Anti-CD33 immunotoxins are artificial molecules that include a toxin linked to a CD33 binding domain. In particular embodiments, immunotoxins selectively deliver an effective dose of a cytotoxin to non-genetically modified CD33-expressing cells.

(160) To prepare immunotoxins, linker-cytotoxin conjugates can be made by conventional methods analogous to those described by Doronina et al. (Bioconjugate Chem. 17: 114-124, 2006).

Immunotoxins containing CD33 binding domains can be prepared by standard methods for cysteine conjugation, such as by methods analogous to that described in Hamblett et al., Clin. Cancer Res. 10:7063-7070, 2004; Doronina et al., Nat. Biotechnol. 21(7): 778-784, 2003; and Francisco et al., Blood 102:1458-1465. 2003.

(161) Immunotoxins with multiple (e.g., four) cytotoxins per binding domain can be prepared by partial reduction of the binding domain with an excess of a reducing reagent such as dithiothreitol (DTT) or tris(2-carboxyethyl)phosphine (TCEP) at 37° C. for 30 min, then the buffer can be exchanged by elution through SEPHADEX G-25 resin with 1 mM DTPA (diethylene triamine penta-acetic acid) in Dulbecco's phosphate-buffered saline (DPBS). The eluent can be diluted with further DPBS, and the thiol concentration of the binding domain can be measured using 5,5'-dithiobis(2-nitrobenzoic acid) [Ellman's reagent]. An excess, for example 5-fold, of the linker-cytotoxin conjugate can be added at 4° C. for 1 hr, and the conjugation reaction can be quenched by addition of a substantial excess, for example 20-fold, of cysteine. The resulting immunotoxin mixture can be purified on SEPHADEX G-25 equilibrated in PBS to remove unreacted linker-cytotoxin conjugate, desalted if desired, and purified by size-exclusion chromatography. The

resulting immunotoxin can then be sterile filtered, for example, through a 0.2 µm filter, and can be lyophilized if desired for storage.

(162) Frequently used plant toxin drugs are divided into two classes: (1) holotoxins (or class II ribosome inactivating proteins), such as ricin, abrin, mistletoe lectin, and modeccin, and (2) hemitoxins (class I ribosome inactivating proteins), such as pokeweed antiviral protein (PAP), saporin, Bryodin 1, bouganin, and gelonin. Commonly used bacterial toxins include diphtheria toxin (DT) and *Pseudomonas* exotoxin (PE). Kreitman, Current Pharmaceutical Biotechnology 2:313-325 (2001). The toxin may also be an antibody or other peptide. Anti-CD33 ADCs include a CD33 binding domain linked to a cytotoxic drug that results in the bound cell's destruction. ADCs allow for the targeted delivery of a drug moiety to a selected cell, and, in particular embodiments intracellular accumulation therein, where systemic administration of unconjugated drugs may result in unacceptable levels of toxicity to normal cells (Polakis P. (2005) Current Opinion in Pharmacology 5:382-387).

(163) ADC can include targeted drugs which combine properties of both antibodies and cytotoxic drugs by targeting potent cytotoxic drugs to antigen-expressing cells (Teicher, B. A. (2009) Current Cancer Drug Targets 9:982-1004), thereby enhancing the therapeutic index by maximizing efficacy and minimizing off-target toxicity (Carter, P. J. and Senter P. D. (2008) The Cancer Jour. 14(3):154-169; Chari, R. V. (2008) Acc. Chem. Res. 41:98-107). See also Kamath & Iyer (Pharm Res. 32(11): 3470-3479, 2015), which describes considerations for the development of ADCs.

(164) ADC compounds of the disclosure include those with anti-CD33 cell activity. In particular embodiments, the ADC compounds include a CD33 binding domain conjugated, i.e. covalently attached, to the drug moiety.

(165) Examples of drugs useful to include within the ADC format include taxol, taxane, cytochalasin B, gramicidin D, ethidium bromide, emetine, mitomycin, etoposide, tenoposide, vincristine, vinblastine, colchicin, doxorubicin, daunorubicin, dihydroxy anthracenedione, mitoxantrone, mithramycin, actinomycin D, 1-dehydrotestosterone, glucocorticoids, procaine, tetracaine, lidocaine, propranolol, and puromycin and analogs or homologs thereof. Other appropriate toxins include, for example, CC-1065 and analogues thereof, the duocarmycins. Additional examples include maytansinoid (including monomethyl auristatin E [MMAE]; vedotin), dolastatin, auristatin, calicheamicin, pyrrolobenzodiazepine (PBD) dimer, indolino-benzodiazepine dimer, nemorubicin and its derivatives, PNU-159682, anthracycline, *vinca* alkaloid, trichothecene, camptothecin, elinafide, and stereoisomers, isosteres, analogs, and derivatives thereof that have cytotoxic activity.

(166) The drug may be obtained from essentially any source; it may be synthetic or a natural product isolated from a selected source, e.g., a plant, bacterial, insect, mammalian or fungal source. The drug may also be a synthetically modified natural product or an analogue of a natural product.

(167) Exemplary ADCs that target CD33 include GO (which includes the recombinant humanized IgG4 anti-CD33 hP67.6 antibody linked to the cytotoxic antitumor antibiotic calicheamicin; U.S. Pat. No. 5,773,001), lintuzumab (SGN-33; HuM195; Caron et al., Can. Res. 52:6761-6767, 1992), SGN-CD33A (the antibody portion of which is h2H12EC a.k.a h2H12d; see US 2013/0309223), and IMG779.

(168) Anti-CD33 antibody-radioisotope conjugates include a CD33 binding domain linked to a cytotoxic radioisotope for use in nuclear medicine. Nuclear medicine refers to the diagnosis and/or treatment of conditions by administering radioactive isotopes (radioisotopes or radionuclides) to a subject. Therapeutic nuclear medicine is often referred to as radiation therapy or radioimmunotherapy (RIT).

(169) Examples of radioactive isotopes that can be conjugated to CD33 binding domains include iodine-131, indium-111, yttrium-90, and lutetium-177, as well as alpha-emitting radionuclides such as astatine-211 or bismuth-212, bismuth-213, or actinium-225. Methods for preparing radioimmunoconjugates are established in the art. Examples of radioimmunotoxins are

commercially available, including Zevalin® (RIT Oncology, Seattle, WA), and similar methods can be used to prepare radioimmunotoxins using the binding domains of the disclosure.

(170) Examples of radionuclides that are useful for radiation therapy include ^{225}Ac and ^{227}Th . ^{225}Ac is a radionuclide with the half-life of ten days. As ^{225}Ac decays the daughter isotopes ^{221}Fr , ^{213}Bi , and ^{209}Pb are formed. ^{227}Th has a half-life of 19 days and forms the daughter isotope ^{223}Ra .

(171) Additional examples of useful radioisotopes include ^{228}Ac , ^{111}Ag , ^{124}Am , ^{74}As , ^{209}At , ^{194}Au , ^{128}Ba , ^7Be , ^{206}Bi , ^{245}Bk , ^{246}Bk , ^{76}Br , ^{11}C , ^{47}Ca , ^{254}Cf , ^{242}Cm , ^{51}Cr , ^{67}Cu , ^{153}Dy , ^{157}Dy , ^{159}Dy , ^{165}Dy , ^{166}Dy , ^{171}Er , ^{250}Es , ^{254}Es , ^{147}Eu , ^{157}Eu , ^{52}Fe , ^{59}Fe , ^{251}Fm , ^{252}Fm , ^{253}Fm , ^{66}Ga , ^{72}Ga , ^{146}Gd , ^{153}Gd , ^{68}Ge , ^{170}Hf , ^{171}Hf , ^{193}Hg , $^{193\text{m}}\text{Hg}$, $^{160\text{m}}\text{Ho}$, ^{130}I , ^{135}I , $^{114\text{m}}\text{In}$, ^{185}Ir , ^{42}K , ^{43}K , ^{76}Kr , ^{79}Kr , $^{81\text{m}}\text{Kr}$, ^{132}La , ^{262}Lr , ^{169}Lu , $^{174\text{m}}\text{Lu}$, $^{176\text{m}}\text{Lu}$, ^{257}Md , ^{260}Md , ^{28}Mg , ^{52}Mn , ^{90}Mo , ^{24}Na , ^{95}Nb , ^{138}Nd , ^{57}Ni , ^{66}Ni , ^{234}Np , ^{15}O , ^{182}Os , $^{189\text{m}}\text{Os}$, ^{191}Os , ^{32}P , ^{201}Pb , ^{101}Pd , ^{143}Pr , ^{191}Pt , ^{243}Pu , ^{225}Ra , ^{81}Rb , ^{188}Re , ^{105}Rh , ^{211}Rn , ^{103}Ru , ^{35}S , ^{44}Sc , ^{72}Se , ^{153}Sm , ^{125}Sn , ^{91}Sr , ^{173}Ta , ^{154}Tb , ^{127}Te , ^{234}Th , ^{45}Ti , ^{166}Tm , ^{230}U , ^{237}U , ^{240}U , ^{48}V , ^{178}W , ^{181}W , ^{188}W , ^{125}Xe , ^{127}Xe , ^{133}Xe , $^{133\text{m}}\text{Xe}$, ^{135}Xe , $^{85\text{m}}\text{Y}$, ^{86}Y , ^{93}Y , ^{169}Yb , ^{175}Yb , ^{65}Zn , $^{71\text{m}}\text{Zn}$, ^{86}Zr , ^{95}Zr , and/or ^{97}Zr .

(V-C-II) ANTI-CD33 BISPECIFIC & TRISPECIFIC ANTIBODIES

(172) Anti-CD33 bispecific antibodies bind at least two epitopes wherein at least one of the epitopes is located on CD33. Anti-CD33 trispecific antibodies bind at least 3 epitopes, wherein at least one of the epitopes is located on CD33.

(173) Some examples of bispecific antibodies have two heavy chains (each having three heavy chain CDRs, followed by (N-terminal to C-terminal) a CH1 domain, a hinge, a CH2 domain, and a CH3 domain), and two immunoglobulin light chains that confer antigen-binding specificity through association with each heavy chain. However, additional architectures can be used, including bispecific antibodies in which the light chain(s) associate with each heavy chain but do not (or minimally) contribute to antigen-binding specificity, or that can bind one or more of the epitopes bound by the heavy chain antigen-binding regions, or that can associate with each heavy chain and enable binding of one or both of the heavy chains to one or both epitopes. Other forms of bispecific antibodies include the single chain “Janusins” described in Traunecker et al. (Embo Journal, 10, 3655-3659, 1991).

(174) Bispecific antibodies can be prepared as full-length antibodies or antibody fragments (for example, F(ab')_2 bispecific antibodies).

(175) Methods for making bispecific antibodies are also described in Millstein et al. Nature 305:37-39, 1983; WO 1993/008829; and Traunecker et al., EMBO J. 10:3655-3659, 1991. In particular embodiments, bispecific antibodies can be prepared using chemical linkage. For example, Brennan et al. (Science 229: 81, 1985) describes a procedure wherein intact antibodies are proteolytically cleaved to generate F(ab')_2 fragments. These fragments are reduced in the presence of the dithiol complexing agent, sodium arsenite, to stabilize vicinal dithiols and prevent intermolecular disulfide formation. The Fab' fragments generated then are converted to thionitrobenzoate (TNB) derivatives. One of the Fab' -TNB derivatives then is reconverted to the Fab' -thiol by reduction with mercaptoethylamine and is mixed with an equimolar amount of the other Fab' -TNB derivative to form the bispecific antibody.

(176) In particular embodiments, CD33-targeting agents include bi- or trispecific immune cell engaging antibody constructs. An example of a bi- or trispecific immune cell engaging antibody

construct includes those which bind both CD33 and an immune cell (e.g., T-cell) activating epitope, with the goal of bringing immune cells to CD33-expressing cells to destroy the CD33-expressing cells. See, for example, US 2008/0145362. Such constructs are referred to herein as immune-activating bi- or tri-specifics or I-ABTS). In particular embodiments, I-ABTS include AMG330, AMG673, and AMV-564. BiTEs® are one form of I-ABTS. Immune cells that can be targeted for localized activation by I-ABTS within the current disclosure include, for example, T-cells, natural killer (NK) cells, and macrophages which are discussed in more detail herein. Bispecific immune cell engaging antibody constructs, including I-ABTS utilize bispecific binding domains, such as bispecific antibodies to target CD33-expressing cells and immune cells. The binding domain that binds CD33 and the binding domain that binds and activates an immune cell may be joined through a linker, as described elsewhere herein.

(177) T-cell activation can be mediated by two distinct signals: those that initiate antigen-dependent primary activation and provide a T-cell receptor like signal (primary cytoplasmic signaling sequences) and those that act in an antigen-independent manner to provide a secondary or co-stimulatory signal (secondary cytoplasmic signaling sequences). I-ABTS disclosed herein can target any T-cell activating epitope that upon binding induces T-cell activation. Examples of such T-cell activating epitopes are on T-cell markers including CD2, CD3, CD7, CD27, CD28, CD30, CD40, CD83, 4-1BB (CD 137), OX40, lymphocyte function-associated antigen-1 (LFA-1), LIGHT, NKG2C, and B7-H3.

(178) Several different subsets of T-cells have been discovered, each with a distinct function. For example, a majority of T-cells have a TCR existing as a complex of several proteins. The actual T-cell receptor is composed of two separate peptide chains, which are produced from the independent T-cell receptor α and β (TCR α and TCR β) genes and are called α - and β -TCR chains.

(179) CD3 is a primary signal transduction element of T-cell receptors. CD3 is composed of a group of invariant proteins called gamma (γ), delta (δ), epsilon (ϵ), zeta (ζ) and eta (η) chains. The γ , δ , and ϵ chains are structurally-related, each containing an Ig-like extracellular constant domain followed by a transmembrane region and a cytoplasmic domain of more than 40 amino acids. The ζ and η chains have a distinctly different structure: both have a very short extracellular region of only 9 amino acids, a transmembrane region and a long cytoplasmic tail including 113 and 115 amino acids in the ζ and η chains, respectively. The invariant protein chains in the CD3 complex associate to form noncovalent heterodimers of the ϵ chain with a γ chain ($\epsilon\gamma$) or with a δ chain ($\epsilon\delta$) or of the ζ and η chain ($\zeta\eta$), or a disulfide-linked homodimer of two ζ chains ($\zeta\zeta$). 90% of the CD3 complex incorporate the $\zeta\zeta$ homodimer.

(180) The cytoplasmic regions of the CD3 chains include a motif designated the immunoreceptor tyrosine-based activation motif (ITAM). This motif is found in a number of other receptors including the Ig- α /Ig- β heterodimer of the B-cell receptor complex and Fc receptors for IgE and IgG. The ITAM sites associate with cytoplasmic tyrosine kinases and participate in signal transduction following TCR-mediated triggering. In CD3, the γ , δ and ϵ chains each contain a single copy of ITAM, whereas the ζ and η chains harbor three ITAMs in their long cytoplasmic regions. Indeed, the ζ and η chains have been ascribed a major role in T-cell activation signal transduction pathways.

(181) In particular embodiments, the CD3 binding domain (e.g., scFv) of an I-ABTS is derived from the OKT3 antibody (also utilized in blinatumomab). The OKT3 antibody is described in detail in U.S. Pat. No. 5,929,212. It includes a variable light chain including a CDRL1 sequence including SASSSVSYMN (SEQ ID NO: 55), a CDRL2 sequence including RWIYDTSKLAS (SEQ ID NO: 56), and a CDRL3 sequence including QQWSSNPFT (SEQ ID NO: 57). In particular embodiments, the CD3 T-cell activating epitope binding domain is a human or humanized binding domain (e.g., scFv) including a variable heavy chain including a CDRH1 sequence including KASGYTFTRYTMH (SEQ ID NO: 58), a CDRH2 sequence including INPSRGYTNYNQKFKD (SEQ ID NO: 59), and a CDRH3 sequence including YYDDHYCLDY

(SEQ ID NO: 60).

(182) The following sequence is an scFv derived from OKT3 which retains the capacity to bind CD3:

(183) TABLE-US-00005 (SEQ ID NO: 61)

QVQLQQSGAELARPGASVKMSCKASGYTFTRYTMHWVKQRPGQGLEWIG
YINPSRGYTNYNQKFKDKATLTDDKSSSTAYMQLSSLTSEDSAVYYCAR
YYDDHYCLDYWGQGTTTLTVSSSGGGGGSGGGGSGGGGSGQIVLTQSPAIMS
ASPGEKVTMTCSASSSVSYMNWYQQKSGTSPKRWIYDTSKLASGVPAHF
RSGSGGTSYSLTISGMEAEDAATYYCQQWSSNPFTFGSGTKLEINR.

It may also be used as a CD3 binding domain.

(184) In particular embodiments, the CD3 T-cell activating epitope binding domain is a human or humanized binding domain (e.g., scFv) including a variable light chain including a CDRL1 sequence including QSLVHNNGNTY (SEQ ID NO: 62), a CDRL2 sequence including KVS, and a CDRL3 sequence including GQGTQYPFT (SEQ ID NO: 63). In particular embodiments, the CD3 T-cell activating epitope binding domain is a human or humanized binding domain (e.g., scFv) including a variable heavy chain including a CDRH1 sequence including GFTFTKAW (SEQ ID NO: 64), a CDRH2 sequence including IKDKSNSYAT (SEQ ID NO: 65), and a CDRH3 sequence including RGVYYALSPFDY (SEQ ID NO: 66). These reflect CDR sequences of the 20G6-F3 antibody.

(185) In particular embodiments, the CD3 T-cell activating epitope binding domain is a human or humanized binding domain (e.g., scFv) including a variable light chain including a CDRL1 sequence including QSLVHDNGNTY (SEQ ID NO: 67), a CDRL2 sequence including KVS, and a CDRL3 sequence including GQGTQYPFT (SEQ ID NO: 63). In particular embodiments, the CD3 T-cell activating epitope binding domain is a human or humanized binding domain (e.g., scFv) including a variable heavy chain including a CDRH1 sequence including GFTFSNAW (SEQ ID NO: 69), a CDRH2 sequence including IKARSNNYAT (SEQ ID NO: 70), and a CDRH3 sequence including RGTYYASKPFDY (SEQ ID NO: 71). These reflect CDR sequences of the 4B4-D7 antibody.

(186) In particular embodiments, the CD3 T-cell activating epitope binding domain is a human or humanized binding domain (e.g., scFv) including a variable light chain including a CDRL1 sequence including QSLEHNNGNTY (SEQ ID NO: 72), a CDRL2 sequence including KVS, and a CDRL3 sequence including GQGTQYPFT (SEQ ID NO: 63). In particular embodiments, the CD3 T-cell activating epitope binding domain is a human or humanized binding domain (e.g., scFv) including a variable heavy chain including a CDRH1 sequence including GFTFSNAW (SEQ ID NO: 69), a CDRH2 sequence including IKDKSNNYAT (SEQ ID NO: 75), and a CDRH3 sequence including RYVHYGIGYAMDA (SEQ ID NO: 76). These reflect CDR sequences of the 4E7-C9 antibody.

(187) In particular embodiments, the CD3 T-cell activating epitope binding domain is a human or humanized binding domain (e.g., scFv) including a variable light chain including a CDRL1 sequence including QSLVHTNGNTY (SEQ ID NO: 77), a CDRL2 sequence including KVS, and a CDRL3 sequence including GQGTHYPFT (SEQ ID NO: 78). In particular embodiments, the CD3 T-cell activating epitope binding domain is a human or humanized binding domain (e.g., scFv) including a variable heavy chain including a CDRH1 sequence including GFTFTNAW (SEQ ID NO: 79), a CDRH2 sequence including KDKSNNYAT (SEQ ID NO: 80), and a CDRH3 sequence including RYVHYRFAYALDA (SEQ ID NO: 81). These reflect CDR sequences of the 18F5-H10 antibody.

(188) Additional examples of anti-CD3 antibodies, binding domains, and CDRs can be found in WO2016/116626. TR66 may also be used. WO 2015/036583 describes a bispecific antibody construct that binds to CD33 and CD3.

(189) CD28 is a surface glycoprotein present on 80% of peripheral T-cells in humans and is present

on both resting and activated T-cells. CD28 binds to B7-1 (CD80) and B7-2 (CD86) and is the most potent of the known co-stimulatory molecules (June et al., Immunol. Today 15:321, 1994; Linsley et al., Ann. Rev. Immunol. 11:191, 1993). In particular embodiments, the CD28 binding domain (e.g., scFv) is derived from CD80, CD86 or the 9D7 antibody. Additional antibodies that bind CD28 include 9.3, KOLT-2, 15E8, 248.23.2, and EX5.3D10. Further, 1YJD provides a crystal structure of human CD28 in complex with the Fab fragment of a mitogenic antibody (5.11A1). In particular embodiments, antibodies that do not compete with 9D7 are selected.

(190) In particular embodiments, a CD28 binding domain is derived from TGN1412. In particular embodiments, the variable heavy chain of TGN1412 includes:

(191) TABLE-US-00006 (SEQ ID NO: 37)

QVQLVQSGAEVKKPGASVKVSCKASGYTFTSYIHWVRQAPGQGLEWIG
CIYPGNVNTNYNEKFKDRATLTVDTSISTAYMELSRLRSDDTAVYFCTR
SHYGLDWNFDVWGQGTTTVTVSS

and the variable light chain of TGN1412 includes:

(192) TABLE-US-00007 (SEQ ID NO: 38)

DIQMTQSPSSLSASVGDRVTITCHASQNIYVWLNWYQQKPGKAPKLLIY
KASNLHTGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQGQTPYPTF GGGTKVEIK.

(193) In particular embodiments, the CD28 binding domain includes a variable light chain including a CDRL1 sequence including HASQNIYVWLN (SEQ ID NO: 68), CDRL2 sequence including KASNLHT (SEQ ID NO: 73), and CDRL3 sequence including QQGQTPYPT (SEQ ID NO: 74), a variable heavy chain including a CDRH1 sequence including GYTFTSYIHW (SEQ ID NO: 90), a CDRH2 sequence including CIYPGNVNTNYNEK (SEQ ID NO: 91), and a CDRH3 sequence including SHYGLDWNFDV (SEQ ID NO: 92).

(194) In particular embodiments, the CD28 binding domain including a variable light chain including a CDRL1 sequence including HASQNIYVWLN (SEQ ID NO: 68), a CDRL2 sequence including KASNLHT (SEQ ID NO: 73), and a CDRL3 sequence including QQGQTPYPT (SEQ ID NO: 74) and a variable heavy chain including a CDRH1 sequence including SYIHW (SEQ ID NO: 49), a CDRH2 sequence including CIYPGNVNTNYNEKFKD (SEQ ID NO: 94), and a CDRH3 sequence including SHYGLDWNFDV (SEQ ID NO: 92).

(195) Activated T-cells express 4-1BB (CD137). In particular embodiments, the 4-1BB binding domain includes a variable light chain including a CDRL1 sequence including RASQSVS (SEQ ID NO: 95), a CDRL2 sequence including ASNRAT (SEQ ID NO: 108), and a CDRL3 sequence including QRSNWPPALT (SEQ ID NO: 109) and a variable heavy chain including a CDRH1 sequence including YYWS (SEQ ID NO: 110), a CDRH2 sequence including INH, and a CDRH3 sequence including YGPGNYDWYFDL (SEQ ID NO: 111).

(196) In particular embodiments, the 4-1BB binding domain includes a variable light chain including a CDRL1 sequence including SGDNIQDQYAH (SEQ ID NO: 112), a CDRL2 sequence including QDKNRPS (SEQ ID NO: 113), and a CDRL3 sequence including ATYTGFGLSLAV (SEQ ID NO: 114) and a variable heavy chain including a CDRH1 sequence including GYSFSTYWIS (SEQ ID NO: 115), a CDRH2 sequence including KIYPGDSYTNYS (SEQ ID NO: 116) and a CDRH3 sequence including GYGIFDY (SEQ ID NO: 35).

(197) Particular embodiments disclosed herein include immune cell binding domains that bind epitopes on CD8. In particular embodiments, the CD8 binding domain (e.g., scFv) is derived from the OKT8 antibody. For example, in particular embodiments, the CD8 T-cell activating epitope binding domain is a human or humanized binding domain (e.g., scFv) including a variable light chain including a CDRL1 sequence including RTSRSISQYLA (SEQ ID NO: 82), a CDRL2 sequence including SGSTLQS (SEQ ID NO: 83), and a CDRL3 sequence including QQHNENPLT (SEQ ID NO: 84). In particular embodiments, the CD8 T-cell activating epitope binding domain is a human or humanized binding domain (e.g., scFv) including a variable heavy chain including a CDRH1 sequence including GFNIKD (SEQ ID NO: 85), a CDRH2 sequence including

RIDPANDNT (SEQ ID NO: 86), and a CDRH3 sequence including GYGYYVFDH (SEQ ID NO: 87). These reflect CDR sequences of the OKT8 antibody.

(198) In particular embodiments, an immune cell binding domain is a scTCR including V.sub.α/β and C.sub.α/β chains (e.g., V.sub.α-C.sub.α, V.sub.β-C.sub.β, V.sub.α-V.sub.β) or including V.sub.α-C.sub.α, V.sub.β-C.sub.β, V.sub.α-V.sub.β pair specific for a target epitope of interest. In particular embodiments, T-cell activating epitope binding domains can be derived from or based on a Vα, Vβ, Cα, or Cβ of a known TCR (e.g., a high-affinity TCR).

(199) Natural killer cells (also known as NK cells, K cells, and killer cells) are activated in response to interferons or macrophage-derived cytokines. They serve to contain viral infections while the adaptive immune response is generating antigen-specific cytotoxic T cells that can clear the infection. NK cells express CD8, CD16 and CD56 but do not express CD3.

(200) In particular embodiments NK cells are targeted for localized activation by I-ABTS. NK cells can induce apoptosis or cell lysis by releasing granules that disrupt cellular membranes and can secrete cytokines to recruit other immune cells.

(201) Examples of activating proteins expressed on the surface of NK cells include NKG2D, CD8, CD16, KIR2DL4, KIR2DS1, KIR2DS2, KIR3DS1, NKG2C, NKG2E, NKG2D, and several members of the natural cytotoxicity receptor (NCR) family. Examples of NCRs that activate NK cells upon ligand binding include NKp30, NKp44, NKp46, NKp80, and DNAM-1.

(202) Examples of commercially available antibodies that bind to an NK cell receptor and induce and/or enhance activation of NK cells include: 5C6 and 1D11, which bind and activate NKG2D (available from BioLegend® San Diego, CA); mAb 33, which binds and activates KIR2DL4 (available from BioLegend®); P44-8, which binds and activates NKp44 (available from BioLegend®); SKI, which binds and activates CD8; and 3G8 which binds and activates CD16.

(203) In particular embodiments, the I-ABTS can bind to and block an NK cell inhibitory receptor to enhance NK cell activation. Examples of NK cell inhibitory receptors that can be bound and blocked include KIR2DL1, KIR2DL2/3, KIR3DL1, NKG2A, and KLRG1. In particular embodiments, a binding domain that binds and blocks the NK cell inhibitory receptors KIR2DL1 and KIR2DL2/3 includes a variable light chain region of the sequence:

(204) TABLE-US-00008 (SEQ ID NO: 88)

EIVLTQSPVTLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIY
DASNRATGIPARFSGSGGTDFTLTISLEPEDFAVYYCQQRSNWMYTF GQGTEKLEIKRT
and a variable heavy chain region of the sequence:

(205) TABLE-US-00009 (SEQ ID NO: 89)

QVQLVQSGAEVKKPGSSVKVSCKASGGTFSFYAISWVRQAPGQGLEWMG
GFIPIFGAANYAQKFQGRVTITADESTSTAYMELSSLRSDDTAVYYCAR
IPSGSYYYDYDMDVWGQGTTVTVSS.

Additional NK cell activating antibodies are described in WO/2005/0003172 and U.S. Pat. No. 9,415,104.

(206) Macrophages (and their precursors, monocytes) reside in every tissue of the body (in certain instances as microglia, Kupffer cells and osteoclasts) where they engulf apoptotic cells, pathogens and other non-self-components.

(207) The I-ABTS can be designed to bind to a protein expressed on the surface of macrophages. Examples of activating proteins expressed on the surface of macrophages (and their precursors, monocytes) include CD11b, CD11c, CD64, CD68, CD119, CD163, CD206, CD209, F4/80, IFGR2, Toll-like receptors (TLRs) 1-9, IL-4Rα, and MARCO. Commercially available antibodies that bind to proteins expressed on the surface of macrophages include M1/70, which binds and activates CD11b (available from BioLegend); KP1, which binds and activates CD68 (available from ABCAM, Cambridge, United Kingdom); and ab87099, which binds and activates CD163 (available from ABCAM).

(208) Anti-CD33 tri-specific antibodies are artificial proteins that simultaneously bind to three

different types of antigens, wherein at least one of the antigens is CD33. Tri-specific antibodies are described in, for example, WO2016/105450, WO 2010/028796; WO 2009/007124; WO 2002/083738; US 2002/0051780; and WO 2000/018806.

(209) When CD33-targeting agents are based on antibodies, binding domains, or similar proteins derived therefrom, modifications that provide different administration benefits can be useful. Exemplary administration benefits can include (1) reduced susceptibility to proteolysis, (2) reduced susceptibility to oxidation, (3) altered binding affinity for forming protein complexes, (4) altered binding affinities, (5) reduced immunogenicity; and/or (6) extended half-life. While the disclosure below describes these modifications in terms of their application to antibodies, when applicable to another particular anti-CD33 binding domain format (e.g., an scFv, bispecific antibodies), the modifications can also be applied to these other formats.

(210) In particular embodiments the antibodies can be mutated to increase the half-life of the antibodies in serum. M428L/N434S is a pair of mutations that increase the half-life of antibodies in serum, as described in Zalevsky et al., *Nature Biotechnology* 28, 157-159, 2010.

(211) In particular embodiments the antibodies can be mutated to increase their affinity for Fc receptors. Exemplary mutations that increase the affinity for Fc receptors include: G236A/S239D/A330L/1332E (GASDALIE). Smith et al., *Proceedings of the National Academy of Sciences of the United States of America*, 109(16), 6181-6186, 2012. In particular embodiments, an antibody variant includes an Fc region with one or more amino acid substitutions which improve ADCC, e.g., substitutions at positions 298, 333, and/or 334 of the Fc region (EU numbering of residues). In particular embodiments, alterations are made in the Fc region that result in altered C1q binding and/or Complement Dependent Cytotoxicity (CDC), e.g., as described in U.S. Pat. No. 6,194,551, WO 99/51642, and Idusogie et al., *J. Immunol.* 164: 4178-4184, 2000.

(212) Antibody variants are provided having a carbohydrate structure that lacks fucose attached (directly or indirectly) to an Fc region. For example, the amount of fucose in such antibody may be from 1% to 80%, from 1% to 65%, from 5% to 65% or from 20% to 40%. The amount of fucose is determined by calculating the average amount of fucose within the sugar chain at Asn297, relative to the sum of all glycostructures attached to Asn 297 (e.g. complex, hybrid and high mannose structures) as measured by MALDI-TOF mass spectrometry, as described in WO 2008/077546, for example. Asn297 refers to the asparagine residue located at position 297 in the Fc region (Eu numbering of Fc region residues); however, Asn297 may also be located ± 3 amino acids upstream or downstream of position 297, i.e., between positions 294 and 300, due to minor sequence variations in antibodies. Such fucosylation variants may have improved ADCC function. See, e.g., WO2000/61739; WO 2001/29246; WO2002/031140; US2002/0164328; WO2003/085119; WO2003/084570; US2003/0115614; US2003/0157108; US2004/0093621; US2004/0110704; US2004/0132140; US2004/0110282; US2004/0109865; WO2005/035586; WO2005/035778; WO2005/053742; Okazaki et al. *J. Mol. Biol.* 336:1239-1249 (2004); and Yamane-Ohnuki et al. *Biotech. Bioeng.* 87: 614 (2004). Examples of cell lines capable of producing defucosylated antibodies include Lec13 CHO cells deficient in protein fucosylation (Ripka et al. *Arch. Biochem. Biophys.* 249:533-545, 1986, and knockout cell lines, such as alpha-1,6-fucosyltransferase gene, FUT8, knockout CHO cells (see, e.g., Yamane-Ohnuki et al., *Biotech. Bioeng.* 87: 614, 2004; Kanda et al., *Biotechnol. Bioeng.*, 94(4):680-688, 2006; and WO2003/085107).

(213) In particular embodiments, modified antibodies include those wherein one or more amino acids have been replaced with a non-amino acid component, or where the amino acid has been conjugated to a functional group or a functional group has been otherwise associated with an amino acid. The modified amino acid may be, e.g., a glycosylated amino acid, a PEGylated amino acid, a farnesylated amino acid, an acetylated amino acid, a biotinylated amino acid, an amino acid conjugated to a lipid moiety, or an amino acid conjugated to an organic derivatizing agent. Amino acid(s) can be modified, for example, co-translationally or post-translationally during recombinant production (e.g., N-linked glycosylation at N—X—S/T motifs during expression in mammalian

cells) or modified by synthetic means. The modified amino acid can be within the sequence or at the terminal end of a sequence. Modifications also include nitrated constructs.

(214) PEGylation particularly is a process by which polyethylene glycol (PEG) polymer chains are covalently conjugated to other molecules such as proteins. Several methods of PEGylating proteins have been reported in the literature. For example, N-hydroxy succinimide (NHS)-PEG was used to PEGylate the free amine groups of lysine residues and N-terminus of proteins; PEGs bearing aldehyde groups have been used to PEGylate the amino-termini of proteins in the presence of a reducing reagent; PEGs with maleimide functional groups have been used for selectively PEGylating the free thiol groups of cysteine residues in proteins; and site-specific PEGylation of acetyl-phenylalanine residues can be performed.

(215) Covalent attachment of proteins to PEG has proven to be a useful method to increase the half-lives of proteins in the body (Abuchowski, A. et al., *Cancer Biochem. Biophys.*, 1984, 7:175-186; Hershfield, M. S. et al., *N. Engl. J. Medicine*, 1987, 316:589-596; and Meyers, F. J. et al., *Clin. Pharmacol. Ther.*, 49:307-313, 1991). The attachment of PEG to proteins not only protects the molecules against enzymatic degradation, but also reduces their clearance rate from the body. The size of PEG attached to a protein has significant impact on the half-life of the protein. The ability of PEGylation to decrease clearance is generally not a function of how many PEG groups are attached to the protein, but the overall molecular weight of the altered protein. Usually the larger the PEG is, the longer the in vivo half-life of the attached protein. In addition, PEGylation can also decrease protein aggregation (Suzuki et al., *Biochem. Biophys. Acta* 788:248, 1984), alter protein immunogenicity (Abuchowski et al., *J. Biol. Chem.* 252: 3582, 1977), and increase protein solubility as described, for example, in PCT Publication No. WO 92/16221).

(216) Several sizes of PEGs are commercially available (Nektar Advanced PEGylation Catalog 2005-2006; and NOF DDS Catalogue Ver 7.1), which are suitable for producing proteins with targeted circulating half-lives. A variety of active PEGs have been used including mPEG succinimidyl succinate, mPEG succinimidyl carbonate, and PEG aldehydes, such as mPEG-propionaldehyde.

(V-C-III) ANTI-CD33 CARS OR TCRS

(217) CD33-targeting agents also include immune cells expressing CAR or TCR that specifically bind CD33. Methods to genetically modify cells to express an exogenous gene are described above in section (IV).

(218) CAR refer to proteins including several distinct subcomponents. The subcomponents include at least an extracellular component, a transmembrane domain, and an intracellular component. Within the current disclosure, the extracellular component includes a binding domain that binds CD33. When the binding domain binds CD33, the intracellular component signals the immune cell to destroy the bound cell. Binding domains that specifically bind CD33 are described above.

(219) The intracellular or otherwise the cytoplasmic signaling components of CAR are responsible for activation of the cell in which the CAR is expressed. The term “intracellular signaling components” or “intracellular components” is thus meant to include any portion of the intracellular domain sufficient to transduce an activation signal. Intracellular components of expressed CAR can include effector domains. An effector domain is an intracellular portion of a fusion protein or receptor that can directly or indirectly promote a biological or physiological response in a cell when receiving the appropriate signal. In certain embodiments, an effector domain is part of a protein or protein complex that receives a signal when bound, or it binds directly to a target molecule, which triggers a signal from the effector domain. An effector domain may directly promote a cellular response when it contains one or more signaling domains or motifs, such as an immunoreceptor tyrosine-based activation motif (ITAM). In other embodiments, an effector domain will indirectly promote a cellular response by associating with one or more other proteins that directly promote a cellular response, such as co-stimulatory domains.

(220) Effector domains can provide for activation of at least one function of a modified cell upon

binding to the cellular marker expressed by a CD33-expressing cell. Activation of the modified cell can include one or more of differentiation, proliferation and/or activation or other effector functions. In particular embodiments, an effector domain can include an intracellular signaling component including a T cell receptor and a co-stimulatory domain which can include the cytoplasmic sequence from a co-receptor or co-stimulatory molecule.

(221) An effector domain can include one, two, three or more receptor signaling domains, intracellular signaling components (e.g., cytoplasmic signaling sequences), co-stimulatory domains, or combinations thereof. Exemplary effector domains include signaling and stimulatory domains selected from: 4-1BB (CD137), CARD11, CD3 γ , CD3 δ , CD3 ϵ , CD3 ζ , CD27, CD28, CD79A, CD79B, DAP10, FcR α , FcR β (Fc ϵ R1b), FcR γ , Fyn, HVEM (LIGHTR), ICOS, LAG3, LAT, Lck, LRP, NKG2D, NOTCH1, pT α , PTCH2, OX40, ROR2, Ryk, SLAMF1, Slp76, TCR α , TCR β , TRIM, Wnt, Zap70, or any combination thereof. In particular embodiments, exemplary effector domains include signaling and co-stimulatory domains selected from: CD86, Fc γ RIIa, DAP12, CD30, CD40, PD-1, lymphocyte function-associated antigen-1 (LFA-1), CD2, CD7, LIGHT, NKG2C, B7-H3, a ligand that specifically binds with CD83, CDS, ICAM-1, GITR, BAFFR, SLAMF7, NKp80 (KLRF1), CD127, CD160, CD19, CD4, CD8 α , CD8 β , IL2R β , IL2R γ , IL7R α , ITGA4, VLA1, CD49a, IA4, CD49D, ITGA6, VLA-6, CD49f, ITGAD, CD11d, ITGAE, CD103, ITGAL, CD11a, ITGAM, CD11b, ITGAX, CD11c, ITGB1, CD29, ITGB2, CD18, ITGB7, TNFR2, TRANCE/RANKL, DNAM1 (CD226), SLAMF4 (CD244, 2B4), CD84, CD96 (Tactile), CEACAM1, CRTAM, Ly9 (CD229), PSGL1, CD100 (SEMA4D), CD69, SLAMF6 (NTB-A, Ly108), SLAM (CD150, IPO-3), BLAME (SLAMF8), SELPLG (CD162), LTBR, GADS, PAG/Cbp, NKp44, NKp30, or NKp46.

(222) Intracellular signaling component sequences that act in a stimulatory manner may include iTAMs. Examples of iTAMs including primary cytoplasmic signaling sequences include those derived from CD3 γ , CD3 δ , CD3 ϵ , CD3 ζ , CD5, CD22, CD66d, CD79a, CD79b, and common FcR γ (FCER1G), Fc γ RIIa, FcR β (Fc ϵ Rib), DAP10, and DAP12. In particular embodiments, variants of CD3 ζ retain at least one, two, three, or all ITAM regions.

(223) Additional examples of intracellular signaling components include the cytoplasmic sequences of the CD3 ζ chain, and/or co-receptors that act in concert to initiate signal transduction following binding domain engagement.

(224) A co-stimulatory domain is a domain whose activation can be required for an efficient lymphocyte response to cellular marker binding. Some molecules are interchangeable as intracellular signaling components or co-stimulatory domains. Examples of costimulatory domains include CD27, CD28, 4-1BB (CD 137), OX40, CD30, CD40, PD-1, ICOS, lymphocyte function-associated antigen-1 (LFA-1), CD2, CD7, LIGHT, NKG2C, B7-H3, and a ligand that specifically binds with CD83. Further examples of such co-stimulatory domain molecules include CDS, ICAM-1, GITR, BAFFR, HVEM (LIGHTR), SLAMF7, NKp80 (KLRF1), NKp44, NKp30, NKp46, CD160, CD19, CD4, CD8 α , CD8 β , IL2R β , IL2R γ , IL7R α , ITGA4, VLA1, CD49a, ITGA4, IA4, CD49D, ITGA6, VLA-6, CD49f, ITGAD, CD11d, ITGAE, CD103, ITGAL, CD11a, ITGAM, CD11b, ITGAX, CD11c, ITGB1, CD29, ITGB2, CD18, ITGB7, TNFR2, TRANCE/RANKL, DNAM1 (CD226), SLAMF4 (CD244, 2B4), CD84, CD96 (Tactile), NKG2D, CEACAM1, CRTAM, Ly9 (CD229), PSGL1, CD100 (SEMA4D), CD69, SLAMF6 (NTB-A, Ly108), SLAM (SLAMF1, CD150, IPO-3), BLAME (SLAMF8), SELPLG (CD162), LTBR, LAT, GADS, SLP-76, PAG/Cbp, and CD19a.

(225) In particular embodiments, the intracellular signaling component includes (i) all or a portion of the signaling domain of CD3, (ii) all or a portion of the signaling domain of 4-1BB, or (iii) all or a portion of the signaling domain of CD3 and 4-1BB.

(226) Intracellular components may also include one or more of a protein of a Wnt signaling pathway (e.g., LRP, Ryk, or ROR2), NOTCH signaling pathway (e.g., NOTCH1, NOTCH2, NOTCH3, or NOTCH4), Hedgehog signaling pathway (e.g., PTCH or SMO), receptor tyrosine

kinases (RTKs) (e.g., epidermal growth factor (EGF) receptor family, fibroblast growth factor (FGF) receptor family, hepatocyte growth factor (HGF) receptor family, insulin receptor (IR) family, platelet-derived growth factor (PDGF) receptor family, vascular endothelial growth factor (VEGF) receptor family, tropomyosin receptor kinase (Trk) receptor family, ephrin (Eph) receptor family, AXL receptor family, leukocyte tyrosine kinase (LTK) receptor family, tyrosine kinase with immunoglobulin-like and EGF-like domains 1 (TIE) receptor family, receptor tyrosine kinase-like orphan (ROR) receptor family, discoidin domain (DDR) receptor family, rearranged during transfection (RET) receptor family, tyrosine-protein kinase-like (PTK7) receptor family, related to receptor tyrosine kinase (RYK) receptor family, or muscle specific kinase (MuSK) receptor family); G-protein-coupled receptors, GPCRs (Frizzled or Smoothened); serine/threonine kinase receptors (BMPR or TGFR); or cytokine receptors (IL1R, IL2R, IL7R, or IL15R).

(227) As indicated, transmembrane domains within a CAR molecule, often serving to connect the extracellular component and intracellular component through the cell membrane. The transmembrane domain can anchor the expressed molecule in the modified cell's membrane.

(228) The transmembrane domain can be derived either from a natural and/or a synthetic source. When the source is natural, the transmembrane domain can be derived from any membrane-bound or transmembrane protein. Transmembrane domains can include at least the transmembrane region(s) of the α , β or ζ chain of a T-cell receptor, CD28, CD27, CD3 epsilon, CD45, CD4, CD5, CD8, CD9, CD16, CD22; CD33, CD37, CD64, CD80, CD86, CD134, CD137 and CD154. In particular embodiments, a transmembrane domain may include at least the transmembrane region(s) of, e.g., KIRDS2, OX40, CD2, CD27, LFA-1 (CD 11a, CD18), ICOS (CD278), 4-1BB (CD137), GITR, CD40, BAFFR, HVEM (LIGHTR), SLAMF7, NKp80 (KLRF1), NKp44, NKp30, NKp46, CD160, CD19, IL2R β , IL2R γ , IL7R α , ITGA1, VLA1, CD49a, ITGA4, IA4, CD49D, ITGA6, VLA-6, CD49f, ITGAD, CDI Id, ITGAE, CD103, ITGAL, CDI Ia, ITGAM, CDI Ib, ITGAX, CDI Ic, ITGB1, CD29, ITGB2, CD18, ITGB7, TNFR2, DNAM1(CD226), SLAMF4 (CD244, 2B4), CD84, CD96 (Tactile), CEACAM1, CRT AM, Ly9(CD229), PSGL1, CD100 (SEMA4D), SLAMF6 (NTB-A, Lyl08), SLAM (SLAMF1, CD150, IPO-3), BLAME (SLAMF8), SELPLG (CD162), LTBR, PAG/Cbp, NKG2D, or NKG2C. In particular embodiments, a variety of human hinges can be employed as well including the human Ig (immunoglobulin) hinge (e.g., an IgG4 hinge, an IgD hinge), a GS linker (e.g., a Gly-Ser linker described herein), a KIR2DS2 hinge or a CD8a hinge.

(229) In particular embodiments, a transmembrane domain has a three-dimensional structure that is thermodynamically stable in a cell membrane, and generally ranges in length from 15 to 30 amino acids. The structure of a transmembrane domain can include an α helix, a β barrel, a β sheet, a β helix, or any combination thereof.

(230) A transmembrane domain can include one or more additional amino acids adjacent to the transmembrane region, e.g., one or more amino acid within the extracellular region of the CAR (e.g., up to 15 amino acids of the extracellular region) and/or one or more additional amino acids within the intracellular region of the CAR (e.g., up to 15 amino acids of the intracellular components). In one aspect, the transmembrane domain is from the same protein that the signaling domain, co-stimulatory domain or the hinge domain is derived from. In another aspect, the transmembrane domain is not derived from the same protein that any other domain of the CAR is derived from. In some instances, the transmembrane domain can be selected or modified by amino acid substitution to avoid binding of such domains to the transmembrane domains of the same or different surface membrane proteins to minimize interactions with other unintended members of the receptor complex. In one aspect, the transmembrane domain is capable of homodimerization with another CAR on the cell surface of a CAR-expressing cell. In a different aspect, the amino acid sequence of the transmembrane domain may be modified or substituted so as to minimize interactions with the binding domains of the native binding partner present in the same CAR-expressing cell.

(231) CAR and TCR expressed by genetically modified immune cells often additionally include spacer regions. Spacer regions can position the binding domain away from the immune cell (e.g., T cell) surface to enable proper cell/cell contact, antigen binding and activation (Patel et al., *Gene Therapy* 6: 412-419 (1999)). As indicated, an extracellular spacer region of a fusion binding protein is generally located between a hydrophobic portion or transmembrane domain and the extracellular binding domain, and the spacer region length may be varied to maximize antigen recognition (e.g., tumor recognition) based on the selected target molecule, selected binding epitope, or antigen-binding domain size and affinity (see, e.g., Guest et al., *J. Immunother.* 28:203-11 (2005); PCT Publication No. WO 2014/031687).

(232) Junction amino acids can be a linker which can be used to connect the sequences of CAR domains when the distance provided by a spacer is not needed and/or wanted. Junction amino acids are short amino acid sequences that can be used to connect co-stimulatory intracellular signaling components. In particular embodiments, junction amino acids are 9 amino acids or less.

(233) In particular embodiments, CAR targeting CD33-expressing cells includes the sequence set forth in SEQ ID NOs: 144 and 145. Exemplary methods to produce CD33 CAR T-cells are described in WO2018/US34743.

(234) In particular embodiments, cells genetically modified to express a CAR or TCR can additionally express one or more tag cassettes, transduction markers, and/or suicide switches. In some embodiments, the transduction marker and/or suicide switch is within the same construct but is expressed as a separate molecule on the cell surface. Tag cassettes and transduction markers can be used to activate, promote proliferation of, detect, enrich for, isolate, track, deplete and/or eliminate cells genetically modified to express a CAR or TCR in vitro, in vivo and/or ex vivo. "Tag cassette" refers to a unique synthetic peptide sequence affixed to, fused to, or that is part of a CD33-targeting agent, to which a cognate binding molecule (e.g., ligand, antibody, or other binding partner) is capable of specifically binding where the binding property can be used to activate, promote proliferation of, detect, enrich for, isolate, track, deplete and/or eliminate the tagged protein and/or cells expressing the tagged protein. Transduction markers can serve the same purposes but are derived from naturally occurring molecules and are often expressed using a skipping element that separates the transduction marker from the rest of the CD33-targeting agent.

(235) Tag cassettes that bind cognate binding molecules include, for example, His tag (SEQ ID NO: 146), Flag tag (SEQ ID NO: 147), Xpress tag (SEQ ID NO: 148), Avi tag (SEQ ID NO: 149), Calmodulin tag (SEQ ID NO: 150), Polyglutamate tag, HA tag (SEQ ID NO: 151), Myc tag (SEQ ID NO: 152), Softag 1 (SEQ ID NO: 153), Softag 3 (SEQ ID NO: 154), and V5 tag (SEQ ID NO: 155).

(236) Conjugate binding molecules that specifically bind tag cassette sequences disclosed herein are commercially available. For example, His tag antibodies are commercially available from suppliers including Life Technologies, Pierce Antibodies, and GenScript. Flag tag antibodies are commercially available from suppliers including Pierce Antibodies, GenScript, and Sigma-Aldrich. Xpress tag antibodies are commercially available from suppliers including Pierce Antibodies, Life Technologies and GenScript. Avi tag antibodies are commercially available from suppliers including Pierce Antibodies, IsBio, and Genecopoeia. Calmodulin tag antibodies are commercially available from suppliers including Santa Cruz Biotechnology, Abcam, and Pierce Antibodies. HA tag antibodies are commercially available from suppliers including Pierce Antibodies, Cell Signaling Technology and Abcam. Myc tag antibodies are commercially available from suppliers including Santa Cruz Biotechnology, Abcam, and Cell Signaling Technology.

(237) Transduction markers may be selected from at least one of a truncated CD19 (tCD19; see Budde et al., *Blood* 122: 1660, 2013); a truncated human epidermal growth factor (tEGFR; see Wang et al., *Blood* 118: 1255, 2011); an extracellular domain of human CD34; and/or RQR8 which combines target epitopes from CD34 (see Fehse et al, *Mol. Therapy* 1(5 Pt 1): 448-456, 2000) and CD20 antigens (see Philip et al, *Blood* 124: 1277-1278).

(238) In particular embodiments, a polynucleotide encoding an iCaspase9 construct (iCasp9) may be inserted into a CD33-targeting agent nucleotide construct as a suicide switch.

(239) Control features may be present in multiple copies or can be expressed as distinct molecules with the use of a skipping element. For example, a CAR can have one, two, three, four or five tag cassettes and/or one, two, three, four, or five transduction markers could also be expressed. For example, embodiments can include a CD33-targeting agent having two Myc tag cassettes, or a His tag and an HA tag cassette, or a HA tag and a Softag 1 tag cassette, or a Myc tag and a SBP tag cassette. In particular embodiments, a transduction marker includes tEFGR. Exemplary transduction markers and cognate pairs are described in U.S. Ser. No. 13/463,247.

(240) One advantage of including at least one control feature in cells genetically modified to express a CAR or TCR is that, if necessary or beneficial, the cells can be depleted following administration to a subject using the cognate binding molecule to a tag cassette.

(241) In certain embodiments, CD33-targeting agents may be detected or tracked in vivo by using antibodies that bind with specificity to a control feature (e.g., anti-Tag antibodies), or by other cognate binding molecules that specifically bind the control feature, which binding partners for the control feature are conjugated to a fluorescent dye, radio-tracer, iron-oxide nanoparticle or other imaging agent known in the art for detection by X-ray, CT-scan, MRI-scan, PET-scan, ultrasound, flow-cytometry, near infrared imaging systems, or other imaging modalities (see, e.g., Yu, et al., *Theranostics* 2:3, 2012).

(242) Thus, CD33-targeting agents expressing at least one control feature can be more readily identified, isolated, sorted, tracked, and/or eliminated as compared to a CD33-targeting agent without a tag cassette.

(VI) CELL FORMULATIONS AND CD33-TARGETING AGENT COMPOSITIONS

(243) Therapeutic cell formulations and CD33-targeting agent compositions can be formulated for administration to subjects. In particular embodiments, cell-based formulations are administered to subjects as soon as reasonably possible following their initial formulation. In particular embodiments, formulations and/or compositions can be frozen (e.g., cryopreserved or lyophilized) prior to administration to a subject.

(244) For example, as is understood by one of ordinary skill in the art, the freezing of cells can be destructive (see Mazur, P., 1977, *Cryobiology* 14:251-272) but there are numerous procedures available to prevent such damage. For example, damage can be avoided by (a) use of a cryoprotective agent, (b) control of the freezing rate, and/or (c) storage at a temperature sufficiently low to minimize degradative reactions. Exemplary cryoprotective agents include dimethyl sulfoxide (DMSO) (Lovelock and Bishop, 1959, *Nature* 183:1394-1395; Ashwood-Smith, 1961, *Nature* 190:1204-1205), glycerol, polyvinylpyrrolidone (Rinfret, 1960, *Ann. N.Y. Acad. Sci.* 85:576), polyethylene glycol (Sloviter and Ravdin, 1962, *Nature* 196:548), albumin, dextran, sucrose, ethylene glycol, i-erythritol, D-ribitol, D-mannitol (Rowe et al., 1962, *Fed. Proc.* 21:157), D-sorbitol, i-inositol, D-lactose, choline chloride (Bender et al., 1960, *J. Appl. Physiol.* 15:520), amino acids (Phan The Tran and Bender, 1960, *Exp. Cell Res.* 20:651), methanol, acetamide, glycerol monoacetate (Lovelock, 1954, *Biochem. J.* 56:265), and inorganic salts (Phan The Tran and Bender, 1960, *Proc. Soc. Exp. Biol. Med.* 104:388; Phan The Tran and Bender, 1961, in *Radiobiology, Proceedings of the Third Australian Conference on Radiobiology*, Ilbery ed., Butterworth, London, p. 59). In particular embodiments, DMSO can be used. Addition of plasma (e.g., to a concentration of 20-25%) can augment the protective effects of DMSO. After addition of DMSO, cells can be kept at 0° C. until freezing, because DMSO concentrations of 1% can be toxic at temperatures above 4° C.

(245) In the cryopreservation of cells, slow controlled cooling rates can be critical and different cryoprotective agents (Rapatz et al., 1968, *Cryobiology* 5(1): 18-25) and different cell types have different optimal cooling rates (see e.g., Rowe and Rinfret, 1962, *Blood* 20:636; Rowe, 1966, *Cryobiology* 3(1):12-18; Lewis, et al., 1967, *Transfusion* 7(1):17-32; and Mazur, 1970, *Science*

168:939-949 for effects of cooling velocity on survival of stem cells and on their transplantation potential). The heat of fusion phase where water turns to ice should be minimal. The cooling procedure can be carried out by use of, e.g., a programmable freezing device or a methanol bath procedure. Programmable freezing apparatuses allow determination of optimal cooling rates and facilitate standard reproducible cooling.

(246) In particular embodiments, DMSO-treated cells can be pre-cooled on ice and transferred to a tray containing chilled methanol which is placed, in turn, in a mechanical refrigerator (e.g., Harris® (Thermo Fisher Scientific Inc., Waltham, MA) or Revco® (Thermo Fisher Scientific Inc., Waltham, MA)) at -80°C . Thermocouple measurements of the methanol bath and the samples indicate a cooling rate of 1° to $3^{\circ}\text{C}/\text{minute}$ can be preferred. After at least two hours, the specimens can have reached a temperature of -80°C . and can be placed directly into liquid nitrogen (-196°C).

(247) After thorough freezing, the cells can be rapidly transferred to a long-term cryogenic storage vessel. In a preferred embodiment, samples can be cryogenically stored in liquid nitrogen (-196°C) or vapor (-1°C). Such storage is facilitated by the availability of highly efficient liquid nitrogen refrigerators.

(248) Further considerations and procedures for the manipulation, cryopreservation, and long-term storage of cells, can be found in the following exemplary references: U.S. Pat. Nos. 4,199,022; 3,753,357; and 4,559,298; Gorin, 1986, Clinics In Haematology 15(1):19-48; Bone-Marrow Conservation, Culture and Transplantation, Proceedings of a Panel, Moscow, Jul. 22-26, 1968, International Atomic Energy Agency, Vienna, pp. 107-186; Livesey and Linner, 1987, Nature 327:255; Linner et al., 1986, J. Histochem. Cytochem. 34(9):1123-1135; Simione, 1992, J. Parenter. Sci. Technol. 46(6):226-32).

(249) Following cryopreservation, frozen cells can be thawed for use in accordance with methods known to those of ordinary skill in the art. Frozen cells are preferably thawed quickly and chilled immediately upon thawing. In particular embodiments, the vial containing the frozen cells can be immersed up to its neck in a warm water bath; gentle rotation will ensure mixing of the cell suspension as it thaws and increase heat transfer from the warm water to the internal ice mass. As soon as the ice has completely melted, the vial can be immediately placed on ice.

(250) In particular embodiments, methods can be used to prevent cellular clumping during thawing. Exemplary methods include: the addition before and/or after freezing of DNase (Spitzer et al., 1980, Cancer 45:3075-3085), low molecular weight dextran and citrate, hydroxyethyl starch (Stiff et al., 1983, Cryobiology 20:17-24), etc.

(251) As is understood by one of ordinary skill in the art, if a cryoprotective agent that is toxic to humans is used, it should be removed prior to therapeutic use. DMSO has no serious toxicity.

(252) Exemplary carriers and modes of administration of cells are described at pages 14-15 of U.S. Patent Publication No. 2010/0183564. Additional pharmaceutical carriers are described in Remington: The Science and Practice of Pharmacy, 21st Edition, David B. Troy, ed., Lippicott Williams & Wilkins (2005).

(253) In particular embodiments, cells can be harvested from a culture medium, and washed and concentrated into a carrier in a therapeutically-effective amount. Exemplary carriers include saline, buffered saline, physiological saline, water, Hanks' solution, Ringer's solution, Nonnosol-R (Abbott Labs, Chicago, IL), Plasma-Lyte A® (Baxter Laboratories, Inc., Morton Grove, IL), glycerol, ethanol, and combinations thereof.

(254) In particular embodiments, carriers can be supplemented with human serum albumin (HSA) or other human serum components or fetal bovine serum. In particular embodiments, a carrier for infusion includes buffered saline with 5% HAS or dextrose. Additional isotonic agents include polyhydric sugar alcohols including trihydric or higher sugar alcohols, such as glycerin, erythritol, arabitol, xylitol, sorbitol, or mannitol.

(255) Carriers can include buffering agents, such as citrate buffers, succinate buffers, tartrate

buffers, fumarate buffers, gluconate buffers, oxalate buffers, lactate buffers, acetate buffers, phosphate buffers, histidine buffers, and/or trimethylamine salts.

(256) Stabilizers refer to a broad category of excipients which can range in function from a bulking agent to an additive which helps to prevent cell adherence to container walls. Typical stabilizers can include polyhydric sugar alcohols; amino acids, such as arginine, lysine, glycine, glutamine, asparagine, histidine, alanine, ornithine, L-leucine, 2-phenylalanine, glutamic acid, and threonine; organic sugars or sugar alcohols, such as lactose, trehalose, stachyose, mannitol, sorbitol, xylitol, ribitol, myoinositol, galactitol, glycerol, and cyclitols, such as inositol; PEG; amino acid polymers; sulfur-containing reducing agents, such as urea, glutathione, thiocetic acid, sodium thioglycolate, thioglycerol, alpha-monothioglycerol, and sodium thiosulfate; low molecular weight polypeptides (i.e., <10 residues); proteins such as HSA, bovine serum albumin, gelatin or immunoglobulins; hydrophilic polymers such as polyvinylpyrrolidone; monosaccharides such as xylose, mannose, fructose and glucose; disaccharides such as lactose, maltose and sucrose; trisaccharides such as raffinose, and polysaccharides such as dextran.

(257) Where necessary or beneficial, compositions or formulations can include a local anesthetic such as lidocaine to ease pain at a site of injection.

(258) Exemplary preservatives include phenol, benzyl alcohol, meta-cresol, methyl paraben, propyl paraben, octadecyldimethylbenzyl ammonium chloride, benzalkonium halides, hexamethonium chloride, alkyl parabens such as methyl or propyl paraben, catechol, resorcinol, cyclohexanol, and 3-pentanol.

(259) Therapeutically effective amounts of cells within cell-based formulations can be greater than 10×10^2 cells, greater than 10×10^3 cells, greater than 10×10^4 cells, greater than 10×10^5 cells, greater than 10×10^6 cells, greater than 10×10^7 cells, greater than 10×10^8 cells, greater than 10×10^9 cells, greater than 10×10^{10} cells, or greater than 10×10^{11} cells. In cell-based formulations disclosed herein, cells are generally in a volume of a liter or less, 500 ml or less, 250 ml or less, or 100 ml or less. Hence the density of administered cells is typically greater than 10×10^4 cells/ml, 10×10^7 cells/ml or 10×10^8 cells/ml.

(260) Therapeutically effective amounts of protein-based compounds within CD33 targeting compositions can include 0.1 to 5 pg or $\mu\text{g/mL}$ or L, or from 0.5 to 1 pg or $\mu\text{g/mL}$ or L. In other examples, a dose can include 1 pg or $\mu\text{g/mL}$ or L, 15 pg or $\mu\text{g/mL}$ or L, 30 pg or $\mu\text{g/mL}$ or L, 50 pg or $\mu\text{g/mL}$ or L, 55 pg or $\mu\text{g/mL}$ or L, 70 pg or $\mu\text{g/mL}$ or L, 90 pg or $\mu\text{g/mL}$ or L, 150 pg or $\mu\text{g/mL}$ or L, 350 pg or $\mu\text{g/mL}$ or L, 500 pg or $\mu\text{g/mL}$ or L, 750 pg or $\mu\text{g/mL}$ or L, 1000 pg or $\mu\text{g/mL}$ or L, 0.1 to 5 mg/mL or L or from 0.5 to 1 mg/mL or L. In other examples, a dose can include 1 mg/mL or L, 10 mg/mL or L, 30 mg/mL or L, 50 mg/mL or L, 70 mg/mL or L, 100 mg/mL or L, 300 mg/mL or L, 500 mg/mL or L, 700 mg/mL or L, 1000 mg/mL or L or more.

(261) Cell formulations and CD33 targeting compositions can be prepared for administration by, for example, injection, infusion, perfusion, or lavage. CD33-targeting agent compositions can also be prepared as oral, inhalable, or implantable compositions.

(VII) METHODS OF USE

(262) The formulations and compositions disclosed herein can be used for treating subjects (humans, veterinary animals (dogs, cats, reptiles, birds, etc.), livestock (horses, cattle, goats, pigs, chickens, etc.), and research animals (monkeys, rats, mice, fish, etc.). In particular embodiments, subjects are human patients. Treating subjects includes delivering therapeutically effective amounts. Therapeutically effective amounts include those that provide effective amounts, prophylactic treatments, and/or therapeutic treatments.

(263) An “effective amount” is the amount of a formulation necessary to result in a desired physiological change in a subject. Effective amounts are often administered for research purposes.

(264) A “prophylactic treatment” includes a treatment administered to a subject who does not display signs or symptoms of a condition to be treated or displays only early signs or symptoms of the condition to be treated such that treatment is administered for the purpose of diminishing,

preventing, or decreasing the risk of developing the condition. Thus, a prophylactic treatment functions as a preventative treatment against a condition.

(265) A “therapeutic treatment” includes a treatment administered to a subject who displays symptoms or signs of a condition and is administered to the subject for the purpose of reducing the severity or progression of the condition.

(266) The actual dose and amount of a therapeutic formulation and/or composition administered to a particular subject can be determined by a physician, veterinarian, or researcher taking into account parameters such as physical and physiological factors including target; body weight; type of condition; severity of condition; upcoming relevant events, when known; previous or concurrent therapeutic interventions; idiopathy of the subject; and route of administration, for example. In addition, in vitro and in vivo assays can optionally be employed to help identify optimal dosage ranges.

(267) Therapeutically effective amounts of cell-based compositions can include $10^{4.4}$ to $10^{9.9}$ cells/kg body weight, or $10^{3.3}$ to $10^{11.1}$ cells/kg body weight. Exemplary doses may include greater than $10^{2.2}$ cells, greater than $10^{3.3}$ cells, greater than $10^{4.4}$ cells, greater than $10^{5.5}$ cells, greater than $10^{6.6}$ cells, greater than $10^{7.7}$ cells, greater than $10^{8.8}$ cells, greater than $10^{9.9}$ cells, greater than $10^{10.0}$ cells, or greater than $10^{11.1}$ cells.

(268) Therapeutically effective amounts of protein-based compounds within CD33 targeting compositions can include 0.1 to 5 pg or $\mu\text{g/kg}$, or from 0.5 to 1 $\mu\text{g/kg}$. In other examples, a dose can include 1 pg or $\mu\text{g/kg}$, 15 pg or $\mu\text{g/kg}$, 30 pg or $\mu\text{g/kg}$, 50 pg or $\mu\text{g/kg}$, 55 pg or $\mu\text{g/kg}$, 70 pg or $\mu\text{g/kg}$, 90 pg or $\mu\text{g/kg}$, 150 pg or $\mu\text{g/kg}$, 350 pg or $\mu\text{g/kg}$, 500 pg or $\mu\text{g/kg}$, 750 pg or $\mu\text{g/kg}$, 1000 pg or $\mu\text{g/kg}$, 0.1 to 5 mg/kg or from 0.5 to 1 mg/kg. In other examples, a dose can include 1 mg/kg, 10 mg/kg, 30 mg/kg, 50 mg/kg, 70 mg/kg, 100 mg/kg, 300 mg/kg, 500 mg/kg, 700 mg/kg, 1000 mg/kg or more.

(269) Therapeutically effective amounts can be administered through any appropriate administration route such as by, injection, infusion, perfusion, and more particularly by administration by one or more of bone marrow, intravenous, intradermal, intraarterial, intranodal, intralymphatic, intraperitoneal injection, infusion, or perfusion). Administration of CD33-targeting agents can additionally be through oral administration, inhalation, or implantation.

(270) Therapeutically effective amounts can be achieved by administering single or multiple doses during the course of a treatment regimen, depending on, for example, the particular treatment protocol being implemented. In particular embodiments, the treatment protocol may be dictated by a clinical trial protocol or an FDA-approved treatment protocol.

(271) In particular embodiments, methods of the present disclosure can be used to treat acquired thrombocytopenia, acute lymphoblastic leukemia (ALL), acute myelogenous leukemia (AML), adrenoleukodystrophy, agnogenic myeloid metaplasia, AIDS, amegakaryocytosis/congenital thrombocytopenia, aplastic anemia, ataxia telangiectasia, β -thalassemia major, Chediak-Higashi syndrome, chronic granulomatous disease, chronic lymphocytic leukemia (CLL), chronic myelogenous leukemia (CML), chronic myelomonocytic leukemia, common variable immune deficiency (CVID), complement disorders, congenital agammaglobulinemia, Diamond Blackfan syndrome, diffuse large B-cell lymphoma, Fabry disease (alpha-galactosidase A), familial erythrophagocytic lymphohistiocytosis, Fanconi's anemia, fetal maternal incompatibility, follicular lymphoma, Gaucher disease (glucocerebrosidase), hemolytic anemia, Hodgkin's lymphoma, Hurler's syndrome, hyper IgM, IgG subclass deficiency, hypogammaglobulinemia, immune thrombocytopenia purpura, juvenile myelomonocytic leukemia, leukemia, lymphoma, May-Hegglin syndrome, metachromatic leukodystrophy, mucopolysaccharidoses, mucopolysaccharidosis type I (alpha-L-Iduronidase), multiple myeloma (MM), myelodysplastic syndrome (MDS or myelodysplasia), myelofibrosis, non-Hodgkin's lymphoma (NHL), paroxysmal nocturnal hemoglobinuria (PNH), Pompe disease, primary immunodeficiency diseases with

antibody deficiency, pure red cell aplasia, refractory anemia, SCID, selective IgA deficiency, severe aplastic anemia, Shwachmann-Diamond-Blackfan anemia, sickle cell disease, specific antibody deficiency, systemic lupus erythematosus (SLE), thrombocytopenia, Wiskott-Aldridge syndrome, and X-linked agammaglobulinemia (XLA).

(272) Additional exemplary cancers that may be treated include solid tumors, astrocytoma, atypical teratoid rhabdoid tumor, brain and central nervous system (CNS) cancer, breast cancer, carcinosarcoma, chondrosarcoma, chordoma, choroid plexus carcinoma, choroid plexus papilloma, clear cell sarcoma of soft tissue, gastrointestinal stromal tumor, glioblastoma, HBV-induced hepatocellular carcinoma, head and neck cancer, kidney cancer, lung cancer, malignant rhabdoid tumor, medulloblastoma, melanoma, meningioma, mesothelioma, neuroglial tumor, not otherwise specified (NOS) sarcoma, oligoastrocytoma, oligodendroglioma, osteosarcoma, ovarian cancer, ovarian clear cell adenocarcinoma, ovarian endometrioid adenocarcinoma, ovarian serous adenocarcinoma, pancreatic cancer, pancreatic ductal adenocarcinoma, pancreatic endocrine tumor, pineoblastoma, prostate cancer, renal cell carcinoma, renal medullo carcinoma, rhabdomyosarcoma, sarcoma, schwannoma, skin squamous cell carcinoma, and stem cell cancer.

(273) As indicated previously, FA is an inherited genetic disease characterized by fragile bone marrow cells and the inability to repair DNA damage, which accumulates in repopulating stem cells, resulting in eventual bone marrow failure. This disease can arise through mutations in any of a family of Fanconi-associated genes, with the most common of these mutations occurring in either the FANCA, FANCC, or FANCG genes. The current treatment protocol for patients is a bone marrow transplant from a matched donor, ideally from a sibling. However, the majority of patients will not have an appropriately matched sibling donor, and transplants from alternative donors are still associated with substantial toxicity and morbidity. For these patients, ongoing trials use an autologous transplant combined with new gene therapy approaches to introduce a corrected form of the mutated gene through collection and modification of the patient's own hematopoietic stem cells (e.g., NCT01331018).

(274) Another compounding problem for transplant recipients is the conditioning regimen used to prepare the marrow compartment for infused cells to engraft. Since chemotherapy or other DNA damaging agents are not well tolerated in these patients due to their underlying disease condition, in autologous transplants, gene modified cells are re-administered without prior conditioning. While safer, avoidance of conditioning potentially prevents efficient engraftment of corrected cells into the marrow niche where they can begin contributing to hematopoietic development. In the allogeneic transplant setting initial conditioning regimens included TBI and cyclophosphamide (Cy), however, significant mortality was observed secondary to graft-versus-host disease (GVHD) and Cy toxicity including hemorrhagic cystitis, mucositis, and cardiac failure. For this reason, reduced-intensity conditioning (RIC) is now used for FA patients which includes low-dose Cy, fludarabine, and anti-thymocyte globulin. Although overall survival has improved using RIC, late complications continue to be an issue whether associated with conditioning, GVHD, or from disease-related complications.

(275) For the reasons noted above, FA is an ideal candidate for autologous gene therapy, wherein the patient's own HSC can be supplied a functional FA gene, thereby diminishing GVHD risk. Importantly, the rationale for autologous genetic correction, even in a small number of cells, is supported by the spontaneous correction of the mutated FA gene documented in a few FA patients and resulting improvement in hematologic parameters. This “somatic mosaicism” occurs in single cell clones that can then sustain hematopoiesis over years without the requirement for marrow conditioning. A number of preclinical studies have demonstrated in vitro gene delivery by viral vectors, resulting in FA phenotype correction as demonstrated by protection from DNA crosslinking agents, such as mitomycin C (MMC). Integrating retroviral vectors encoding FANCA or FANCC cDNA were used to transduce FA murine hematopoietic progenitor cells, restore resistance of colony forming cells to MMC, and repopulate murine homozygous deficient models.

As a result, several clinical trials have been conducted. All of these trials have attempted collection of FA patient HSPC by selecting CD34^{sup}.+ cells for ex vivo gene transfer and subsequent reinfusion to limit off-target transduction for reasons of both safety and efficacy. One important remaining obstacle with autologous gene therapy is the presence of residual FA hematopoiesis that can result in myeloid malignancy, a scenario that could be minimized with the inclusion of the disclosed strategy to eliminate non-corrected or host FA cells. The current disclosure and treatment methods address this concern.

(276) In particular embodiments, therapeutic efficacy can be observed through mouse models of FA transplantation that have been used to study ex vivo gene therapy of HSPCs. One such model includes a functional knockout of the FANCA gene, resulting in fragile marrow of these mice that are thus unable to form healthy colonies when bone marrow is plated in outgrowth assays in the presence of even low levels of MMC, a DNA damaging agent. Healthy heterozygote littermates exhibit bone marrow colony forming potential regardless of MMC presence, whereas FANCA mice are demonstrated to have a significant decrease in colony forming potential with increasing MMC concentration. This mimics the clinical setting where patient stem cells exhibit a similar phenotype when exposed to DNA damaging agents. The use of low-dose Cy for bone marrow transplant in this mouse model has been demonstrated. Without some form of pre-conditioning prior to transplant, donor cells can home to the bone marrow; however, they do not contribute to peripheral hematopoiesis. This underscores the need to both clear FANCA-deficient stem cell populations and promote engraftment and hematopoietic development of transplanted donor or gene-corrected autologous cell populations.

(277) In particular embodiments, therapeutic efficacy for FA (and other immune deficiency disorders) can be observed through lymphocyte reconstitution, improved clonal diversity and thymopoiesis, reduced infections, and/or improved patient outcome. Therapeutic efficacy can also be observed through one or more of weight gain and growth, improved gastrointestinal function (e.g., reduced diarrhea), reduced upper respiratory symptoms, reduced fungal infections of the mouth (thrush), reduced incidences and severity of pneumonia, reduced meningitis and blood stream infections, and reduced ear infections. In particular embodiments, treating FA with methods of the present disclosure include increasing resistance of BM derived cells to mitomycin C (MMC). In particular embodiments, the resistance of BM derived cells to MMC can be measured by a cell survival assay in methylcellulose and MMC.

(278) In particular embodiments, methods of the present disclosure can be used to treat SCID-X1. In particular embodiments, methods of the present disclosure can be used to treat SCID (e.g., JAK 3 kinase deficiency SCID, purine nucleoside phosphorylase (PNP) deficiency SCID, adenosine deaminase (ADA) deficiency SCID, MHC class II deficiency or recombinaise activating gene (RAG) deficiency SCID). In particular embodiments, therapeutic efficacy can be observed through lymphocyte reconstitution, improved clonal diversity and thymopoiesis, reduced infections, and/or improved patient outcome. Therapeutic efficacy can also be observed through one or more of weight gain and growth, improved gastrointestinal function (e.g., reduced diarrhea), reduced upper respiratory symptoms, reduced fungal infections of the mouth (thrush), reduced incidences and severity of pneumonia, reduced meningitis and blood stream infections, and reduced ear infections. In particular embodiments, treating SCID-X1 with methods of the present disclosure include restoring functionality to the γ C-dependent signaling pathway. The functionality of the γ C-dependent signaling pathway can be assayed by measuring tyrosine phosphorylation of effector molecules STAT3 and/or STAT5 following in vitro stimulation with IL-21 and/or IL-2, respectively. Tyrosine phosphorylation of STAT3 and/or STAT5 can be measured by intracellular antibody staining.

(279) Particular embodiments include treatment of secondary, or acquired, immune deficiencies such as immune deficiencies caused by trauma, viruses, chemotherapy, toxins, and pollution. As previously indicated, acquired immunodeficiency syndrome (AIDS) is an example of a secondary

immune deficiency disorder caused by a virus, the human immunodeficiency virus (HIV), in which a depletion of T lymphocytes renders the body unable to fight infection. Thus, as another example, a gene can be selected to provide a therapeutically effective response against an infectious disease. In particular embodiments, the infectious disease is human immunodeficiency virus (HIV). The therapeutic gene may be, for example, a gene rendering immune cells resistant to HIV infection, or which enables immune cells to effectively neutralize the virus via immune reconstruction, polymorphisms of genes encoding proteins expressed by immune cells, genes advantageous for fighting infection that are not expressed in the patient, genes encoding an infectious agent, receptor or coreceptor; a gene encoding ligands for receptors or coreceptors; viral and cellular genes essential for viral replication including; a gene encoding ribozymes, antisense RNA, small interfering RNA (siRNA) or decoy RNA to block the actions of certain transcription factors; a gene encoding dominant negative viral proteins, intracellular antibodies, intrakines and suicide genes. Exemplary therapeutic genes and gene products include $\alpha 2\beta 1$; $\alpha \nu \beta 3$; $\alpha \nu \beta 5$; $\alpha \nu \beta 63$; BOB/GPR15; Bonzo/STRL-33/TYMSTR; CCR2; CCR3; CCR5; CCR8; CD4; CD46; CD55; CXCR4; aminopeptidase-N; HHV-7; ICAM; ICAM-1; PRR2/HveB; HveA; α -dystroglycan; LDLR/ $\alpha 2$ MR/LRP; PVR; PRR1/HveC; and laminin receptor. A therapeutically effective amount for the treatment of HIV, for example, may increase the immunity of a subject against HIV, ameliorate a symptom associated with AIDS or HIV, or induce an innate or adaptive immune response in a subject against HIV. An immune response against HIV may include antibody production and result in the prevention of AIDS and/or ameliorate a symptom of AIDS or HIV infection of the subject or decrease or eliminate HIV infectivity and/or virulence.

(280) In particular embodiments, methods of the present disclosure can be used to treat hypogammaglobulinemia. Hypogammaglobulinemia is caused by a lack of B-lymphocytes and is characterized by low levels of antibodies in the blood. Hypogammaglobulinemia can occur in patients with chronic lymphocytic leukemia (CLL), multiple myeloma (MM), non-Hodgkin's lymphoma (NHL) and other relevant malignancies as a result of both leukemia-related immune dysfunction and therapy-related immunosuppression. Patients with acquired hypogammaglobulinemia secondary to such hematological malignancies, and those patients receiving post-HSPC transplantation are susceptible to bacterial infections. The deficiency in humoral immunity is largely responsible for the increased risk of infection-related morbidity and mortality in these patients, especially by encapsulated microorganisms. For example, *Streptococcus pneumoniae*, *Haemophilus influenzae*, and *Staphylococcus aureus*, as well as *Legionella* and *Nocardia* spp. are frequent bacterial pathogens that cause pneumonia in patients with CLL. Opportunistic infections such as *Pneumocystis carinii*, fungi, viruses, and mycobacteria also have been observed. The number and severity of infections in these patients can be significantly reduced by administration of immune globulin (Griffiths H et al. (1989) Blood 73: 366-368; Chapel H M et al. (1994) Lancet 343: 1059-1063).

(281) In particular embodiments, a therapeutically effective treatment induces or increases expression of fetal hemoglobin (HbF), induces or increases production of hemoglobin and/or induces or increases production of β -globin. In particular embodiments, a therapeutically effective treatment improves blood cell function, and/or increases oxygenation of cells.

(282) In the context of cancers, therapeutically effective amounts have an anti-cancer effect. An anti-cancer effect can be quantified by observing a decrease in the number of cancer cells, a decrease in the number of metastases, a decrease in cancer volume, an increase in life expectancy, induction of apoptosis of cancer cells, induction of cancer cell death, inhibition of cancer cell proliferation, inhibition of tumor (e.g., solid tumor) growth, prevention of metastasis, prolongation of a subject's life, and/or reduction of relapse or re-occurrence of the cancer following treatment.

(283) In particular embodiments, methods of the present disclosure can restore BM function in a subject in need thereof. In particular embodiments, restoring BM function can include improving BM repopulation with gene corrected cells as compared to a subject in need thereof that is not

administered a therapy described herein. Improving BM repopulation with gene corrected cells can include increasing the percentage of cells that are gene corrected. In particular embodiments, the cells are selected from white blood cells and BM derived cells. In particular embodiments, the percentage of cells that are gene corrected can be measured using an assay selected from quantitative real time PCR and flow cytometry.

(284) In particular embodiments, methods of the present disclosure can restore T-cell mediated immune responses in a subject in need thereof. Restoration of T-cell mediated immune responses can include restoring thymic output and/or restoring normal T lymphocyte development.

(285) In particular embodiments, methods of the present disclosure can improve the kinetics and/or clonal diversity of lymphocyte reconstitution in a subject in need thereof. In particular embodiments, improving the kinetics of lymphocyte reconstitution can include increasing the number of circulating T lymphocytes to within a range of a reference level derived from a control population. In particular embodiments, improving the kinetics of lymphocyte reconstitution can include increasing the absolute CD3⁺ lymphocyte count to within a range of a reference level derived from a control population. A range of can be a range of values observed in or exhibited by normal (i.e., non-immuno-compromised) subjects for a given parameter. In particular embodiments, improving the kinetics of lymphocyte reconstitution can include reducing the time required to reach normal lymphocyte counts as compared to a subject in need thereof not administered a therapy described herein. In particular embodiments, improving the kinetics of lymphocyte reconstitution can include increasing the frequency of gene corrected lymphocytes as compared to a subject in need thereof not administered a therapy described herein. In particular embodiments, improving the kinetics of lymphocyte reconstitution can include increasing diversity of clonal repertoire of gene corrected lymphocytes in the subject as compared to a subject in need thereof not administered a gene therapy described herein. Increasing diversity of clonal repertoire of gene corrected lymphocytes can include increasing the number of unique retroviral integration site (RIS) clones as measured by a RIS analysis.

(286) In particular embodiments, restoring thymic output can include restoring the frequency of CD3⁺ T cells expressing CD45RA in peripheral blood to a level comparable to that of a reference level derived from a control population. In particular embodiments, restoring thymic output can include restoring the number of T cell receptor excision circles (TRECs) per 10^{sup.6} maturing T cells to a level comparable to that of a reference level derived from a control population. The number of TRECs per 10^{sup.6} maturing T cells can be determined as described in Kennedy D R et al. (2011) Vet Immunol Immunopathol 142: 36-48.

(287) In particular embodiments, restoring normal T lymphocyte development includes restoring the ratio of CD4⁺ cells: CD8⁺ cells to 2. In particular embodiments, restoring normal T lymphocyte development includes detecting the presence of $\alpha\beta$ TCR in circulating T-lymphocytes. The presence of $\alpha\beta$ TCR in circulating T-lymphocytes can be detected, for example, by flow cytometry using antibodies that bind an α and/or β chain of a TCR. In particular embodiments, restoring normal T lymphocyte development includes detecting the presence of a diverse TCR repertoire comparable to that of a reference level derived from a control population. TCR diversity can be assessed by TCRV β spectratyping, which analyzes genetic rearrangement of the variable region of the TCR β gene. Robust, normal spectratype profiles can be characterized by a Gaussian distribution of fragments sized across 17 families of TCRV β segments. In particular embodiments, restoring normal T lymphocyte development includes restoring T-cell specific signaling pathways. Restoration of T-cell specific signaling pathways can be assessed by lymphocyte proliferation following exposure to the T cell mitogen phytohemagglutinin (PHA). In particular embodiments, restoring normal T lymphocyte development includes restoring white blood cell count, neutrophil cell count, monocyte cell count, lymphocyte cell count, and/or platelet cell count to a level comparable to a reference level derived from a control population.

(288) In particular embodiments, methods of the present disclosure can normalize primary and

secondary antibody responses to immunization in a subject in need thereof. Normalizing primary and secondary antibody responses to immunization can include restoring B-cell and/or T-cell cytokine signaling programs functioning in class switching and memory response to an antigen. Normalizing primary and secondary antibody responses to immunization can be measured by a bacteriophage immunization assay. In particular embodiments, restoration of B-cell and/or T-cell cytokine signaling programs can be assayed after immunization with the T-cell dependent neoantigen bacteriophage ψ X174. In particular embodiments, normalizing primary and secondary antibody responses to immunization can include increasing the level of IgA, IgM, and/or IgG in a subject in need thereof to a level comparable to a reference level derived from a control population. In particular embodiments, normalizing primary and secondary antibody responses to immunization can include increasing the level of IgA, IgM, and/or IgG in a subject in need thereof to a level greater than that of a subject in need thereof not administered a gene therapy described herein. The level of IgA, IgM, and/or IgG can be measured by, for example, an immunoglobulin test. In particular embodiments, the immunoglobulin test includes antibodies binding IgG, IgA, IgM, kappa light chain, lambda light chain, and/or heavy chain. In particular embodiments, the immunoglobulin test includes serum protein electrophoresis, immunoelectrophoresis, radial immunodiffusion, nephelometry and turbidimetry. Commercially available immunoglobulin test kits include MININEPH™ (Binding site, Birmingham, UK), and immunoglobulin test systems from Dako (Glostrup, Denmark) and Dade Behring (Marburg, Germany). In particular embodiments, a sample that can be used to measure immunoglobulin levels includes a blood sample, a plasma sample, a cerebrospinal fluid sample, and a urine sample.

(289) In particular embodiments, therapeutically effective amounts may provide function to immune and other blood cells, reduce or eliminate an immune-mediated condition; and/or reduce or eliminate a symptom of the immune-mediated condition.

(290) In particular embodiments, particular methods of use include the treatment of conditions wherein corrected cells have a selective advantage over non-corrected cells. For example, in FA and SCID, corrected cells have an advantage and only transducing the therapeutic gene into a “few” HSPCs is sufficient for therapeutic efficacy.

(VIII) REFERENCE LEVELS DERIVED FROM CONTROL POPULATIONS

(291) Obtained values for parameters associated with a therapy described herein can be compared to a reference level derived from a control population, and this comparison can indicate whether a therapy described herein is effective for a subject in need thereof. Reference levels can be obtained from one or more relevant datasets from a control population. A “dataset” as used herein is a set of numerical values resulting from evaluation of a sample (or population of samples) under a desired condition. The values of the dataset can be obtained, for example, by experimentally obtaining measures from a sample and constructing a dataset from these measurements. As is understood by one of ordinary skill in the art, the reference level can be based on e.g., any mathematical or statistical formula useful and known in the art for arriving at a meaningful aggregate reference level from a collection of individual data points; e.g., mean, median, median of the mean, etc.

Alternatively, a reference level or dataset to create a reference level can be obtained from a service provider such as a laboratory, or from a database or a server on which the dataset has been stored.

(292) A reference level from a dataset can be derived from previous measures derived from a control population. A “control population” is any grouping of subjects or samples of like specified characteristics. The grouping could be according to, for example, clinical parameters, clinical assessments, therapeutic regimens, disease status, severity of condition, etc. In particular embodiments, the grouping is based on age range (e.g., 0-2 years) and non-immunocompromised status. In particular embodiments, a normal control population includes individuals that are age-matched to a test subject and non-immune compromised. In particular embodiments, age-matched includes, e.g., 0-6 months old; 0-2 years old; 0-10 years old; 10-15 years old, 60-65 years old, 70-85 years old, etc., as is clinically relevant under the circumstances. In particular embodiments, a

control population can include those that have an immune deficiency and have not been administered a therapeutically effective amount

(293) In particular embodiments, the relevant reference level for values of a particular parameter associated with a therapy described herein is obtained based on the value of a particular corresponding parameter associated with a therapy in a control population to determine whether a therapy disclosed herein has been therapeutically effective for a subject in need thereof.

(294) In particular embodiments, conclusions are drawn based on whether a sample value is statistically significantly different or not statistically significantly different from a reference level. A measure is not statistically significantly different if the difference is within a level that would be expected to occur based on chance alone. In contrast, a statistically significant difference or increase is one that is greater than what would be expected to occur by chance alone. Statistical significance or lack thereof can be determined by any of various methods well-known in the art. An example of a commonly used measure of statistical significance is the p-value. The p-value represents the probability of obtaining a given result equivalent to a particular data point, where the data point is the result of random chance alone. A result is often considered significant (not random chance) at a p-value less than or equal to 0.05. In particular embodiments, a sample value is “comparable to” a reference level derived from a normal control population if the sample value and the reference level are not statistically significantly different.

(IX) EXEMPLARY EMBODIMENTS

(295) 1. A genetic construct encoding a CD33 blocking molecule that reduces expression of CD33. 2. A genetic construct of embodiment 1 wherein the CD33 blocking molecule results in RNA-interference. 3. A genetic construct of embodiment 1 or 2 wherein the CD33 blocking molecule includes shRNA or siRNA. 4. A genetic construct of any of embodiments 1-3 wherein the CD33 blocking molecule includes shRNA encoded by SEQ ID NO: 8 or SEQ ID NO: 9. 5. A genetic construct of any of embodiments 1-3 wherein the CD33 blocking molecule includes siRNA encoded by SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, or SEQ ID NO: 15. 6. A genetic construct of any of embodiments 1-5, wherein the CD33 blocking molecule includes a wobble base pair. 7. A genetic construct of any of embodiments 1-6 wherein the genetic construct is within a viral vector. 8. A genetic construct of any of embodiments 1-7, wherein the viral vector is a lentiviral vector, a foamy viral vector, or an adenoviral vector that optionally includes a PGK promoter. 9. A genetic construct of any of embodiments 1-8, further including a therapeutic gene. 10. A genetic construct of embodiment 9, wherein the therapeutic gene includes FancA, FancB, FancC, FancD1, FancD2, FancE, FancF, FancG, FancI, FancJ, FancL, FancM, FancN, FancO, FancP, FancQ, FancR, FancS, FancT, FancU, FancV, or FancW or encodes a checkpoint inhibitor, a gene editing molecule, a chimeric antigen receptor that specifically binds a cellular antigen (e.g. a cancer antigen or a viral antigen), and/or a T-cell receptor that specifically binds a cellular antigen (e.g. a cancer antigen or a viral antigen). 11. A genetic construct of embodiment 9, wherein the therapeutic gene includes γ C, JAK3, IL7RA, RAG1, RAG2, DCLRE1C, PRKDC, LIG4, NHEJ1, CD3D, CD3E, CD3Z, CD3G, PTPRC, ZAP70, LCK, AK2, ADA, PNP, WHN, CHD7, ORAI1, STIM1, CORO1A, CIITA, RFXANK, RFX5, RFXAP, RMRP, DKC1, TERT, TINF2, DCLRE1B, or SLC46A1. 12. A genetic construct of embodiment 9, wherein the therapeutic gene includes factor VIII (FVIII), FVII, von Willebrand factor (VWF), FI, FII, FV, FX, FXI, or FXIII. 13. A genetic construct of embodiment 9, wherein the therapeutic gene includes F8 or F9. 14. A genetic construct of embodiment 9, wherein the therapeutic gene includes γ -globin; soluble CD40; CTLA; Fas L; antibodies to CD4, CD5, CD7, CD52, etc.; antibodies to IL1, IL2, IL6; an antibody to TCR specifically present on autoreactive T cells; 1L4; IL10; IL12; IL13; IL1Ra, sIL1RI, sIL1RII; sTNFRI; sTNFRII; antibodies to TNF; P53, PTPN22, and DRB1*1501/DQB1*0602; globin family genes; WAS; phox; dystrophin; pyruvate kinase (PK); CLN3; ABCD1; arylsulfatase A (ARSA); SFTPB; SFTPC; NLX2.1; ABCA3; GATA1; ribosomal protein genes; TERC; CFTR; LRRK2; PARK2; PARK7; PINK1; SNCA; PSEN1; PSEN2; APP;

SOD1; TDP43; FUS; ubiquitin 2; or C9ORF72. 15. A genetic construct of embodiment 9, wherein the therapeutic gene includes ABLI, AKT1, APC, ARSB, BCL11A, BLC1, BLC6, BRCA1, BRCA2, BRIP1, C46, CAS9, C-CAM, CBFAI, CBL, CCR5, CD19, CDA, C-MYC, CRE, CSC4, CSFIR, CTS-I, CYB5R3, DCC, DHFR, DLL1, DMD, EGFR, ERBA, ERBB, EBRB2, ETSI, ETS2, ETV6, FCC, FGR, FOX, FUSI, FYN, GALNS, GLB1, GNS, GUSB, HBB, HBD, HBE1, HBG1, HBG2, HCR, HGSNAT, HOXB4, HRAS, HYAL1, ICAM-1, iCaspase, IDUA, IDS, JUN, KLF4, KRAS, LYN, MCC, MDM2, MGMT, MLL, MMACI, MYB, MEN-I, MEN-11, MYC, NAGLU, NANOG, NF-1, NF-2, NKX2.1, NOTCH, OCT4, p16, p21, p27, p57, p73, PALB2, RAD51C, ras, at least one of RPL3 through RPL40, RPLP0, RPLP1, RPLP2, at least one of RPS2 through RPS30, RPSA, SGSH, SLX4, SOX2, VHL, or WT-1. 16. A genetic construct of any of embodiments 1-15 including a nucleotide encoding an shRNA or siRNA CD33 blocking molecule (e.g., SEQ ID NOs: 8 or 9) cloned between SEQ ID NO: 18 and SEQ ID NO: 19. 17. A cell genetically modified by the genetic construct of any of embodiments 1-16. 18. A cell of embodiment 17 wherein the cell is a hematopoietic stem and progenitor cell (HSPC). 19. A cell of embodiment 17 wherein the cell is a CD34.sup.+CD45RA.sup.-CD90.sup.+ HSC. 20. A population of cells genetically modified by a genetic construct of any of embodiments 1-16. 21. A population of embodiment 20, wherein cells in the population are HSPC and/or CD34.sup.+CD45RA.sup.-CD90.sup.+ HSC. 22. A cell formulation including a cell or population of any of embodiments 17-21 and a pharmaceutically acceptable carrier. 23. A kit including a genetic construct, cell, population of cells or cell formulation according to any of embodiments 1-22 and a CD33-targeting agent. 24. A kit of embodiment 23 wherein the CD33-targeting agent includes an anti-CD33 antibody, an anti-CD33 immunotoxin, an anti-CD33 antibody-drug conjugate, an anti-CD33 antibody-radioisotope conjugate, an anti-CD33 bispecific antibody, an anti-CD33 bispecific immune cell engaging antibody, an anti-CD33 trispecific antibody, and/or an anti-CD33 chimeric antigen receptor (CAR) with one or more binding domains. 25. A kit of embodiment 23 or 24 wherein the CD33-targeting agent includes Hp67.6, lintuzumab, SGN-CD33A, and/or AMG 330. 26. A kit of any of embodiments 23-25 wherein the CD33-targeting agent includes a binding domain derived from Hp67.6, lintuzumab, SGN-CD33A, and/or AMG 330. 27. A kit of any of embodiments 23-26 wherein the CD33-targeting agent includes the CDRs of Hp67.6, lintuzumab, SGN-CD33A, and/or AMG 330 and/or a sequence combination of a variable light chain including SEQ ID NO: 39 and a variable heavy chain including SEQ ID NO: 40; a variable light chain including SEQ ID NO: 47 and a variable heavy chain including SEQ ID NO: 48; a variable light chain including a CDRL1 of SEQ ID NO: 41, a CDRL2 of SEQ ID NO: 42, and a CDRL3 of SEQ ID NO: 43 and a variable heavy chain including a CDRH1 of SEQ ID NO: 44, a CDRH2 of SEQ ID NO: 45, and a CDRH3 of SEQ ID NO: 46; a variable light chain including a CDRL1 of SEQ ID NO: 49, a CDRL2 of SEQ ID NO: 50, and a CDRL3 of SEQ ID NO: 51 and a variable heavy chain including a CDRH1 of SEQ ID NO: 52, a CDRH2 of SEQ ID NO: 53, and a CDRH3 of SEQ ID NO: 54; and/or a variable light chain including a CDRL1 of SEQ ID NO: 98, a CDRL2 of SEQ ID NO: 99, and a CDRL3 of SEQ ID NO: 100 and a variable heavy chain including a CDRH1 of SEQ ID NO: 101, a CDRH2 of SEQ ID NO: 102, and a CDRH3 of SEQ ID NO: 103. 28. A kit of any of embodiments 23-27 wherein the CD33-targeting agent includes an antibody-drug conjugate or an antibody-radioisotope conjugate wherein the drug or radioisotope are selected from taxol, taxane, cytochalasin B, gramicidin D, ethidium bromide, emetine, mitomycin, etoposide, teniposide, vincristine, vinblastine, colchicin, doxorubicin, daunorubicin, dihydroxy anthracenedione, mitoxantrone, mithramycin, maytansinoid, dolastatin, auristatin, calicheamicin, pyrrolobenzodiazepine, nemorubicin PNU-159682, anthracycline, vinca alkaloid, trichothecene, CC1065, camptothecin, elinafide, actinomycin D, 1-dehydrotestosterone, glucocorticoids, procaine, tetracaine, lidocaine, propranolol, puromycin, ricin, CC-1065, duocarmycin, diphtheria toxin, snake venom, cobra venom, mistletoe lectin, modeccin, pokeweed antiviral protein, saporin, Bryodin 1, bouganin, gelonin, *Pseudomonas* exotoxin, iodine-131,

indium-111, yttrium-90, lutetium-177, astatine-211, bismuth-212, and/or bismuth-213 and/or wherein the antibody-drug conjugate includes GO. 29. A kit of any of embodiments 23-28 wherein the CD33-targeting agent includes a linker. 30. A kit of any of embodiments 23-29 wherein the CD33-targeting agent includes a bispecific antibody including a combination of binding variable chains or a binding CDR combination of Hp67.6, lintuzumab, SGN-CD33A, and/or AMG 330 and/or a sequence combination of a variable light chain including SEQ ID NO: 39 and a variable heavy chain including SEQ ID NO: 40; a variable light chain including SEQ ID NO: 47 and a variable heavy chain including SEQ ID NO: 48; a variable light chain including a CDRL1 of SEQ ID NO: 41, a CDRL2 of SEQ ID NO: 42, and a CDRL3 of SEQ ID NO: 43 and a variable heavy chain including a CDRH1 of SEQ ID NO: 44, a CDRH2 of SEQ ID NO: 45, and a CDRH3 of SEQ ID NO: 46; a variable light chain including a CDRL1 of SEQ ID NO: 49, a CDRL2 of SEQ ID NO: 50, and a CDRL3 of SEQ ID NO: 51 and a variable heavy chain including a CDRH1 of SEQ ID NO: 52, a CDRH2 of SEQ ID NO: 53, and a CDRH3 of SEQ ID NO: 54; and/or a variable light chain including a CDRL1 of SEQ ID NO: 98, a CDRL2 of SEQ ID NO: 99, and a CDRL3 of SEQ ID NO: 100 and a variable heavy chain including a CDRH1 of SEQ ID NO: 101, a CDRH2 of SEQ ID NO: 102, and a CDRH3 of SEQ ID NO: 103. 31. A kit of any of embodiments 23-30 wherein the CD33-targeting agent includes a bispecific antibody including at least one binding domain that activates an immune cell. 32. A kit of embodiment 31, wherein the immune cell is a T-cell, natural killer (NK) cell, or a macrophage. 33. A kit of embodiment 31 or 32, wherein the binding domain that activates an immune cell binds CD3, CD28, CD8, NKG2D, CD8, CD16, KIR2DL4, KIR2DS1, KIR2DS2, KIR3DS1, NKG2C, NKG2E, NKG2D, NKp30, NKp44, NKp46, NKp80, DNAM-1, CD11b, CD11c, CD64, CD68, CD119, CD163, CD206, CD209, F4/80, IFGR2, Toll-like receptors 1-9, IL-4R α , or MARCO. 34. A kit of any of embodiments 31-33, wherein the binding domain that activates an immune cell includes a variable light chain including a CDRL1 of SEQ ID NO: 55, a CDRL2 of SEQ ID NO: 56, and a CDRL3 sequence of SEQ ID NO: 57 and a variable heavy chain including a CDRH1 of SEQ ID NO: 58, a CDRH2 of SEQ ID NO: 59, and a CDRH3 of SEQ ID NO: 60. 35. A kit of any of embodiments 31-34, wherein the binding domain that activates an immune cell includes SEQ ID NO: 61. 36. A kit of any of embodiments 31-35, wherein the binding domain that activates an immune cell includes a variable light chain including a CDRL1 of SEQ ID NO: 62, a CDRL2 of KVS, and a CDRL3 sequence of SEQ ID NO: 63 and a variable heavy chain including a CDRH1 of SEQ ID NO: 64, a CDRH2 of SEQ ID NO: 65, and a CDRH3 of SEQ ID NO: 66. 37. A kit of any of embodiments 31-36, wherein the binding domain that activates an immune cell includes a variable light chain including a CDRL1 of SEQ ID NO: 67, a CDRL2 of KVS, and a CDRL3 sequence of SEQ ID NO: 63 and a variable heavy chain including a CDRH1 of SEQ ID NO: 69, a CDRH2 of SEQ ID NO: 70, and a CDRH3 of SEQ ID NO: 71. 38. A kit of any of embodiments 31-37, wherein the binding domain that activates an immune cell includes a variable light chain including a CDRL1 of SEQ ID NO: 72, a CDRL2 of KVS, and a CDRL3 sequence of SEQ ID NO: 63 and a variable heavy chain including a CDRH1 of SEQ ID NO: 69, a CDRH2 of SEQ ID NO: 75, and a CDRH3 of SEQ ID NO: 76. 39. A kit of any of embodiments 31-38, wherein the binding domain that activates an immune cell includes a variable light chain including a CDRL1 of SEQ ID NO: 77, a CDRL2 of KVS, and a CDRL3 sequence of SEQ ID NO: 78 and a variable heavy chain including a CDRH1 of SEQ ID NO: 79, a CDRH2 of SEQ ID NO: 80, and a CDRH3 of SEQ ID NO: 81. 40. A kit of any of embodiments 31-39, wherein the binding domain that activates an immune cell includes a variable light chain including a CDRL1 of SEQ ID NO: 82, a CDRL2 of SEQ ID NO: 83, and a CDRL3 sequence of SEQ ID NO: 84 and a variable heavy chain including a CDRH1 of SEQ ID NO: 85, a CDRH2 of SEQ ID NO: 86, and a CDRH3 of SEQ ID NO: 87. 41. A kit of any of embodiments 31-40, wherein the binding domain that activates an immune cell includes a TCR. 42. A kit of any of embodiments 31-41, wherein the binding domain that activates an immune cell includes a variable light chain including SEQ ID NO: 88 and a variable heavy chain including SEQ ID NO: 89. 43. A

kit of any of embodiments 31-42, wherein the binding domains of the bispecific antibody are joined through a linker. 44. A kit of any of embodiments 23-43, wherein the CD33-targeting agent includes a chimeric antigen receptor (CAR) including one or more binding domains. 45. A kit of embodiment 44, wherein the CAR includes an anti-CD33 binding domain and a binding domain of any of embodiments 31-42. 46. A kit of embodiment 44 or 45, wherein the effector domain of the CAR is selected from 4-1BB, CD3 ϵ , CD3 δ , CD3 ζ , CD27, CD28, CD79A, CD79B, CARD11, DAP10, FcR α , FcR β , FcR γ , Fyn, HVEM, ICOS, Lck, LAG3, LAT, LRP, NOTCH1, Wnt, NKG2D, OX40, ROR2, Ryk, SLAMF1, Slp76, pT α , TCR α , TCR β , TRIM, Zap70, PTCH2, or any combination thereof. 47. A kit of any of embodiments 44-46, wherein the CAR includes a cytoplasmic signaling sequence derived from CD3 zeta, FcR gamma, CD3 gamma, CD3 delta, CD3 epsilon, CD5, CD22, CD79a, CD79b, or CD66d. 48. A kit of any of embodiments 44-47, wherein the CAR includes an intracellular signaling domain and a costimulatory signaling region. 49. A kit of embodiment 48, wherein the costimulatory signaling region includes the intracellular domain of CD27, CD28, 4-1BB, OX40, CD30, CD40, lymphocyte function-associated antigen-1, CD2, CD7, LIGHT, NKG2C, or B7-H3. 50. A kit of any of embodiments 44-49, wherein the CAR includes a spacer region. 51. A kit of any of embodiments 44-50, wherein the CAR includes a transmembrane domain. 52. A method of genetically-modifying a cell to provide a therapeutic gene and to have reduced CD33 expression including exposing the cell to an effective amount of a genetic construct of any of embodiments 1-16. 53. A method of protecting a cell from an anti-CD33 treatment including genetically-modifying the cell with a genetic construct of any of embodiments 1-16. 54. A method of embodiment 53, wherein the cell is a therapeutic cell. 55. A method of embodiment 53 or 54, wherein the protecting is in vivo. 56. A method for treating a subject in need thereof with a cell formulation of embodiment 22 including administering a therapeutically effective amount of the cell formulation to the subject thereby treating the subject. 57. A method of embodiment 56, wherein the treating provides a therapeutically effective treatment against a primary immune deficiency. 58. A method of embodiment 56, wherein the treating provides a therapeutically effective treatment against a secondary immune deficiency. 59. A method of embodiment 56, wherein the treating provides a therapeutically effective treatment for a disorder including: FA, SCID, Pompe disease, Gaucher disease, Fabry disease, Mucopolysaccharidosis type I, familial apolipoprotein E deficiency and atherosclerosis (ApoE), viral infections, and cancer. 60. A method of any of embodiments 56-59 further including administering to the subject a CD33-targeting agent. 61. A method of embodiment 60, wherein the CD33-targeting agent includes an anti-CD33 antibody, an anti-CD33 immunotoxin, an anti-CD33 antibody-drug conjugate, an anti-CD33 antibody-radioisotope conjugate, an anti-CD33 bispecific antibody, an anti-CD33 bispecific immune cell engaging antibody, an anti-CD33 trispecific antibody, and/or an anti-CD33 chimeric antigen receptor (CAR) described in any of the preceding exemplary embodiments.

(X) EXPERIMENTAL EXAMPLES

(296) A novel strategy to selectively protect therapeutic cells by reducing CD33 expression in the therapeutic cells and targeting non-therapeutic cells with anti-CD33 therapy is described. The selective protection results in the enrichment of the therapeutic cells while simultaneously targeting and reducing diseased, malignant, and/or non-therapeutic CD33 expressing cells within a subject. (297) Results. CD33 shRNA4 and shRNA5 strongly reduce CD33 surface expression in the ML1 cell line. To identify a CD33 shRNA that is capable of downregulating CD33 surface expression, five shRNA sequences with predicted interference activity against CD33 coding region (FIG. 3) were cloned into lentiviral transfer plasmid pLentiLox 3.7 (FIG. 16). ML1 cells were transduced with the resulting VSV-G pseudotyped lentiviral vectors at different multiplicity of infections (MOIs). Transduced cells were analyzed by flow cytometry 3 to 7 days post treatment to determine transduction efficiency as measured by GFP expression and CD33 knockdown as determined by surface antibody staining. As shown in FIG. 4, treatment with CD33 shRNA4 and shRNA5 drastically reduced CD33 expression in ML1 cells while CD33 shRNA1 and shRNA6 did not show

any activity. In summary, these results validated the activity of two CD33 shRNA molecules in the ML1 myeloid cell line.

(298) In vitro protection of CD33 shRNA-modified ML1 cells from GO cytotoxicity. To determine if cells in which surface expression of CD33 was efficiently downregulated by the shRNA could be protected from cytotoxicity of the CD33-directed drug GO, ML1 cells modified with the control lentiviral vector or with shRNA1, shRNA4 or shRNA5 were treated with GO in vitro and cytotoxicity was evaluated by staining with a live/dead dye. Effective protection was found for cells treated with shRNA4 (FIG. 8) as compared to shRNA5, which correlated with more effective CD33 downregulation (FIG. 7). In contrast, ML1 cells left untreated or treated with the control vector were effectively killed by GO.

(299) shRNA-mediated CD33 knockdown in human CD34⁺ cells. Having validated the activity of CD33 shRNA in the ML1 cell line, CD33 shRNA activity was assessed in human CD34⁺ HSPCs, which have important therapeutic relevance. Similar to the results from ML1 cells, both shRNA4 and shRNA5 successfully reduced CD33 surface expression in human fetal liver (FL) CD34⁺ cells in a dose dependent manner (FIGS. 5, 6, and 9B-9C). shRNA4 was selected for further studies for the modification of human adult CD34⁺ cells. While transduction efficiency was lower in adult CD34⁺ as compared to FLCD34⁺ (FIG. 9A bottom row), the levels of CD33 knockdown achieved in transduced (i.e. GFP⁺) cells were comparable between both cell types (FIG. 9C).

(300) In vitro selection of CD33 shRNA-modified CD34⁺ cells following GO treatment. Since shRNA-modified cells are protected from GO-cytotoxicity, it was hypothesized that this CD33-based strategy can be used to select for gene-modified cells in vitro or in vivo. shRNA-modified human CD34⁺ cells were thus treated with GO to determine if they have a selective advantage as compared to non-modified cells under these conditions. An increase in GFP⁺ cells was found in both shRNA-treated adult and FL CD34⁺ cells indicating that gene modified cells could be selected in vitro (FIG. 10, line with triangles). In contrast, no selection was seen for cells modified with the control vector (FIG. 10, line with squares), confirming that this effect is dependent on shRNA-mediated CD33 knockdown activity.

(301) Efficient engraftment of CD33 shRNA-modified human CD34⁺ cells in the mouse xenotransplantation model. To assess engraftment and the multilineage differentiation potential of shRNA-modified human stem cells, NSG neonate mice were transplanted with human FL CD34⁺ cells transduced with the control lentiviral vector or with the CD33 shRNA vector (n=6 for each group, FIGS. 11A, 11B). Engraftment and multilineage differentiation were tracked from peripheral blood of transplanted mice within a time span of 26 weeks and no difference in engrafted human CD45⁺ cells was observed between each experimental group (FIG. 12A). Importantly, the frequency of gene-modified, GFP⁺, human CD45⁺ cells was also comparable between the control and shRNA-treated group (FIG. 12B), indicating that expression of the CD33 shRNA in CD34⁺ stem cells did not impact engraftment in this model. CD14⁺ monocytes derived from engrafted CD34⁺ cells were produced at comparable frequencies averaging 5% to 15% of total human cells in both experimental groups (FIG. 13A). Similarly, the frequencies of T- and B-lymphocytes were comparable between both groups, confirming that expression of the shRNA did not affect CD34⁺ stem cell multilineage differentiation potential. As expected, CD33 expression within CD14⁺ monocytes was significantly reduced in transduced monocytes from the shRNA group as compared to the control group (FIG. 13B) for the entire duration of the experiment. In summary, these results demonstrate normal engraftment and differentiation of shRNA-modified human CD34⁺, with persistent knockdown of CD33 surface expression in the mouse transplantation model.

(302) In vivo selection of CD33 shRNA-modified cells after GO treatment. The mouse transplantation model offers a unique opportunity to determine if CD33-directed drugs can select for CD33 shRNA-modified cells in vivo. GO was administered in two mice per group at 17 weeks and 22 weeks post transplantation (FIGS. 11A, 11B), and the frequency of GFP⁺ human CD45⁺ cells was monitored weekly after each administration. GFP⁺ cells steadily increased in the shRNA

group from 55%-70% in one animal and from 30-45% in the other animal (FIG. 14B). In contrast, the frequency of GFP+ cells remained constant in the two animals from the control group (FIG. 14A). These results confirmed that cells expressing CD33 shRNA have a selective advantage after GO treatment in vivo. This selection strategy has the potential to increase the number of cells containing the gene-modification in vivo, which have important therapeutic benefits and can be used for the treatment of a variety of hematopoietic and immune disorders after stem cell gene therapy.

(303) Methods. Cloning of CD33 shRNA target sequence into pLentiLox 3.7 and generation of corresponding VSV-G lentiviral vector. Single stranded oligonucleotides corresponding to the shRNA target sequences were ordered through Integrated DNA Technologies (IDT, Coralville IA) with 5' phosphate modification and PAGE purified (FIG. 3). The format of the sense oligo is 5'T-(GN.sub.18)-(TTCAAGAGA)-(N.sub.18C)-TTTTTTC (SEQ ID NO: 93) and the antisense oligo is the sequence complement of the sense but with additional nucleotides at the 5' end to generate a XhoI restriction site overhang. The N18 represents an 18-nucleotide sequence corresponding to the shRNA target sequence. Sense and antisense oligonucleotides were annealed to form double stranded DNA molecules and cloned in the pLentiLox 3.7 plasmid immediately following the U6 promoter using XhoI/HpaI restriction sites.

(304) Second-generation LVs were produced by three plasmid-polyethylenimine transfection in HEK 293T-cells. HEK 293T-cells were cultured in Dulbecco's modified Eagle medium (DMEM) supplemented with 10% Hyclone Cosmic Calf serum (Thermo Fischer Scientific, Waltham, MA), 1% sodium pyruvate, nonessential aminoacids, L-Glutamine, and 1% penicillin/streptomycin. For LV production in 15-cm plates, cells were plated on 0.1% gelatin at a density of 1.8×10^7 cells/plate and transfected with 27 μ g transfer vector construct (pLentiLox 3.7 containing the different CD33 shRNA sequences), 6 μ g pMDLg-pRRE, 12 μ g pRSC-Rev, and 6 μ g pMD2.G for VSV-G pseudotyped LV. The next day, cells were washed with $1 \times$ Dulbecco's phosphate buffered saline (Thermo Fischer Scientific) and treated with 15 ml media containing 10 mmol/l sodium butyrate (Sigma-Aldrich, St. Louis, MO) for 8 hours. Cell supernatant was harvested and combined with two additional harvests carried out over a time span of 48 hours. The supernatant was filtered through a 0.8 μ m-pore-size filter, concentrated 100-fold by centrifugation for 15-20 hours at 4° C. at $5,000 \times g$, and stored at -80° C. The titer of the vector preparations was determined by adding different amounts of LV to the human fibrosarcoma cell line HT1080. HT1080 cells were grown in DMEM supplemented with 10% fetal bovine serum (FBS), 1% penicillin/streptomycin and were plated at 1×10^5 cells/ml in a 12-well plate one day before transduction. For $100 \times$ concentrated LV, volumes used in the transduction were 1, 0.3, 0.1, 0.03 μ l following serial dilutions of the vector. Protamine sulfate was added to the cells at a final concentration of 8 μ g/ml. Since the pLentiLox 3.7 transfer vector contains a GFP reporter, cells were analyzed by flow cytometry 3 days post transduction and the percentage of GFP-expressing cells was used to calculate the number of infectious units (IU) per ml of vector.

(305) Transduction of ML1 cells and human CD34+ cells with CD33 shRNA lentiviral vectors. All LV transductions were performed in presence of 8 μ g/ml protamine sulfate. ML1 cells were cultured in RPMI medium with 10% FBS, 1% penicillin/streptomycin. Adult human CD34.sup.+ cells were collected from volunteers under an institutional review board-approved protocol. Human fetal liver CD34+ cells were enriched by immunomagnetic separation from tissue obtained from Advance Bioscience Resources Inc. (ABR, Alameda, CA). CD34+ cells were cultured in StemSpan™ serum-free expansion medium II (SFEM II) (StemCell Technologies, Vancouver, Canada) supplemented with penicillin/streptomycin (Life Technologies, Carlsbad, CA), Stem cell factor (PeproTech, Rocky Hill, NJ), Thrombopoietin (PeproTech), and FLT3-L (Milenyi Biotec, Auburn, CA). For transduction, cells were plated on CH-296 fibronectin (Takara, New York, NY) at 2 μ g/ml, and exposed to the vector at various MOI as determined by HT1080 titer.

(306) In vitro cytotoxicity and selection assays. Drug-induced cytotoxicity was quantified as

described previously by Laszlo et al., Oncotarget, 7:43281-43294 (2016). Briefly, parental and shRNA-modified ML1 cells were incubated in 96-well round bottom plates with and without gemtuzumab ozogamicin (GO, Pfizer, New York, NY) for 3 days followed by flow cytometric quantification of cell numbers and cell viability, using 4',6-diamidino-2-phenylindole (DAPI) to detect non-viable cells. In the selection assay, human CD34⁺ cells were grown as described above and were treated with 10 ng/ul GO for 12 hours.

(307) Engraftment of NSG mice with shRNA-modified human CD34⁺ cells and in vivo selection following GO treatment. For in vivo assessment of engineered HSPCs, NOD.CgPrkdcscidII2rgtm1WjI/SzJ (NOD SCID gamma/, NSG) neonate mice were infused with 5.0×10^5 human CD34⁺ cells and peripheral blood and tissue samples were collected and processed as described by Haworth et al., Mol Ther. Methods Clin. Dev., 6: 17-30 (2017). Flow cytometry staining was performed with human CD45-PerCP (Clone 2D1), mouse CD45.1/CD45.2-V500 (Clone 30-F11), CD3-FITC or -APC (Clone UCHT1), CD4-V450 (Clone RPA-T4), CD20-PE (Clone 2H7), CD14-APC or -PE-Cy7 (Clone M5E2), CD34-APC (Clone 581) (all from BD Biosciences, San Jose, CA), and CD33-PE (Clone AC104.3E3, Miltenyi Biotec). In vivo selection of CD33 shRNA-modified cells with GO was assessed by injecting mice intravenously with GO at a dose of 0.1 mg/kg.

(XI) CLOSING PARAGRAPHS

(308) Variants of the sequences disclosed and referenced herein are also included. Guidance in determining which amino acid residues can be substituted, inserted, or deleted without abolishing biological activity can be found using computer programs well known in the art, such as DNASTAR™ (Madison, Wisconsin) software. Preferably, amino acid changes in the protein variants disclosed herein are conservative amino acid changes, i.e., substitutions of similarly charged or uncharged amino acids. A conservative amino acid change involves substitution of one of a family of amino acids which are related in their side chains.

(309) In a peptide or protein, suitable conservative substitutions of amino acids are known to those of skill in this art and generally can be made without altering a biological activity of a resulting molecule. Those of skill in this art recognize that, in general, single amino acid substitutions in non-essential regions of a polypeptide do not substantially alter biological activity (see, e.g., Watson et al. Molecular Biology of the Gene, 4th Edition, 1987, The Benjamin/Cummings Pub. Co., p. 224). Naturally occurring amino acids are generally divided into conservative substitution families as follows: Group 1: Alanine (Ala), Glycine (Gly), Serine (Ser), and Threonine (Thr); Group 2: (acidic): Aspartic acid (Asp), and Glutamic acid (Glu); Group 3: (acidic; also classified as polar, negatively charged residues and their amides): Asparagine (Asn), Glutamine (Gln), Asp, and Glu; Group 4: Gln and Asn; Group 5: (basic; also classified as polar, positively charged residues): Arginine (Arg), Lysine (Lys), and Histidine (His); Group 6 (large aliphatic, nonpolar residues): Isoleucine (Ile), Leucine (Leu), Methionine (Met), Valine (Val) and Cysteine (Cys); Group 7 (uncharged polar): Tyrosine (Tyr), Gly, Asn, Gln, Cys, Ser, and Thr; Group 8 (large aromatic residues): Phenylalanine (Phe), Tryptophan (Trp), and Tyr; Group 9 (non-polar): Proline (Pro), Ala, Val, Leu, Ile, Phe, Met, and Trp; Group 11 (aliphatic): Gly, Ala, Val, Leu, and Ile; Group 10 (small aliphatic, nonpolar or slightly polar residues): Ala, Ser, Thr, Pro, and Gly; and Group 12 (sulfur-containing): Met and Cys. Additional information can be found in Creighton (1984) Proteins, W.H. Freeman and Company.

(310) In making such changes, the hydropathic index of amino acids may be considered. The importance of the hydropathic amino acid index in conferring interactive biologic function on a protein is generally understood in the art (Kyte and Doolittle, 1982, J. Mol. Biol. 157(1), 105-32). Each amino acid has been assigned a hydropathic index on the basis of its hydrophobicity and charge characteristics (Kyte and Doolittle, 1982). These values are: Ile (+4.5); Val (+4.2); Leu (+3.8); Phe (+2.8); Cys (+2.5); Met (+1.9); Ala (+1.8); Gly (−0.4); Thr (−0.7); Ser (−0.8); Trp (−0.9); Tyr (−1.3); Pro (−1.6); His (−3.2); Glutamate (−3.5); Gln (−3.5); aspartate (−3.5); Asn

(-3.5); Lys (-3.9); and Arg (-4.5).

(311) It is known in the art that certain amino acids may be substituted by other amino acids having a similar hydropathic index or score and still result in a protein with similar biological activity, i.e., still obtain a biological functionally equivalent protein. In making such changes, the substitution of amino acids whose hydropathic indices are within ± 2 is preferred, those within ± 1 are particularly preferred, and those within ± 0.5 are even more particularly preferred. It is also understood in the art that the substitution of like amino acids can be made effectively on the basis of hydrophilicity.

(312) As detailed in U.S. Pat. No. 4,554,101, the following hydrophilicity values have been assigned to amino acid residues: Arg (+3.0); Lys (+3.0); aspartate (+3.0 $\pm$ 1); glutamate (+3.0 $\pm$ 1); Ser (+0.3); Asn (+0.2); Gln (+0.2); Gly (0); Thr (-0.4); Pro (-0.5 $\pm$ 1); Ala (-0.5); His (-0.5); Cys (-1.0); Met (-1.3); Val (-1.5); Leu (-1.8); Ile (-1.8); Tyr (-2.3); Phe (-2.5); Trp (-3.4). It is understood that an amino acid can be substituted for another having a similar hydrophilicity value and still obtain a biologically equivalent, and in particular, an immunologically equivalent protein. In such changes, the substitution of amino acids whose hydrophilicity values are within ± 2 is preferred, those within ± 1 are particularly preferred, and those within ± 0.5 are even more particularly preferred.

(313) As outlined above, amino acid substitutions may be based on the relative similarity of the amino acid side-chain substituents, for example, their hydrophobicity, hydrophilicity, charge, size, and the like.

(314) As indicated elsewhere, variants of gene sequences can include codon optimized variants, sequence polymorphisms, splice variants, and/or mutations that do not affect the function of an encoded product to a statistically-significant degree.

(315) Variants of the protein, nucleic acid, and gene sequences disclosed herein also include sequences with at least 70% sequence identity, 80% sequence identity, 85% sequence identity, 90% sequence identity, 95% sequence identity, 96% sequence identity, 97% sequence identity, 98% sequence identity, or 99% sequence identity to the protein, nucleic acid, or gene sequences disclosed herein.

(316) "% sequence identity" refers to a relationship between two or more sequences, as determined by comparing the sequences. In the art, "identity" also means the degree of sequence relatedness between protein, nucleic acid, or gene sequences as determined by the match between strings of such sequences. "Identity" (often referred to as "similarity") can be readily calculated by known methods, including those described in: Computational Molecular Biology (Lesk, A. M., ed.) Oxford University Press, NY (1988); Biocomputing: Informatics and Genome Projects (Smith, D. W., ed.) Academic Press, NY (1994); Computer Analysis of Sequence Data, Part I (Griffin, A. M., and Griffin, H. G., eds.) Humana Press, N J (1994); Sequence Analysis in Molecular Biology (Von Heijne, G., ed.) Academic Press (1987); and Sequence Analysis Primer (Gribskov, M. and Devereux, J., eds.) Oxford University Press, NY (1992). Preferred methods to determine identity are designed to give the best match between the sequences tested. Methods to determine identity and similarity are codified in publicly available computer programs. Sequence alignments and percent identity calculations may be performed using the Megalign program of the LASERGENE bioinformatics computing suite (DNASTAR, Inc., Madison, Wisconsin). Multiple alignment of the sequences can also be performed using the Clustal method of alignment (Higgins and Sharp CABIOS, 5, 151-153 (1989) with default parameters (GAP PENALTY=10, GAP LENGTH PENALTY=10). Relevant programs also include the GCG suite of programs (Wisconsin Package Version 9.0, Genetics Computer Group (GCG), Madison, Wisconsin); BLASTP, BLASTN, BLASTX (Altschul, et al., J. Mol. Biol. 215:403-410 (1990); DNASTAR (DNASTAR, Inc., Madison, Wisconsin); and the FASTA program incorporating the Smith-Waterman algorithm (Pearson, Comput. Methods Genome Res., [Proc. Int. Symp.] (1994), Meeting Date 1992, 111-20. Editor(s): Suhai, Sandor. Publisher: Plenum, New York, N.Y. Within the context of this disclosure it will be understood that where sequence analysis software is used for analysis, the results of the

analysis are based on the “default values” of the program referenced. As used herein “default values” will mean any set of values or parameters, which originally load with the software when first initialized.

(317) Variants also include nucleic acid molecules that hybridizes under stringent hybridization conditions to a sequence disclosed herein and provide the same function as the reference sequence. Exemplary stringent hybridization conditions include an overnight incubation at 42° C. in a solution including 50% formamide, 5×SSC (750 mM NaCl, 75 mM trisodium citrate), 50 mM sodium phosphate (pH 7.6), 5×Denhardt's solution, 10% dextran sulfate, and 20 µg/ml denatured, sheared salmon sperm DNA, followed by washing the filters in 0.1×SSC at 50° C. Changes in the stringency of hybridization and signal detection are primarily accomplished through the manipulation of formamide concentration (lower percentages of formamide result in lowered stringency); salt conditions, or temperature. For example, moderately high stringency conditions include an overnight incubation at 37° C. in a solution including 6×SSPE (20×SSPE=3M NaCl; 0.2M NaH₂PO₄; 0.02M EDTA, pH 7.4), 0.5% SDS, 30% formamide, 100 µg/ml salmon sperm blocking DNA; followed by washes at 50° C. with 1×SSPE, 0.1% SDS. In addition, to achieve even lower stringency, washes performed following stringent hybridization can be done at higher salt concentrations (e.g. 5×SSC). Variations in the above conditions may be accomplished through the inclusion and/or substitution of alternate blocking reagents used to suppress background in hybridization experiments. Typical blocking reagents include Denhardt's reagent, BLOTTO, heparin, denatured salmon sperm DNA, and commercially available proprietary formulations. The inclusion of specific blocking reagents may require modification of the hybridization conditions described above, due to problems with compatibility.

(318) In particular embodiments, the disclosure provides proteins that bind with a cognate binding molecule with an association rate constant or k_{on} rate of not more than 10⁷ M⁻¹ s⁻¹, less than 5×10⁶ M⁻¹ s⁻¹, less than 2.5×10⁶ M⁻¹ s⁻¹, less than 2×10⁶ M⁻¹ s⁻¹, less than 1.5×10⁶ M⁻¹ s⁻¹, less than 10⁶ M⁻¹ s⁻¹, less than 5×10⁵ M⁻¹ s⁻¹, less than 2.5×10⁵ M⁻¹ s⁻¹, less than 2×10⁵ M⁻¹ s⁻¹, less than 1.5×10⁵ M⁻¹ s⁻¹, less than 10⁵ M⁻¹ s⁻¹, less than 5×10⁴ M⁻¹ s⁻¹, less than 2.5×10⁴ M⁻¹ s⁻¹, less than 2×10⁴ M⁻¹ s⁻¹, less than 1.5×10⁴ M⁻¹ s⁻¹, less than 10⁴ M⁻¹ s⁻¹, less than 10³ M⁻¹ s⁻¹, less than 10² M⁻¹ s⁻¹, or in a range of 10² M⁻¹ s⁻¹ to 10⁷ M⁻¹ s⁻¹, in a range of 10³ M⁻¹ s⁻¹ to 10⁶ M⁻¹ s⁻¹, in a range of 10⁴ M⁻¹ s⁻¹ to 10⁵ M⁻¹ s⁻¹, or in a range of 10³ M⁻¹ s⁻¹ to 10⁷ M⁻¹ s⁻¹.

(319) In particular embodiments, the disclosure provides proteins that bind with a cognate binding molecule a k_{off} rate of not less than 0.5 s⁻¹, not less than 0.25 s⁻¹, not less than 0.2 s⁻¹, not less than 0.1 s⁻¹, not less than 5×10⁻² s⁻¹, not less than 2.5×10⁻² s⁻¹, not less than 2×10⁻² s⁻¹, not less than 1.5×10⁻² s⁻¹, not less than 10⁻² s⁻¹, not less than 5×10⁻³ s⁻¹, not less than 2.5×10⁻³ s⁻¹, not less than 2×10⁻³ s⁻¹, not less than 1.5×10⁻³ s⁻¹, not less than 10⁻³ s⁻¹, not less than 5×10⁻⁴ s⁻¹, not less than 2.5×10⁻⁴ s⁻¹, not less than 2×10⁻⁴ s⁻¹, not less than 1.5×10⁻⁴ s⁻¹, not less than 10⁻⁴ s⁻¹, not less than 5×10⁻⁵ s⁻¹, not less than 2.5×10⁻⁵ s⁻¹, not less than 2×10⁻⁵ s⁻¹, not less than 1.5×10⁻⁵ s⁻¹, not less than 10⁻⁵ s⁻¹, not less than 5×10⁻⁶ s⁻¹, not less than 2.5×10⁻⁶ s⁻¹, not less than 2×10⁻⁶ s⁻¹, not less than 1.5×10⁻⁶ s⁻¹, not less than 10⁻⁶ s⁻¹, or in a range of 0.5 to 10⁻⁶ s⁻¹, in a range of 10⁻² s⁻¹ to 10⁻⁵ s⁻¹, or in a range of 10⁻³ s⁻¹ to 10⁻⁴ s⁻¹.

(320) In particular embodiments, the disclosure provides proteins that bind with a cognate binding

molecule with an affinity constant or $K_{\text{sub.d}}$ ($k_{\text{sub.off}}/k_{\text{sub.on}}$) of, either before and/or after modification, less than 10^{+6} M, less than $5 \times 10^{+5}$ M, less than $2.5 \times 10^{+5}$ M, less than $2 \times 10^{+5}$ M, less than $1.5 \times 10^{+5}$ M, less than 10^{+5} M, less than $5 \times 10^{+4}$ M, less than $2.5 \times 10^{+4}$ M, less than $2 \times 10^{+4}$ M, less than $1.5 \times 10^{+4}$ M, less than 10^{+4} M, less than $5 \times 10^{+3}$ M, less than $2.5 \times 10^{+3}$ M, less than $2 \times 10^{+3}$ M, less than $1.5 \times 10^{+3}$ M, less than 10^{+3} M, less than 500 M, less than 250 M, less than 200 M, less than 150 M, less than 100 M, less than 50 M, less than 25 M, less than 20 M, less than 15 M, or less than 10 M, or in a range of 10^{+2} M to 10^{+6} M, in a range of 10^{+2} M to 10^{+5} M, or in a range of 10^{+3} M to $1 \times 10^{+4}$ M.

(321) In particular embodiments, the disclosure provides proteins that bind with a cognate binding molecule with a dissociation constant or $K_{\text{sub.d}}$ ($k_{\text{sub.off}}/k_{\text{sub.on}}$) of, either before and/or after modification, not less than 0.05 M, not less than 0.025 M, not less than 0.02 M, not less than 0.01 M, not less than 5×10^{-3} M, not less than 2.5×10^{-3} M, not less than 2×10^{-3} M, not less than 1.5×10^{-3} M, not less than 10^{-3} M, not less than 5×10^{-4} M, not less than 2.5×10^{-4} M, not less than 2×10^{-4} M, not less than 1.5×10^{-4} M, not less than 10^{-4} M, not less than 5×10^{-5} M, not less than 2.5×10^{-5} M, not less than 2×10^{-5} M, not less than 1.5×10^{-5} M, not less than 10^{-5} M, not less than 5×10^{-6} M, not less than 2.5×10^{-6} M, not less than 2×10^{-6} M, not less than 1.5×10^{-6} M, not less than 10^{-6} M, or not less than 10^{-7} M, or in a range of 0.05 M to 10^{-7} M, in a range of 5×10^{-3} M to 10^{-6} M, or in a range of 10^{-4} M to 10^{-7} M.

(322) When antibody residues are provided, the assignment of amino acids to each domain is in accordance with Kabat Sequences of Proteins of Immunological Interest (National Institutes of Health, Bethesda, Md. (1987 and 1991)) unless otherwise specified.

(323) Unless otherwise indicated, aspects of the present disclosure can employ conventional techniques of immunology, molecular biology, microbiology, cell biology and recombinant DNA. These methods are described in the following publications. See, e.g., Sambrook, et al. Molecular Cloning: A Laboratory Manual, 2nd Edition (1989); F. M. Ausubel, et al. eds., Current Protocols in Molecular Biology, (1987); the series Methods IN Enzymology (Academic Press, Inc.); M. MacPherson, et al., PCR: A Practical Approach, IRL Press at Oxford University Press (1991); MacPherson et al., eds. PCR 2: Practical Approach, (1995); Harlow and Lane, eds. Antibodies, A Laboratory Manual, (1988); and R. I. Freshney, ed. Animal Cell Culture (1987).

(324) As will be understood by one of ordinary skill in the art, each embodiment disclosed herein can comprise, consist essentially of or consist of its particular stated element, step, ingredient or component. Thus, the terms “include” or “including” should be interpreted to recite: “comprise, consist of, or consist essentially of.” The transition term “comprise” or “comprises” means includes, but is not limited to, and allows for the inclusion of unspecified elements, steps, ingredients, or components, even in major amounts. The transitional phrase “consisting of” excludes any element, step, ingredient or component not specified. The transition phrase “consisting essentially of” limits the scope of the embodiment to the specified elements, steps, ingredients or components and to those that do not materially affect the embodiment. A material effect would cause a statistically-significant reduction in resistance to a CD33 targeting therapy in cells genetically modified with a viral vector including a therapeutic gene and a CD33 blocking molecule as disclosed herein.

(325) Unless otherwise indicated, all numbers expressing quantities of ingredients, properties such as molecular weight, reaction conditions, and so forth used in the specification and claims are to be understood as being modified in all instances by the term “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the specification and attached claims are approximations that may vary depending upon the desired properties sought to be obtained by the

present invention. At the very least, and not as an attempt to limit the application of the doctrine of equivalents to the scope of the claims, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. When further clarity is required, the term “about” has the meaning reasonably ascribed to it by a person skilled in the art when used in conjunction with a stated numerical value or range, i.e. denoting somewhat more or somewhat less than the stated value or range, to within a range of $\pm 20\%$ of the stated value; $\pm 19\%$ of the stated value; $\pm 18\%$ of the stated value; $\pm 17\%$ of the stated value; $\pm 16\%$ of the stated value; $\pm 15\%$ of the stated value; $\pm 14\%$ of the stated value; $\pm 13\%$ of the stated value; $\pm 12\%$ of the stated value; $\pm 11\%$ of the stated value; $\pm 10\%$ of the stated value; $\pm 9\%$ of the stated value; $\pm 8\%$ of the stated value; $\pm 7\%$ of the stated value; $\pm 6\%$ of the stated value; $\pm 5\%$ of the stated value; $\pm 4\%$ of the stated value; $\pm 3\%$ of the stated value; $\pm 2\%$ of the stated value; or $\pm 1\%$ of the stated value.

(326) Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the invention are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the standard deviation found in their respective testing measurements.

(327) The terms “a,” “an,” “the” and similar referents used in the context of describing the invention (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. Recitation of ranges of values herein is merely intended to serve as a shorthand method of referring individually to each separate value falling within the range. Unless otherwise indicated herein, each individual value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein is intended merely to better illuminate the invention and does not pose a limitation on the scope of the invention otherwise claimed. No language in the specification should be construed as indicating any non-claimed element essential to the practice of the invention.

(328) Groupings of alternative elements or embodiments of the invention disclosed herein are not to be construed as limitations. Each group member may be referred to and claimed individually or in any combination with other members of the group or other elements found herein. It is anticipated that one or more members of a group may be included in, or deleted from, a group for reasons of convenience and/or patentability. When any such inclusion or deletion occurs, the specification is deemed to contain the group as modified thus fulfilling the written description of all Markush groups used in the appended claims.

(329) Certain embodiments of this invention are described herein, including the best mode known to the inventors for carrying out the invention. Of course, variations on these described embodiments will become apparent to those of ordinary skill in the art upon reading the foregoing description. The inventor expects skilled artisans to employ such variations as appropriate, and the inventors intend for the invention to be practiced otherwise than specifically described herein. Accordingly, this invention includes all modifications and equivalents of the subject matter recited in the claims appended hereto as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the invention unless otherwise indicated herein or otherwise clearly contradicted by context.

(330) Furthermore, numerous references have been made to patents, printed publications, journal articles and other written text throughout this specification (referenced materials herein). Each of the referenced materials are individually incorporated herein by reference in their entirety for their referenced teaching.

(331) In closing, it is to be understood that the embodiments of the invention disclosed herein are illustrative of the principles of the present invention. Other modifications that may be employed are

within the scope of the invention. Thus, by way of example, but not of limitation, alternative configurations of the present invention may be utilized in accordance with the teachings herein. Accordingly, the present invention is not limited to that precisely as shown and described. (332) The particulars shown herein are by way of example and for purposes of illustrative discussion of the preferred embodiments of the present invention only and are presented in the cause of providing what is believed to be the most useful and readily understood description of the principles and conceptual aspects of various embodiments of the invention. In this regard, no attempt is made to show structural details of the invention in more detail than is necessary for the fundamental understanding of the invention, the description taken with the drawings and/or examples making apparent to those skilled in the art how the several forms of the invention may be embodied in practice.

(333) Definitions and explanations used in the present disclosure are meant and intended to be controlling in any future construction unless clearly and unambiguously modified in the following examples or when application of the meaning renders any construction meaningless or essentially meaningless. In cases where the construction of the term would render it meaningless or essentially meaningless, the definition should be taken from Webster's Dictionary, 3rd Edition or a dictionary known to those of ordinary skill in the art, such as the Oxford Dictionary of Biochemistry and Molecular Biology (Ed. Anthony Smith, Oxford University Press, Oxford, 2004).

Claims

1. A genetic construct comprising a CD33 blocking molecule selected from SEQ ID NO: 8 or SEQ ID NO: 9.
2. The genetic construct of claim 1, wherein the genetic construct is within a viral vector.
3. The genetic construct of claim 2, wherein the viral vector is a lentiviral vector, a foamy viral vector, or an adenoviral vector that optionally comprises a PGK promoter.
4. The genetic construct of claim 1, further comprising a therapeutic gene.
5. The genetic construct of claim 4, wherein the therapeutic gene: (i) comprises FancA, FancB, FancC, FancD1, FancD2, FancE, FancF, FancG, FancI, FancJ, FancL, FancM, FancN, FancO, FancP, FancQ, FancR, FancS, FancT, FancU, FancV, or FancW; or (ii) encodes a checkpoint inhibitor, a gene editing molecule, a chimeric antigen receptor that specifically binds a cellular antigen or a T-cell receptor that specifically binds a cellular antigen; or (iii) comprises γ C, JAK3, IL7RA, RAG1, RAG2, DCLRE1C, PRKDC, LIG4, NHEJ1, CD3D, CD3E, CD3Z, CD3G, PTPRC, ZAP70, LCK, AK2, ADA, PNP, WHN, CHD7, ORAI1, STIM1, CORO1A, CIITA, RFXANK, RFX5, RFXAP, RMRP, DKC1, TERT, TINF2, DCLRE1B, or SLC46A1; or (iv) comprises factor VIII (FVIII), FVII, von Willebrand factor (VWF), FI, FII, FV, FX, FXI, or FXIII); or (v) comprises F8 or F9; or (vi) comprises γ -globin; soluble CD40; CTLA; Fas L; an antibody to CD4, CD5, CD7, CD52; an antibody to IL1, IL2, IL6; an antibody to TCR specifically present on autoreactive T cells; IL4; IL10; IL12; IL13; IL1Ra, sIL1RI, sIL1RII; sTNFRI; sTNFRII; an antibody to TNF; P53, PTPN22, and DRB1*1501/DQB1*0602; globin family genes; WAS; phox; dystrophin; pyruvate kinase (PK); CLN3; ABCD1; arylsulfatase A (ARSA); SFTPB; SFTPC; NLX2.1; ABCA3; GATA1; ribosomal protein genes; TERC; CFTR; LRRK2; PARK2; PARK7; PINK1; SNCA; PSEN1; PSEN2; APP; SOD1; TDP43; FUS; ubiquilin 2; or C9ORF72; or (vii) comprises ABLI, AKT1, APC, ARSB, BCL11A, BLC1, BLC6, BRCA1, BRCA2, BRIP1, C46, CAS9, C-CAM, CBAI, CBL, CCR5, CD19, CDA, C-MYC, CRE, CSCR4, CSFIR, CTS-I, CYB5R3, DCC, DHFR, DLL1, DMD, EGFR, ERBA, ERBB, EBRB2, ETSI, ETS2, ETV6, FCC, FGR, FOX, FUSI, FYN, GALNS, GLB1, GNS, GUSB, HBB, HBD, HBE1, HBG1, HBG2, HCR, HGSNAT, HOXB4, HRAS, HYAL1, ICAM-1, iCaspase, IDUA, IDS, JUN, KLF4, KRAS, LYN, MCC, MDM2, MGMT, MLL, MMACI, MYB, MEN-I, MEN-II, MYC, NAGLU, NANOG, NF-1, NF-2, NKX2.1, NOTCH, OCT4, p16, p21, p27, p57, p73, PALB2, RAD51C, ras, at least one of

RPL3 through RPL40, RPLPO, RPLP1, RPLP2, at least one of RPS2 through RPS30, RPSA, SGSH, SLX4, SOX2, VHL, or WT-I; or (viii) any two or more of (i)-(vii).

6. The genetic construct of claim 1, cloned between SEQ ID NO: 18 and SEQ ID NO: 19.

7. The genetic construct of claim 1, wherein the CD33 blocking molecule comprises a wobble base pair.

8. A kit comprising a genetic construct of claim 1 and a CD33-targeting agent.

9. The kit of claim 8, wherein the CD33-targeting agent comprises: (i) an anti-CD33 antibody, an anti-CD33 immunotoxin, an anti-CD33 antibody-drug conjugate, an anti-CD33 antibody-radioisotope conjugate, an anti-CD33 bispecific antibody, an anti-CD33 bispecific immune cell engaging antibody, an anti-CD33 trispecific antibody, and/or an anti-CD33 chimeric antigen receptor (CAR) with one or more binding domains; or (ii) Hp67.6, lintuzumab, SGN-CD33A, and/or AMG 330; or (iii) a binding domain derived from Hp67.6, lintuzumab, SGN-CD33A, and/or AMG 330; or (iv) the CDRs of Hp67.6, lintuzumab, SGN-CD33A, and/or AMG 330 and/or a sequence combination of a variable light chain comprising SEQ ID NO: 39 and a variable heavy chain comprising SEQ ID NO: 40; a variable light chain comprising SEQ ID NO: 47 and a variable heavy chain comprising SEQ ID NO: 48; a variable light chain comprising a CDRL1 of SEQ ID NO: 41, a CDRL2 of SEQ ID NO: 42, and a CDRL3 of SEQ ID NO: 43 and a variable heavy chain comprising a CDRH1 of SEQ ID NO: 44, a CDRH2 of SEQ ID NO: 45, and a CDRH3 of SEQ ID NO: 46; a variable light chain comprising a CDRL1 of SEQ ID NO: 49, a CDRL2 of SEQ ID NO: 50, and a CDRL3 of SEQ ID NO: 51 and a variable heavy chain comprising a CDRH1 of SEQ ID NO: 52, a CDRH2 of SEQ ID NO: 53, and a CDRH3 of SEQ ID NO: 54; and/or a variable light chain comprising a CDRL1 of SEQ ID NO: 98, a CDRL2 of SEQ ID NO: 99, and a CDRL3 of SEQ ID NO: 100 and a variable heavy chain comprising a CDRH1 of SEQ ID NO: 101, a CDRH2 of SEQ ID NO: 102, and a CDRH3 of SEQ ID NO: 103; or (v) an antibody-drug conjugate or an antibody-radioisotope conjugate wherein the drug or radioisotope are selected from taxol, taxane, cytochalasin B, gramicidin D, ethidium bromide, emetine, mitomycin, etoposide, tenoposide, vincristine, vinblastine, colchicin, doxorubicin, daunorubicin, dihydroxy anthracinedione, mitoxantrone, mithramycin, maytansinoid, dolastatin, auristatin, calicheamicin, pyrrolobenzodiazepine, nemorubicin PNU-159682, anthracycline, vinca alkaloid, trichothecene, CC1065, camptothecin, elinafide, actinomycin D, 1-dehydrotestosterone, glucocorticoids, procaine, tetracaine, lidocaine, propranolol, puromycin, ricin, CC-1065, duocarmycin, diphtheria toxin, snake venom, cobra venom, mistletoe lectin, modeccin, pokeweed antiviral protein, saporin, Bryodin 1, bouganin, gelonin, *Pseudomonas* exotoxin, iodine-131, indium-111, yttrium-90, lutetium-177, astatine-211, bismuth-212, and/or bismuth-213, and/or wherein the antibody-drug conjugate comprises gemtuzumab ozogamicin (GO); or (vi) a linker; or (vii) a bispecific antibody comprising a combination of binding variable chains or a binding CDR combination of Hp67.6, lintuzumab, SGN-CD33A, and/or AMG 330 and/or a sequence combination of a variable light chain comprising SEQ ID NO: 39 and a variable heavy chain comprising SEQ ID NO: 40; a variable light chain comprising SEQ ID NO: 47 and a variable heavy chain comprising SEQ ID NO: 48; a variable light chain comprising a CDRL1 of SEQ ID NO: 41, a CDRL2 of SEQ ID NO: 42, and a CDRL3 of SEQ ID NO: 43 and a variable heavy chain comprising a CDRH1 of SEQ ID NO: 44, a CDRH2 of SEQ ID NO: 45, and a CDRH3 of SEQ ID NO: 46; a variable light chain comprising a CDRL1 of SEQ ID NO: 49, a CDRL2 of SEQ ID NO: 50, and a CDRL3 of SEQ ID NO: 51 and a variable heavy chain comprising a CDRH1 of SEQ ID NO: 52, a CDRH2 of SEQ ID NO: 53, and a CDRH3 of SEQ ID NO: 54; and/or a variable light chain comprising a CDRL1 of SEQ ID NO: 98, a CDRL2 of SEQ ID NO: 99, and a CDRL3 of SEQ ID NO: 100 and a variable heavy chain comprising a CDRH1 of SEQ ID NO: 101, a CDRH2 of SEQ ID NO: 102, and a CDRH3 of SEQ ID NO: 103; or (viii) any two or more of (i)-(vii).

10. The kit of claim 8, wherein the CD33-targeting agent comprises a bispecific antibody comprising at least one binding domain that activates an immune cell.

11. The kit of claim 10, wherein the binding domain that activates an immune cell: (i) binds CD3, CD28, CD8, NKG2D, CD8, CD16, KIR2DL4, KIR2DS1, KIR2DS2, KIR3DS1, NKG2C, NKG2E, NKG2D, NKp30, NKp44, NKp46, NKp80, DNAM-1, CD11b, CD11c, CD64, CD68, CD119, CD163, CD206, CD209, F4/80, IFGR2, Toll-like receptors 1-9, IL-4R α , or MARCO; or (ii) comprises a variable light chain comprising a CDRL1 of SEQ ID NO: 55, a CDRL2 of SEQ ID NO: 56, and a CDRL3 sequence of SEQ ID NO: 57 and a variable heavy chain comprising a CDRH1 of SEQ ID NO: 58, a CDRH2 of SEQ ID NO: 59, and a CDRH3 of SEQ ID NO: 60; or (iii) comprises SEQ ID NO: 61; or (iv) comprises a variable light chain comprising a CDRL1 of SEQ ID NO: 62, a CDRL2 of KVS, and a CDRL3 sequence of SEQ ID NO: 63 and a variable heavy chain comprising a CDRH1 of SEQ ID NO: 64, a CDRH2 of SEQ ID NO: 65, and a CDRH3 of SEQ ID NO: 66; or (v) comprises a variable light chain comprising a CDRL1 of SEQ ID NO: 67, a CDRL2 of KVS, and a CDRL3 sequence of SEQ ID NO: 63 and a variable heavy chain comprising a CDRH1 of SEQ ID NO: 69, a CDRH2 of SEQ ID NO: 70, and a CDRH3 of SEQ ID NO: 71; or (vi) comprises a variable light chain comprising a CDRL1 of SEQ ID NO: 72, a CDRL2 of KVS, and a CDRL3 sequence of SEQ ID NO: 63 and a variable heavy chain comprising a CDRH1 of SEQ ID NO: 69, a CDRH2 of SEQ ID NO: 75, and a CDRH3 of SEQ ID NO: 76; or (vii) comprises a variable light chain comprising a CDRL1 of SEQ ID NO: 77, a CDRL2 of KVS, and a CDRL3 sequence of SEQ ID NO: 78 and a variable heavy chain comprising a CDRH1 of SEQ ID NO: 79, a CDRH2 of SEQ ID NO: 80, and a CDRH3 of SEQ ID NO: 81; or (viii) comprises a variable light chain comprising a CDRL1 of SEQ ID NO: 82, a CDRL2 of SEQ ID NO: 83, and a CDRL3 sequence of SEQ ID NO: 84 and a variable heavy chain comprising a CDRH1 of SEQ ID NO: 85, a CDRH2 of SEQ ID NO: 86, and a CDRH3 of SEQ ID NO: 87; (ix) comprises a TCR; or (x) comprises a variable light chain comprising SEQ ID NO: 88 and a variable heavy chain comprising SEQ ID NO: 89; or (xi) any two or more of (i)-(x).

12. The kit of claim 8, wherein the CD33-targeting agent comprises a chimeric antigen receptor (CAR) comprising one or more binding domains.

13. The kit of claim 12, wherein: (i) the CAR comprises an anti-CD33 binding domain and a binding domain that activates an immune cell; or (ii) the CAR comprises an effector domain selected from 4-1BB, CD3 ϵ , CD3 δ , CD3 ζ , CD27, CD28, CD79A, CD79B, CARD11, DAP10, FcR α , FcR β , FcR γ , Fyn, HVEM, ICOS, Lck, LAG3, LAT, LRP, NOTCH1, Wnt, NKG2D, OX40, ROR2, Ryk, SLAMF1, Slp76, pTa, TCRA, TCRB, TRIM, Zap70, PTCH2, or any combination thereof; or (iii) the CAR comprises a cytoplasmic signaling sequence derived from CD3 zeta, FcR gamma, CD3 gamma, CD3 delta, CD3 epsilon, CD5, CD22, CD79a, CD79b, or CD66d; (iv) the CAR comprises an intracellular signaling domain and a costimulatory signaling region; or (v) the CAR comprises a spacer region; or (vi) the CAR comprises a transmembrane domain; (vii) the CAR comprises a costimulatory signaling region comprises the intracellular domain of CD27, CD28, 4-1BB, OX40, CD30, CD40, lymphocyte function-associated antigen-1, CD2, CD7, LIGHT, NKG2C, or B7-H3; or (viii) any two or more of (i)-(vii).

14. A method of genetically-modifying a cell to provide a therapeutic gene and to have reduced CD33 expression, the method comprising exposing the cell to an effective amount of a genetic construct of claim 4.

15. A method of protecting a cell from an anti-CD33 treatment comprising genetically-modifying the cell with a genetic construct of claim 1.

16. The method of claim 15, wherein: the cell is a therapeutic cell; and/or the protecting is in vivo.
